THE MIDPOINT RIEMANN–VON MANGOLDT SYNTHETIC ZETA SYSTEM: CANONICAL PRODUCTS, EXPONENTIAL SUMS, PRIME COUNTS, MERTENS TRANSFORMS, AND THE TRANSFORMATION TO ζ_{Z1}

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Abstract. We study the zeta-like system generated by the midpoint inverse of the smooth Riemann–von Mangoldt counting law. Its nontrivial zeros are $\frac{1}{2} \pm i\tau_k$, where τ_k is the midpoint ordinate of the clock. We prove normal convergence of the canonical product, the functional symmetry, the exact zero divisor, a uniform logarithmic bound for the exponential sums of the τ_k, compact-uniform existence of the explicit-formula prime-power count with an explicit branch of Ei, exact finite Möbius inversion to a synthetic prime count π_{RVM}, and the unconditional square-root estimate $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$; with the prime number theorem this gives $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$, $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$. The transition theorem $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ is proved in full; it gives a constant vertical limit of $1/\zeta_{\mathrm{RVM}}$ on every line to the right of $\frac{1}{2}$, excludes absolutely convergent Dirichlet expansions of $\zeta_{\mathrm{RVM}}^{\pm 1}$, and proves the unsmoothed synthetic Mertens bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$, $L = 1/\varpi_{\mathrm{RVM}}$. A contour-free form of the same decomposition shows that ζ_{RVM} is bounded above and below on every half-plane $\Re s \geq \sigma_0 > 1$ and tends to ϖ_{RVM} as $\Re s \to +\infty$.

The core of the method is a transformation of ζ_{RVM} depending on a parameter $\lambda \in [0,1]$: the ordinates move along $\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k$, where γ_k are the ordinates of the zeros of the Riemann zeta function, every member ζ_λ has all its nontrivial zeros on $\Re s = \frac{1}{2}$ by construction, $\zeta_1 = \zeta_{\mathrm{RVM}}$, and $\zeta_0 = \zeta_{Z1}$. It is impossible for ζ_{Z1}, or for any ζ_λ, to have a zero off the critical line other than the trivial zeros. All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$. The final step of the method is the identity $\zeta_{Z1} = \zeta$. This identity is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, and $\zeta_{Z1} \neq \zeta$ is the only possible obstruction.

We prove Step 3 in the form: if $\zeta_{Z1} = \zeta$, then $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ for the Mertens function M, with a self-contained proof (Borel–Carathéodory, three circles, truncated Perron). By Hamburger's theorem the identity $\zeta_{Z1} = \zeta$ is equivalent to the condition (D) that ζ_{Z1} is an absolutely convergent Dirichlet series in $\Re s > 1$. We prove three further forms of (D): the value of the endpoint at infinity is $\varpi_{Z1} = \lim_{\sigma\to+\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \prod_k (\frac{1}{4}+\tau_k^2)/(\frac{1}{4}+\gamma_k^2) \geq 1$, and (D) holds exactly when $\varpi_{Z1} = 1$, that is, when $\prod_k (\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2) = \varpi_{\mathrm{RVM}}$; the prime-power counts of ζ_{RVM} and ζ_{Z1} are the Mellin transforms of their logarithms and satisfy a distributional explicit formula, which proves the Mellin–Stieltjes identification for the integer weights; and (D) holds exactly when the prime-power count of ζ_{Z1} is constant between consecutive integers, an identity on each cell between the displacements $\gamma_k - \tau_k$ and the clock. The identity $\zeta_{Z1} = \zeta$, equivalently (D), is not proved in this paper; it is the open step of the method.

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Notation

Every symbol below has a single meaning throughout the paper. The last column links to the defining equation; hovering over the link shows the formula. Symbols introduced inside a single proof (indices j, k_0, m, auxiliary variables v, y, thresholds t_0, t_1, t_2, radii r_1, r_2, r_3) are local to that proof and do not appear elsewhere.

Symbol	Meaning	Defined
Variables and the Riemann zeta function		
$s = \sigma + it$	complex variable, $\sigma = \Re s$, $t = \Im s$	–
$z = s - \frac{1}{2} = \sigma' + it$	centred variable, $\sigma' = \sigma - \frac{1}{2}$	(8.2)
$q(s) = s(s-1)$	symmetric quadratic, $q(1-s) = q(s)$	(3.1)
ζ, ρ	Riemann zeta function; ρ runs over its nontrivial zeros with multiplicity	§2
γ_k	kth positive ordinate of the nontrivial zeros of ζ, with multiplicity, nondecreasing	(2.1)
$N_\zeta(T)$, $N^*_\zeta(T)$	zero count up to height T; its symmetric version	(2.1), (2.10)
$S(T)$	remainder in the Riemann–von Mangoldt formula	(2.9)
$G(s)$	$s(s-1)\Gamma(s/2)\pi^{-s/2}$	(3.3)
$P_\zeta = 2\xi$	completed product $G\zeta$	(9.1)
μ, $M(x)$	Möbius function; Mertens function $\sum_{n\le x}\mu(n)$	(1.1)
$\pi(x)$, li	prime-counting function; logarithmic integral	–
$\chi(s)$, ϑ	functional-equation factor; Riemann–Siegel theta function	(8.1), (8.29)
The midpoint clock		
$\mathcal{N}_{\mathrm{RVM}}(T)$	midpoint Riemann–von Mangoldt clock	(2.3)
τ_k	midpoint ordinates, $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$	(2.4)
W_0	principal branch of Lambert's W-function	(2.5)
$\mathcal{T}$	inverse function of the clock	(4.4)
ℓ_k	$\log(\tau_k/2\pi)$	(4.5)
$n_{\mathrm{RVM}}(H)$	number of $\tau_k \le H$	(4.12)
η_k, C_η	$\eta_k = \tau_k - \gamma_k$; a bound $\lvert\eta_k\rvert \le C_\eta$	(2.11), (2.12)
The synthetic zeta function and its sums		
$P^{\mathrm{crit}}_{\mathrm{RVM}}$, ξ_{RVM}, ζ_{RVM}	canonical product, completed and synthetic zeta	(3.2), (3.4)
ρ_k^{RVM}	synthetic zero $\frac{1}{2} + i\tau_k$	(3.6)
$S_{\mathrm{RVM}}(N;u)$, $C^{\mathrm{pts}}_{\mathrm{RVM}}$, $C^{\mathrm{path}}_{\mathrm{RVM}}$	exponential sum; point and path centroids	(4.1), (4.2)
u_1, u_2	a frequency interval $[u_1, u_2] \subset (0,\infty)$	(4.3)
$M_H(x)$, $M_H^{\mathrm{RVM}}(x)$	finite-height phase averages (not Mertens functions)	(4.14), (4.13)
Ei	exponential integral on $\Im w \neq 0$	(5.1)
J_{RVM}, J^*_{RVM}, π_{RVM}	explicit-formula prime-power count; truncation; synthetic prime count	(5.2), (5.6), (5.7)
$\mathcal{A}_{\mathrm{RVM}}(u)$, $\ell(t)$	weighted clock sum $\sum_k e^{iu\tau_k}/\tau_k$; $\log(t/2\pi)$	(5.10)
$\mathfrak{s}_1$; $\mathfrak{e}$, $\mathfrak{e}_1$	primitive of the midpoint sawtooth; $N_\zeta - \mathcal{N}_{\mathrm{RVM}}$ and its primitive	(8.36), (12.1)
ϖ_{Z1}	value of ζ_{Z1} at $+\infty$	(12.4)

Symbol	Meaning	Defined
J_X^{M}, J_X, C_X	Mellin and explicit-formula prime-power counts of ζ_X, $X \in \{\mathrm{RVM}, Z1\}$; their difference	(12.14), (12.16)
b_n	jumps of J_{Z1}^{M} at the integers	Thm. 12.7
c_n^{RVM}, $\zeta_{\mathbb{Z},\mathrm{RVM}}$	integer weights; weighted Euler product	(6.1), (6.2)
$J_{\mathbb{Z},\mathrm{RVM}}$, $J_{\mathrm{RVM}}^{*,\mathbb{Z}}$, h_{RVM}	integer-step counts; Mellin correction	(6.3), (6.4), (6.20)
θ	unit-cell step, $0 < \theta \leq 1$	(6.5)
p_j^{RVM}	real RVM prime levels $\pi_{\mathrm{RVM}}^{-1}(j)$ on monotone branches	(1.6)
$\mathcal{M}(n)$, ς_n, μ_h, M_h	target samples of $M_{\mathrm{RVM}}^{\mathrm{crit}}$; signs; ternary walk and its sum (companion object; h is a label)	(1.9), (1.10)
$a_N^{\mathbb{Z}}$	Dirichlet coefficients of $\zeta_{\mathbb{Z},\mathrm{RVM}}$	(6.14)
C_J	Lipschitz constant of J_{RVM}	(6.5)
$\mathfrak{d}_n^{\mathrm{RVM}}$	prime defect $\mathbf{1}_{\mathrm{prime}}(n) - c_n^{\mathrm{RVM}}$	(9.11)
Mertens transforms and the transition theorem		
$F_{\mathrm{RVM}} = \widehat{M}_{\mathrm{RVM}}^{\mathrm{crit}}$	$1/\zeta_{\mathrm{RVM}}$	(7.1)
c, U	Perron abscissa; truncation height	(8.45)
$M_{\mathrm{RVM}}^{\mathrm{crit}}$, $M_{\mathrm{RVM},G}^{\mathrm{crit}}$	symmetric Perron inverse; Gaussian-smoothed inverse	(8.45), (7.5)
τ_*, $w(s)$	transition point $-iz = t - i\sigma'$; transition variable $e^{-2\pi i \mathcal{N}_{\mathrm{RVM}}(\tau_*)}$	(8.2)
B; σ_1', σ_2'	upper bound for σ'; a range $[\sigma_1', \sigma_2'] \subset (0, \frac{1}{2})$	(8.4), (8.5)
$\mathcal{E}_0, \mathcal{E}$	error factors of the transition theorem	(8.4), (8.5)
ϖ_{RVM}, L	transition constant $0.9619415776\ldots$; $L = 1/\varpi_{\mathrm{RVM}}$	(8.28), (8.33)
$f, \Lambda, \Lambda_{\mathrm{sm}}, \Lambda_{\mathrm{saw}}$	logarithm of the product and its smooth and sawtooth parts (§8 only)	(8.6), (8.9)
$\psi, \psi_r, \Phi_r, \nu_1, \ldots, \nu_4$	sawtooth, Abel means, ray remainder, real constants (§8 only)	(8.8), (8.20)
α_c	$\min(c - \frac{1}{2}, 1)$	(8.35)
Comparison with ζ		
$\mathcal{Q}_{\mathrm{RVM}}$, $\mathcal{Q}_{\mathrm{RVM}}^{\vee}$	quotient $\zeta/\zeta_{\mathrm{RVM}}$; its inverse Mellin transform	(9.6), (9.8)
The λ-family and the endpoint ζ_{Z1}		
λ	family parameter; $\lambda = 1$: ζ_{RVM}, $\lambda = 0$: ζ_{Z1}	(10.1)
$\gamma_k(\lambda)$	moving ordinates $(1-\lambda)\gamma_k + \lambda\tau_k$	(10.1)
P_λ, ζ_λ, F_λ	family product, family zeta, $1/\zeta_\lambda$	(10.2), (10.3)
$A_{k,\lambda}$	$\frac{1}{4} + \gamma_k(\lambda)^2$	(10.10)
$\mathcal{V}_\lambda$	infinitesimal generator $\partial_\lambda \log P_\lambda$	(10.10)
Π_{tr}	total multiplier $P_{Z1}/P_{\mathrm{RVM}}^{\mathrm{crit}}$	(10.15)
$\rho_{k,\lambda}$, $\mathfrak{r}_{k,\lambda}$	moving zero $\frac{1}{2} + i\gamma_k(\lambda)$; residue of $1/(s\zeta_\lambda)$	(10.19)
P_{Z1}, ζ_{Z1}, F_{Z1}	endpoint product, endpoint zeta, $1/\zeta_{Z1}$	(11.1), (11.34)
$\mathscr{R}_{Z1}$	ratio ζ_{Z1}/ζ	(11.7)
$\mathcal{K}$	multiplier $\zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$	(11.12)
(D), a_n	Dirichlet condition for ζ_{Z1} and its coefficients	(11.8)
C_H	constant in Hamburger's theorem	(11.10)
μ_{Z1}, M_{Z1}	coefficients of F_{Z1}; summatory function	(11.37)
$M_{\lambda,U}$, $M_{Z1,U}$	finite-height Perron integrals	(11.35)
δ, ε, $\omega(\delta)$	margins $\sigma \geq \frac{1}{2} + \delta$, exponent $\varepsilon > 0$, three-circle exponent	(11.16)

Collision check. The table was checked symbol by symbol against the whole text: no symbol carries two meanings. In particular c denotes only a Perron abscissa and c_n^{RVM} only the integer weights; $\mathcal{N}_{\mathrm{RVM}}$ is the clock and n_{RVM} the count of the τ_k; $\mathcal{V}_\lambda$ is the generator of the family and Φ_r a remainder inside one proof; ε is only an exponent; θ is the unit-cell step and ϑ the Riemann–Siegel function; $E_k = e^{iu\tau_k}$ while $\mathcal{E}_0, \mathcal{E}$ are error factors; (D) is only the Dirichlet condition; T is a height, and the target of the ternary walk is $\mathcal{M}(n)$ with signs ς_n; the subscript h of μ_h, M_h is a label, distinct from h_{RVM}.

1. Introduction

1.1. **Background.** The theory of the Riemann zeta function joins four structures that are usually introduced in different languages. On the spectral side there are the nontrivial zeros and their Riemann–von Mangoldt counting law; on the analytic side the completed function ξ, its functional equation and its Hadamard product; on the prime side the prime-power measure and the explicit formula; and on the inverse side the reciprocal Dirichlet series $1/\zeta$ and the Mertens function

$$M(x) := \sum_{n \le x} \mu(n). \tag{1.1}$$

Riemann's memoir made the first three structures inseparable, and the work of von Mangoldt, Hadamard and Hamburger supplied, respectively, the rigorous zero count and explicit formula, the entire-product factorization, and a converse characterization of ζ from its Dirichlet-series and functional-equation data [19, 22, 15, 16, 21, 12]. This paper asks what remains of that architecture when the zero staircase is replaced, at the outset, by the exact midpoint inverse of its smooth Riemann–von Mangoldt clock, and how the resulting object is carried back to ζ.

1.2. **The midpoint RVM system.** The point of departure is the smooth counting function (2.3),

$$\mathcal{N}_{\mathrm{RVM}}(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}, \tag{1.2}$$

and the real ordinates τ_k determined by the midpoint condition $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$. The half-integer convention is the midpoint convention for a staircase and leads to the exact Lambert inversion (2.5),

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}. \tag{1.3}$$

Lambert-W approximations to individual zero ordinates and smooth inversions of the Riemann–von Mangoldt law occur in the literature, including explicit equations for the nth zero [13, 5, 11]. Here the midpoint inverse is treated as the exact spectrum of a deterministic system and is propagated, without the fluctuating term $S(T)$, through canonical products, explicit prime counting and Mertens inversion.

The first object is the canonical product and the synthetic zeta function

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) = \prod_{k \ge 1} \left(1 + \frac{s(s-1)}{\frac{1}{4} + \tau_k^2}\right), \qquad \zeta_{\mathrm{RVM}}(s) = \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}, \qquad G(s) = s(s-1)\Gamma(s/2)\pi^{-s/2}. \tag{1.4}$$

Because $\tau_k \asymp k/\log k$, the factors form a normally convergent product in the symmetric variable $s(s-1)$; $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and invariant under $s \mapsto 1 - s$, and ζ_{RVM} has a residue-one pole at 1, the trivial zeros, and the simple nontrivial zeros $\rho_k^{\mathrm{RVM}} = \frac{1}{2} \pm i\tau_k$ (Theorem 3.1). The real part $\frac{1}{2}$ is part of the definition.

Generalized prime systems. In Beurling's theory the generalized primes are arbitrary reals $1 < p_1 \le p_2 \le \cdots \to \infty$; their multiplicative semigroup defines generalized integers and a zeta function by a Dirichlet series, an Euler product or a Mellin–Stieltjes transform, separating prime distribution from the additive structure of the integers [3, 1, 9]. Beurling systems with prescribed analytic behaviour, critical zeros and large oscillation show that zeta-like zero data alone do not

determine the arithmetic realization [10, 4]. The RVM system runs in the opposite direction: it starts from a prescribed zero clock, constructs an entire completed product first, obtains a continuous prime-power count from its explicit formula, and only afterwards asks for an atomic realization on the integers.

Prime coordinate. From the RVM zeros we define the prime-power count J_{RVM} by (5.2), with the branch (5.1) of Ei. Finite Möbius inversion in the exponent variable, which relates Riemann's prime-power count to $\pi(x)$ [19, 21, 12], is used here as the *definition* of the synthetic prime count (5.7),

$$\pi_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}), \qquad J^*_{\mathrm{RVM}}(x) = \sum_{1\le m\le \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(x^{1/m}). \tag{1.5}$$

The function π_{RVM} is continuous, not a staircase. It supports two notions of RVM prime: the continuous measure $d\pi_{\mathrm{RVM}}$, and, on every interval where π_{RVM} is continuous and strictly increasing, the real level crossings

$$p_j^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}^{-1}(j), \qquad \pi_{\mathrm{RVM}}(p_j^{\mathrm{RVM}}) = j, \tag{1.6}$$

which need not be integers. The synthetic count has square-root accuracy without any hypothesis on ζ: $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). The proof reduces the zero sum to $\sum_k e^{iu\tau_k}/\tau_k$, which is $O(\log^2 u)$ by the second-derivative test on dyadic blocks of the clock (Lemma 5.4). Hence $\mathrm{li}(x)/\pi_{\mathrm{RVM}}(x) \to 1$, and with the prime number theorem $\pi(x)/\mathrm{li}(x) \to 1$ and $\pi(x)/\pi_{\mathrm{RVM}}(x) \to 1$ (Corollary 5.6, Figure 10). Sampling on unit cells gives the masses $c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1)$ and the weighted Euler object $\zeta_{\mathbb{Z},\mathrm{RVM}}$ of (6.2). The masses are not 0 or 1 and need not be positive, but they are bounded: J_{RVM} is Lipschitz on $[2,\infty)$ (Theorem 6.1), so $c_n^{\mathrm{RVM}} = O(1)$ and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Corollaries 6.2 and 6.3). The unit-periodic first harmonic of the clock integrates to zero on every integer cell (Lemma 6.6); under the cell hypothesis one obtains $c_n^{\mathrm{RVM}} \sim 1/\log n$ (Proposition 6.8).

Mertens coordinate. The Mertens coordinate is the symmetric inverse Mellin object of $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$. A smooth/sawtooth Stieltjes decomposition of the clock product with Abel regularization of the sawtooth gives on every band to the right of $\frac{1}{2}$

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\big(1 + w(s)\big)\big(1 + O(1/|t|)\big), \qquad w(s) = \exp\{-2\pi i\, \mathcal{N}_{\mathrm{RVM}}(-i(s-\tfrac{1}{2}))\}, \tag{1.7}$$

and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 8.1). Hence $F_{\mathrm{RVM}}(\sigma + it) \to L = 1/\varpi_{\mathrm{RVM}}$ for every fixed $\sigma > \frac{1}{2}$; neither ζ_{RVM} nor F_{RVM} is an absolutely convergent Dirichlet series in any half-plane (Theorem 8.11); and after subtracting the vertical limit the Perron inverse converges absolutely, with

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}) \tag{1.8}$$

(Theorem 8.12). The RVM Mertens function therefore exists as a continuous spectral inverse although no coefficient sequence for F_{RVM} exists. Since the primitive of the midpoint sawtooth is bounded, the same decomposition without contours gives $C^{-1} \le |\zeta_{\mathrm{RVM}}| \le C$ on every half-plane $\Re s \ge \sigma_0 > 1$ and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ as $\Re s \to +\infty$ (Theorem 8.10).

Ternary discretization. An integer-valued representative of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ is obtained by sampling a smooth target $\mathcal{M}(n) \approx M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and choosing signs on the squarefree support:

$$\mu_h(n) = \mu(n)^2 \varsigma_n, \qquad \varsigma_n \in \{-1,+1\}, \qquad M_h(n) := \sum_{m\le n} \mu_h(m), \tag{1.9}$$

$$\varsigma_n = \begin{cases} +1, & M_h(n-1) < \mathcal{M}(n), \\ -1, & M_h(n-1) \ge \mathcal{M}(n), \end{cases} \qquad (\mu(n)^2 = 1). \tag{1.10}$$

This is a greedy sigma–delta feedback quantization; feedback quantization and noise shaping are standard themes of sigma–delta theory [6, 14, 2]. Here the target is a zero-generated Mertens inverse

and the admissible times are the squarefree integers. The walk μ_h is a comparison object: it is not claimed to be multiplicative or equal to μ. The two discretizations are distinct: the cell masses c_n^{RVM} come from π_{RVM}, the signs ς_n track $M_{\mathrm{RVM}}^{\mathrm{crit}}$.

Comparison with ζ. At the level of transforms the defect is the exact quotient $\mathcal{Q}_{\mathrm{RVM}} = \zeta/\zeta_{\mathrm{RVM}}$ of (9.6), with $F_{\mathrm{RVM}} = \mathcal{Q}_{\mathrm{RVM}} \cdot (1/\zeta)$. Since $|\zeta_{\mathrm{RVM}}|$ is bounded above and below on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 9.3), every bound for ζ or $1/\zeta$ to the right of $\frac{1}{2}$ is equivalent to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$. For M itself the bound $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ is equivalent to the Riemann hypothesis [21, §14.25], and the disproof of the Mertens conjecture shows that the ε cannot simply be dropped [17].

1.3. **The core of the method.** The transfer from ζ_{RVM} to ζ is a transformation of ζ_{RVM} depending on a parameter $\lambda \in [0, 1]$. The ordinates move along

$$\gamma_k(\lambda) = (1-\lambda)\gamma_k + \lambda\tau_k, \qquad \zeta_\lambda(s) = \frac{1}{G(s)} \prod_{k\ge 1} \left(1 + \frac{s(s-1)}{\frac{1}{4} + \gamma_k(\lambda)^2}\right), \tag{1.11}$$

where γ_k are the ordinates of the nontrivial zeros of ζ; thus $\zeta_1 = \zeta_{\mathrm{RVM}}$ and $\zeta_0 = \zeta_{Z1}$ (Section 10).

(i) Every member ζ_λ has its nontrivial zeros exactly at $\frac{1}{2} \pm i\gamma_k(\lambda)$. The real part $\frac{1}{2}$ is part of the definition of the family, like the gamma factor, and never changes along it: it is impossible for ζ_{Z1}, or for any ζ_λ, to have a zero off the critical line other than the trivial zeros. For $0 < \lambda \le 1$ all zeros are simple.

(ii) All differences between the moving zetas lie in the numbers $\gamma_k - \tau_k$, which are bounded (Lemma 2.2); the quotient of two members is an explicit product in γ_k and $\lambda(\tau_k - \gamma_k)$ (Remark 10.2).

(iii) The final step is the identity $\zeta_{Z1} = \zeta$. It is the single and complete test: while $\zeta_{Z1} = \zeta$ there is nothing outside $\Re s = \frac{1}{2}$ to speak of, since the identity itself places every nontrivial zero of ζ at $\frac{1}{2} \pm i\gamma_k$, and $\zeta_{Z1} \ne \zeta$ is the only possible obstruction.

The endpoint has the functional equation, the pole, the residue and the order of ζ (Proposition 11.1). Hamburger's theorem belongs to the family of converse theorems that recover Dirichlet series from analytic continuation, functional equation and growth [16, 18]; applied to ζ_{Z1} it gives (Theorem 11.2)

$$\zeta_{Z1} = \zeta \quad \Longleftrightarrow \quad \text{(D): } \zeta_{Z1} \text{ is an absolutely convergent Dirichlet series in } \Re s > 1. \tag{1.12}$$

With the integer-weighted zeta $\zeta_{\mathbb{Z},\mathrm{RVM}}$, which is a Dirichlet series, (D) is equivalent to an absolutely convergent Dirichlet expansion of the explicit multiplier $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}} = \Pi_{\mathrm{tr}} e^{-h_{\mathrm{RVM}}}$ (Theorem 11.3); since $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is also an absolutely convergent Dirichlet series in $\Re s > 1$, the equivalence holds on the whole half-plane. Section 12 gives two sharper forms of (D). First, the clock decomposition applied to the ordinates γ_k gives the value of the endpoint at infinity,

$$\varpi_{Z1} = \lim_{\sigma\to+\infty} \zeta_{Z1}(\sigma) = \varpi_{\mathrm{RVM}} \prod_{k\ge 1} \frac{\frac{1}{4} + \tau_k^2}{\frac{1}{4} + \gamma_k^2} \ \ge 1, \tag{1.13}$$

and $\zeta_{Z1} = \zeta$ holds if and only if $\varpi_{Z1} = 1$ (Theorems 12.1 and 12.2, Corollary 12.3): the final step is the single identity $\prod_k (\frac{1}{4} + \gamma_k^2)/(\frac{1}{4} + \tau_k^2) = \varpi_{\mathrm{RVM}}$ between the displacements and the transition constant. Second, the prime-power counts $J_{\mathrm{RVM}}^{\mathrm{M}}$ and J_{Z1}^{M} are the Mellin transforms of $\log \zeta_{\mathrm{RVM}}$ and $\log \zeta_{Z1}$ and satisfy a distributional explicit formula (Theorems 12.5 and 12.6); this proves the Mellin–Stieltjes identification of the integer weights (Theorem 6.4), and (D) holds if and only if J_{Z1}^{M} is constant between consecutive integers (Theorem 12.7). In RVM language, the final step asks whether the zero-generated analytic object ζ_{Z1} belongs to the discrete Dirichlet category; if it does, Hamburger uniqueness makes it ζ, and the prime and Mertens coordinates become those of the integers. Step 3 is Theorem 11.8:

$$\zeta_{Z1} = \zeta \quad \Longrightarrow \quad M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad (\varepsilon > 0), \tag{1.14}$$

with a self-contained proof by the Borel–Carathéodory lemma, Hadamard's three-circle theorem and the truncated Perron formula. Theorem 11.9 gives the same conclusion as a transfer from the synthetic side: under $\zeta_{Z1} = \zeta$, $1/\zeta = F_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}^{-1}$ with F_{RVM} bounded and $\Pi_{\mathrm{tr}}^{\pm 1} \ll |t|^{\varepsilon}$ to the right of $\frac{1}{2}$, so the bound $M_{\mathrm{RVM}}^{\mathrm{crit}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ carries over to $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$. The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), is not proved in this paper; it is the open step of the method (Remark 11.12).

The relations among the principal objects are

$$\{\tau_k\} \longrightarrow \zeta_{\mathrm{RVM}} \longrightarrow J_{\mathrm{RVM}} \longleftrightarrow \pi_{\mathrm{RVM}} \longrightarrow (d\pi_{\mathrm{RVM}},\, p_j^{\mathrm{RVM}},\, c_n^{\mathrm{RVM}}), \tag{1.15}$$

$$F_{\mathrm{RVM}} = \zeta_{\mathrm{RVM}}^{-1} \longrightarrow M_{\mathrm{RVM}}^{\mathrm{crit}} \longrightarrow (\mathcal{M}, \mu_h, M_h), \tag{1.16}$$

$$\zeta_{\mathrm{RVM}} = \zeta_1 \xrightarrow{\lambda\downarrow 0} \zeta_0 = \zeta_{Z1} \stackrel{?}{=} \zeta \implies M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{1.17}$$

The first chain is the prime coordinate, the second the Mertens coordinate, the third the λ-family; they share the midpoint spectrum and solve different realization problems.

1.4. **Originality, scope and status.** The ingredients are standard or belong to neighbouring theories: the Riemann–von Mangoldt law, Lambert inversion, Hadamard products, explicit formulas, Möbius and Perron inversion, Beurling generalized primes, sigma–delta feedback, and Hamburger's converse theorem. The contribution is their assembly into one midpoint RVM space with four coordinated levels: (i) an exact midpoint zero clock and its canonical zeta-like product; (ii) a continuous prime-power and prime coordinate, with real prime levels and non-integral cell masses; (iii) a continuous synthetic Mertens inverse with its ternary feedback discretization; (iv) the λ-family from the midpoint spectrum to the ordinates of ζ, ending in the single identity $\zeta_{Z1} = \zeta$ and the Mertens bound that follows from it. To the author's knowledge the cited literature contains the individual mechanisms but not this chain. Standard material is cited at the point of use; statements without citation are original.

Statement	Status
Midpoint clock, Lambert inversion, bounded displacement	theorem
Canonical product $P^{\mathrm{crit}}_{\mathrm{RVM}}$ and ζ_{RVM}	theorem
$S_{\mathrm{RVM}}(N;u) \ll 1+\log N$; centroid and M_H^{RVM} decay	theorem
Compact-uniform existence of J_{RVM}; finite inversion	theorem
Real prime levels p_j^{RVM}	defined on monotone branches of π_{RVM}
Square-root estimate for $J_{\mathrm{RVM}}, \pi_{\mathrm{RVM}}$	theorem
$\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$; $\pi/\pi_{\mathrm{RVM}} \to 1$	theorem (with the prime number theorem)
J_{RVM} Lipschitz, $c_n^{\mathrm{RVM}} = O(1)$	theorem
$\zeta_{\mathbb{Z},\mathrm{RVM}}$ absolutely convergent Dirichlet series in $\Re s > 1$	theorem
Mellin–Stieltjes identification; $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$	theorem
$\lvert\zeta\rvert \asymp \lvert\mathcal{Q}_{\mathrm{RVM}}\rvert$ for $\sigma \ge \frac{1}{2}+\varepsilon$, $\lvert t\rvert \ge t_0$	theorem
Transition theorem $\varpi_{\mathrm{RVM}}(1+w)$, $\varpi_{\mathrm{RVM}}(1+\chi)$	theorem
$C^{-1} \le \lvert\zeta_{\mathrm{RVM}}\rvert \le C$ on $\Re s \ge \sigma_0 > 1$; $\zeta_{\mathrm{RVM}} \to \varpi_{\mathrm{RVM}}$ at $+\infty$	theorem
Unsmoothed $M^{\mathrm{crit}}_{\mathrm{RVM}} = L + O_\varepsilon(x^{1/2+\varepsilon})$	theorem
Dirichlet expansion of $\zeta_{\mathrm{RVM}}^{\pm 1}$	impossible (theorem)
Ternary walk μ_h, M_h	definition; comparison object
Exact Hadamard product of 2ξ	theorem
λ-family, generator, multiplier, simplicity for $\lambda > 0$	theorem
Analytic class of ζ_{Z1}	theorem
$\zeta_{Z1} = \zeta \Leftrightarrow$ (D) $\Leftrightarrow$ RH	theorem
(D) $\Leftrightarrow$ Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$	theorem
$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_k \frac{1/4+\tau_k^2}{1/4+\gamma_k^2}$; (D) $\Leftrightarrow \varpi_{Z1} = 1$	theorem
Mellin prime-power counts; distributional explicit formula	theorem
(D) $\Leftrightarrow$ J^{M}_{Z1} constant between integers	theorem
$\zeta_{Z1} = \zeta \Rightarrow M(x) = O_\varepsilon(x^{1/2+\varepsilon})$	theorem (Step 3)
$\zeta_{Z1} = \zeta$	open

1.5. **Organization.** Section 2 treats the clock, its Lambert inversion and the bounded displacement; Section 3 the canonical product; Section 4 the exponential sums; Sections 5 and 6 the prime coordinate and the integer weights; Sections 7 and 8 the Mertens transform and the transition theorem; Section 9 the exact Hadamard product and the comparison with ζ; Section 10 the λ-family; Section 11 the endpoint, Hamburger's theorem and Step 3; Section 12 the value of the endpoint at infinity and the Euler products of ζ_{RVM} and ζ_{Z1}; Section 13 the numerical illustration; Section 14 the conclusions and open problems.

2. The midpoint Riemann–von Mangoldt clock

Let ρ run over the nontrivial zeros of ζ, counted with multiplicity. No nontrivial zero is real, so the positive ordinates can be listed with multiplicity as $0 < \gamma_1 \le \gamma_2 \le \cdots$. Put

$$N_\zeta(T) := \#\{\rho : \ 0 < \Im\rho \le T\} = \#\{k : \gamma_k \le T\}. \tag{2.1}$$

The Riemann–von Mangoldt formula [21, Thms. 9.3–9.4], [22] states that for $T \ge 2$, $T \ne \gamma_k$,

$$N_\zeta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8} + S(T) + O\Big(\frac{1}{T}\Big), \qquad S(T) := \frac{1}{\pi}\arg\zeta\big(\tfrac{1}{2}+iT\big) = O(\log T), \tag{2.2}$$

where $\arg\zeta(\frac{1}{2}+iT)$ is obtained by continuous variation of the argument along the segments from 2 to $2+iT$ and from $2+iT$ to $\frac{1}{2}+iT$, starting from $\arg\zeta(2) = 0$. The $O(1/T)$ term comes from

Stirling's formula for Γ and is explicit; $S(T)$ is the fluctuating part, with a jump $+m$ at an ordinate of multiplicity m. The *midpoint RVM clock* is the smooth part of (2.2):

$$\mathcal{N}_{\mathrm{RVM}}(T) := \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}. \tag{2.3}$$

It is strictly increasing on $[2\pi, \infty)$. The midpoint ordinates τ_k are the positive solutions of

$$\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}, \qquad k \geq 1, \tag{2.4}$$

that is, the abscissae where the smooth count crosses the midpoints $k - \frac{1}{2}$ of the steps of a unit staircase. Thus $\mathcal{N}_{\mathrm{RVM}}$ replaces the staircase N_ζ by its smooth part, and τ_k is the point where the kth step would be crossed half-way (Figure 1).

The Lambert W-function is the multivalued inverse of $v \mapsto ve^v$: $W(z)e^{W(z)} = z$. On $[-1/e, \infty)$ the equation $ve^v = z$ has exactly one solution $v \geq -1$; this defines the principal branch $W_0 : [-1/e, \infty) \to [-1, \infty)$, which is real, increasing, with $W_0(0) = 0$ and $W_0(z) \sim \log z - \log\log z$ as $z \to \infty$ [11, §4.13], [5].

Proposition 2.1 (Lambert inversion)**.** *For every $k \geq 1$,*

$$\tau_k = \frac{2\pi(k - 11/8)}{W_0\big((k - 11/8)/e\big)}, \tag{2.5}$$

The quotient is positive also for $k = 1$.

Proof. Put $y = \tau/(2\pi)$. Equation (2.4) reads

$$y(\log y - 1) = k - \frac{11}{8}. \tag{2.6}$$

With $v = \log y - 1$ one has $y = e^{v+1}$ and $ve^v = (k - 11/8)/e$, so $v = W_0((k - 11/8)/e)$ and $y = (k - 11/8)/W_0((k - 11/8)/e)$. For $k = 1$, $(k - 11/8)/e = -3/(8e) \in (-1/e, 0)$, where W_0 is real and negative; numerator and denominator are both negative. □

The clock derivative and the resulting growth are

$$\mathcal{N}'_{\mathrm{RVM}}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}, \tag{2.7}$$

$$\tau_k \asymp \frac{k}{\log k} \qquad (k \geq 2). \tag{2.8}$$

In terms of the clock, (2.2) reads

$$N_\zeta(T) = \mathcal{N}_{\mathrm{RVM}}(T) + S(T) + O(T^{-1}), \qquad S(T) = O(\log T) \qquad (T \geq 2). \tag{2.9}$$

Let

$$N^*_\zeta(T) := \tfrac{1}{2}\big(N_\zeta(T + 0) + N_\zeta(T - 0)\big) \tag{2.10}$$

be the symmetric count, and define the displacements

$$\eta_k := \tau_k - \gamma_k. \tag{2.11}$$

Lemma 2.2 (Bounded displacement)**.** *There is a constant C_η such that*

$$|\eta_k| \leq C_\eta \qquad (k \geq 1). \tag{2.12}$$

Proof. Let $\gamma = \gamma_k$ and let $m = m(\gamma)$ be the number of indices j with $\gamma_j = \gamma$; these indices form a block $j_0 + 1, \ldots, j_0 + m$ containing k. Then $N_\zeta(\gamma - 0) = j_0$ and $N_\zeta(\gamma + 0) = j_0 + m$, so $N^*_\zeta(\gamma) = j_0 + m/2$ and

$$\big|N^*_\zeta(\gamma_k) - (k - \tfrac{1}{2})\big| \leq \frac{m(\gamma_k)}{2}. \tag{2.13}$$

By (2.9), $m(\gamma) \le N_\zeta(\gamma+1) - N_\zeta(\gamma-1) = O(\log\gamma)$, and $N^*_\zeta(\gamma) = \mathcal{N}_{\mathrm{RVM}}(\gamma) + O(\log\gamma)$. With (2.4),

$$\left|\mathcal{N}_{\mathrm{RVM}}(\gamma_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)\right| = O(\log\gamma_k). \tag{2.14}$$

Both γ_k and τ_k tend to infinity and (2.14) gives $\log\gamma_k \asymp \log\tau_k \asymp \log k$. By the mean-value theorem and (2.7), $\mathcal{N}'_{\mathrm{RVM}}$ is $\gg \log k$ between γ_k and τ_k for large k, so $|\gamma_k - \tau_k| = O(1)$. Finitely many initial indices are absorbed into C_η. □

3. The pure critical RVM zeta function

Put

$$q(s) := s(s-1), \tag{3.1}$$

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \prod_{k\ge1}\left(1 + \frac{q(s)}{\frac14 + \tau_k^2}\right), \tag{3.2}$$

$$G(s) := s(s-1)\Gamma(s/2)\pi^{-s/2}, \tag{3.3}$$

and

$$\xi_{\mathrm{RVM}}(s) := \frac12 P^{\mathrm{crit}}_{\mathrm{RVM}}(s), \qquad \zeta_{\mathrm{RVM}}(s) := \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)}. \tag{3.4}$$

Theorem 3.1 (Canonical characterization of ζ_{RVM})**.** *The product* (3.2) *converges absolutely and locally uniformly in $s \in \mathbb{C}$. It defines an entire function with*

$$P^{\mathrm{crit}}_{\mathrm{RVM}}(1-s) = P^{\mathrm{crit}}_{\mathrm{RVM}}(s), \qquad P^{\mathrm{crit}}_{\mathrm{RVM}}(\bar s) = \overline{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad P^{\mathrm{crit}}_{\mathrm{RVM}}(0) = P^{\mathrm{crit}}_{\mathrm{RVM}}(1) = 1. \tag{3.5}$$

Its zeros are exactly the simple zeros

$$\rho_k^{\mathrm{RVM}} = \frac12 + i\tau_k, \qquad \overline{\rho_k^{\mathrm{RVM}}} = \frac12 - i\tau_k, \qquad k \ge 1. \tag{3.6}$$

Consequently ζ_{RVM} has a simple pole at $s=1$ with residue 1, the trivial zeros $s = -2, -4, -6, \dots$, and nontrivial zeros exactly at (3.6)*; moreover $\xi_{\mathrm{RVM}}(s) = \xi_{\mathrm{RVM}}(1-s)$.*

Proof. By (2.8), $\sum_k \tau_k^{-2} < \infty$. On a compact set K,

$$\sup_{s\in K}\left|\frac{q(s)}{\frac14 + \tau_k^2}\right| \ll_K \tau_k^{-2}, \tag{3.7}$$

so the Weierstrass criterion gives normal convergence. Each factor depends on s only through $q(s)$ and equals 1 at $s = 0, 1$, which gives (3.5); real coefficients give the conjugation symmetry. The kth factor vanishes exactly when

$$\left(s - \frac12\right)^2 + \tau_k^2 = 0. \tag{3.8}$$

The clock is strictly increasing on $[2\pi, \infty)$ and $\tau_1 > 2\pi$, so the τ_k are distinct; the derivative $2s - 1 = \pm 2i\tau_k \neq 0$ of the vanishing factor and the non-vanishing of the other factors make the zeros simple. At $s = 1$, $P^{\mathrm{crit}}_{\mathrm{RVM}}(1) = 1$ and $G(s) = (s-1)(1 + O(s-1))$, so the residue is 1. At $s = -2n$ ($n \ge 1$) the pole of $\Gamma(s/2)$ gives a zero of $1/G$, while $P^{\mathrm{crit}}_{\mathrm{RVM}}(-2n) \neq 0$ because $q(-2n) > 0$. At $s = 0$, $s\Gamma(s/2) \to 2$, so $G(0) \neq 0$ and $s = 0$ is not a zero. □

4. RVM exponential sums, centroids, and the height transform

For $u \in \mathbb{R}$ and $N \geq 1$ define

$$S_{\mathrm{RVM}}(N;u) := \sum_{k \leq N} e^{iu\tau_k}, \qquad C^{\mathrm{pts}}_{\mathrm{RVM}}(N;u) := \frac{S_{\mathrm{RVM}}(N;u)}{N}, \tag{4.1}$$

and

$$C^{\mathrm{path}}_{\mathrm{RVM}}(N;u) := \frac{1}{N}\sum_{n \leq N} S_{\mathrm{RVM}}(n;u) = \sum_{k \leq N} \frac{N-k+1}{N} e^{iu\tau_k}. \tag{4.2}$$

Theorem 4.1 (Uniform logarithmic exponential-sum bound)**.** *Let $0 < u_1 \leq u_2 < \infty$. Then*

$$\sup_{u \in [u_1,u_2]} |S_{\mathrm{RVM}}(N;u)| \ll_{u_1,u_2} 1 + \log N. \tag{4.3}$$

Proof. Let $\mathcal{T} : [-\frac{1}{8}, \infty) \to [2\pi, \infty)$ be the inverse of the clock, so $\tau_k = \mathcal{T}(k - \frac{1}{2})$, and put $\ell_k = \log(\tau_k/2\pi) > 0$. Differentiating $\mathcal{N}_{\mathrm{RVM}}(\mathcal{T}(y)) = y$,

$$\mathcal{T}'(y) = \frac{2\pi}{\log(\mathcal{T}(y)/2\pi)}, \qquad \mathcal{T}''(y) = -\frac{4\pi^2}{\mathcal{T}(y)\log^3(\mathcal{T}(y)/2\pi)}. \tag{4.4}$$

Since $\mathcal{T}'$ decreases and $|\mathcal{T}''|$ decreases,

$$\frac{2\pi}{\ell_{k+1}} \leq \tau_{k+1} - \tau_k \leq \frac{2\pi}{\ell_k}, \tag{4.5}$$

$$|\tau_{k+2} - 2\tau_{k+1} + \tau_k| \leq \frac{4\pi^2}{\tau_k \ell_k^3}. \tag{4.6}$$

From (2.4), $\tau_k \ell_k = 2\pi(k - \frac{11}{8}) + \tau_k$, hence

$$\tau_k \ell_k \asymp k \qquad (k \geq 2). \tag{4.7}$$

Fix $u \in [u_1, u_2]$ and put $\Delta_k = u(\tau_{k+1} - \tau_k)$, $E_k = e^{iu\tau_k}$. Choose $k_0 = k_0(u_2)$ with $\ell_{k_0} \geq 2u_2$; then $0 < \Delta_k \leq \pi$ for $k \geq k_0$. With $b_k := (e^{i\Delta_k} - 1)^{-1}$ one has $E_{k+1} = e^{i\Delta_k} E_k$, hence

$$E_k = b_k(E_{k+1} - E_k). \tag{4.8}$$

Summing (4.8) over $k_0 \leq k \leq N$ and shifting the index in the first part gives the exact identity

$$\sum_{k=k_0}^{N} E_k = b_N E_{N+1} - b_{k_0} E_{k_0} + \sum_{k=k_0+1}^{N} (b_{k-1} - b_k) E_k. \tag{4.9}$$

For $0 < x \leq \pi$, $|e^{ix} - 1| = 2\sin(x/2) \geq 2x/\pi$. Using (4.5) and (4.6),

$$|b_k| \leq \frac{\pi}{2\Delta_k} \leq \frac{\ell_{k+1}}{4u_1}, \qquad |b_{k-1} - b_k| \leq |\Delta_k - \Delta_{k-1}|\,|b_{k-1}||b_k| \leq \frac{\pi^2 u_2}{4u_1^2}\,\frac{\ell_{k+1}^2}{\tau_{k-1}\ell_{k-1}^3} \ll_{u_1,u_2} \frac{1}{\tau_k \ell_k} \ll \frac{1}{k}, \tag{4.10}$$

the last step by (4.7). Inserting (4.10) into (4.9) gives $|\sum_{k_0}^{N} E_k| \ll \ell_{N+1} + \ell_{k_0+1} + \sum_{k \leq N} k^{-1} \ll 1 + \log N$. The indices $k < k_0$ contribute at most k_0. All constants depend only on u_1, u_2. □

Corollary 4.2 (Point centroid)**.** *Uniformly for $u \in [u_1, u_2] \subset (0, \infty)$,*

$$C^{\mathrm{pts}}_{\mathrm{RVM}}(N;u) = O_{u_1,u_2}\!\left(\frac{1 + \log N}{N}\right) \longrightarrow 0. \tag{4.11}$$

Theorem 4.1 alone does not imply convergence of the path centroid (4.2)*.*

Proof. The first assertion is (4.3) divided by N. For the second, the identity $C^{\mathrm{path}}_{\mathrm{RVM}} = N^{-1}\sum_{n \leq N} S_{\mathrm{RVM}}(n;u)$ with $S_{\mathrm{RVM}}(n;u) = O(\log n)$ gives only $O(\log N)$. □

Define the midpoint count

$$n_{\mathrm{RVM}}(H) := \#\{k : \tau_k \le H\}. \tag{4.12}$$

For $x > 0$ and $n_{\mathrm{RVM}}(H) \ge 1$ define

$$M_H^{\mathrm{RVM}}(x) := \frac{1}{n_{\mathrm{RVM}}(H)} \sum_{\tau_k \le H} x^{i\tau_k}, \tag{4.13}$$

and for the ordinates of ζ

$$M_H(x) := \frac{1}{N_\zeta(H)} \sum_{\gamma_k \le H} x^{i\gamma_k}. \tag{4.14}$$

Corollary 4.3 (Synthetic height-transform decay)**.** *For every fixed $x > 1$, with $N = n_{\mathrm{RVM}}(H)$,*

$$M_H^{\mathrm{RVM}}(x) = O_x\Big(\frac{1 + \log N}{N}\Big) \longrightarrow 0 \qquad (H \to \infty). \tag{4.15}$$

Proof. Take $u = \log x > 0$ in Theorem 4.1; (4.13) is the point centroid at $N = n_{\mathrm{RVM}}(H)$. □

Remark 4.4. Neither (4.13) nor (4.14) is the Mertens function; both are normalized phase averages.

5. The RVM explicit formula, J_{RVM}, and π_{RVM}

For $w = a + ib$ with $b > 0$ define

$$\mathrm{Ei}(w) := \int_{-\infty}^{a} \frac{e^{v+ib}}{v+ib}\, dv, \qquad \mathrm{Ei}(\bar{w}) := \overline{\mathrm{Ei}(w)}. \tag{5.1}$$

The path is the horizontal line $\Im = b$, which avoids 0; this is the branch of $\mathrm{li}(x^\rho) = \mathrm{Ei}(\rho \log x)$ used in Riemann's explicit formula [12]. For $x > 1$ put

$$J_{\mathrm{RVM}}(x) := \mathrm{li}(x) - 2\Re \sum_{k \ge 1} \mathrm{Ei}\Big(\big(\frac{1}{2} + i\tau_k\big)\log x\Big) - \log 2 + \int_x^\infty \frac{dy}{y(y^2-1)\log y}, \tag{5.2}$$

the zero sum being taken in the order of k.

Lemma 5.1 (Asymptotics of Ei off the real axis)**.** *For $w = a + ib$, $b > 0$,*

$$\mathrm{Ei}(w) = \frac{e^w}{w} + \varrho(w), \qquad |\varrho(w)| \le \frac{e^a}{b^2}. \tag{5.3}$$

Proof. Integrate (5.1) by parts: $\int_{-\infty}^a e^{v+ib}(v+ib)^{-1}dv = e^w/w + \int_{-\infty}^a e^{v+ib}(v+ib)^{-2}dv$, the boundary term at $-\infty$ being 0. The last integral is bounded by $\int_{-\infty}^a e^v b^{-2} dv = e^a/b^2$. □

Theorem 5.2 (Compact-uniform existence of the RVM zero sum)**.** *The series $\sum_k \Re\,\mathrm{Ei}((\frac{1}{2} + i\tau_k)u)$ converges uniformly for u in every compact interval $[u_1, u_2] \subset (0, \infty)$. Hence J_{RVM} is a continuous function on $(1, \infty)$.*

Proof. Apply Lemma 5.1 with $w = (\frac{1}{2} + i\tau_k)u$, $a = u/2$, $b = \tau_k u$. Since $\Re\big(e^{iu\tau_k}/(\frac{1}{2} + i\tau_k)\big) = (\frac{1}{2}\cos u\tau_k + \tau_k \sin u\tau_k)/(\frac{1}{4} + \tau_k^2)$,

$$\Re\,\mathrm{Ei}\big((\tfrac{1}{2} + i\tau_k)u\big) = \frac{e^{u/2}}{u}\,\frac{\sin(u\tau_k)}{\tau_k} + O_{u_1,u_2}(\tau_k^{-2}), \tag{5.4}$$

uniformly for $u \in [u_1, u_2]$. The error series converges absolutely and uniformly. For the main series, Abel summation with $S_{\mathrm{RVM}}(K; u) = O_{u_1,u_2}(1 + \log K)$ (Theorem 4.1) and

$$\frac{1}{\tau_K} - \frac{1}{\tau_{K+1}} \le \frac{\tau_{K+1} - \tau_K}{\tau_K^2} \le \frac{2\pi}{\ell_K \tau_K^2} \ll \frac{\log K}{K^2} \tag{5.5}$$

gives a uniformly convergent majorant $\sum_K (\log K)^2 K^{-2}$, while the boundary term $S_{\mathrm{RVM}}(K; u)/\tau_K \to 0$ uniformly. Hence $\sum_k e^{iu\tau_k}/\tau_k$ converges uniformly, and so does its imaginary part. □

Define the support-truncated version

$$J^*_{\mathrm{RVM}}(x) := \begin{cases} J_{\mathrm{RVM}}(x), & x \geq 2, \\ 0, & 0 < x < 2, \end{cases} \tag{5.6}$$

set $\pi_{\mathrm{RVM}}(x) = 0$ for $0 < x < 2$, and for $x \geq 2$

$$\pi_{\mathrm{RVM}}(x) := \sum_{1 \leq m \leq \log_2 x} \frac{\mu(m)}{m} J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.7}$$

Proposition 5.3 (Exact finite prime-power inversion)**.** *For every $x \geq 2$,*

$$J^*_{\mathrm{RVM}}(x) = \sum_{1 \leq m \leq \log_2 x} \frac{1}{m} \pi_{\mathrm{RVM}}(x^{1/m}). \tag{5.8}$$

Proof. All sums are finite because $x^{1/m} < 2$ for $m > \log_2 x$. Substituting (5.7) into the right side of (5.8), the coefficient of $J^*_{\mathrm{RVM}}(x^{1/r})$ is

$$\frac{1}{r} \sum_{d \mid r} \mu(d), \tag{5.9}$$

which is 1 for $r = 1$ and 0 otherwise. □

5.1. **The square-root estimate.** For $u > 0$ put

$$\mathcal{A}_{\mathrm{RVM}}(u) := \sum_{k \geq 1} \frac{e^{iu\tau_k}}{\tau_k}, \tag{5.10}$$

summed in the order of k, and write $\ell(t) = \log(t/2\pi)$, so that $\mathcal{N}'_{\mathrm{RVM}}(t) = \ell(t)/2\pi$ and $\ell(\tau_k) = \ell_k$.

Lemma 5.4 (The weighted clock sum)**.** *For every $u \geq \log 2$ the series* (5.10) *converges, and*

$$|\mathcal{A}_{\mathrm{RVM}}(u)| \leq \frac{\log^2(2+2u)}{4\pi} + \frac{\log(2+2u)}{2\pi} + C_{\mathcal{A}}, \tag{5.11}$$

with an absolute constant $C_{\mathcal{A}}$.

Proof. Put $Y_j = 2^j \cdot 4\pi(1+u)$ for $j \geq 0$.

Head $\tau_k \leq Y_0$. Let $n(t) = \#\{k : \tau_k \leq t\}$. By (2.4), $\tau_k \leq t$ if and only if $k \leq \mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2}$, so $n(t) \leq \mathcal{N}_{\mathrm{RVM}}(t) + \frac{1}{2} = \frac{t}{2\pi}(\ell(t) - 1) + \frac{11}{8} \leq \frac{t\,\ell(t)}{2\pi}$ for $t \geq \tau_1 > \frac{11\pi}{4}$. Stieltjes integration gives

$$\sum_{\tau_k \leq Y_0} \frac{1}{\tau_k} = \frac{n(Y_0)}{Y_0} + \int_{\tau_1}^{Y_0} \frac{n(t)}{t^2}\,dt \leq \frac{\ell(Y_0)}{2\pi} + \int_{\tau_1}^{Y_0} \frac{\ell(t)}{2\pi t}\,dt \leq \frac{\ell(Y_0)}{2\pi} + \frac{\ell(Y_0)^2}{4\pi}, \tag{5.12}$$

and $\ell(Y_0) = \log(2+2u)$.

Blocks. Let $\mathcal{B}_j = \{k : Y_j < \tau_k \leq Y_{j+1}\}$, let $\mathcal{T}$ be the inverse clock of (4.4), and put $\phi(y) = u\,\mathcal{T}(y - \frac{1}{2})/2\pi$, so that $e^{2\pi i\phi(k)} = e^{iu\tau_k}$. By (4.4),

$$\phi''(y) = -\frac{2\pi u}{\mathcal{T}\,\ell(\mathcal{T})^3}, \qquad \mathcal{T} = \mathcal{T}(y - \tfrac{1}{2}). \tag{5.13}$$

On $\mathcal{B}_j$ put $\kappa_j = 2\pi u/(Y_{j+1}\ell(Y_{j+1})^3)$. Since $Y_{j+1} = 2Y_j$ and $\ell(Y_{j+1}) \leq 2\ell(Y_j)$ (because $\ell(Y_j) \geq \ell(Y_0) \geq \log 2$), we have $\kappa_j \leq -\phi'' \leq 16\kappa_j$ there. The number of indices in $\mathcal{B}_j$ is at most $\mathcal{N}_{\mathrm{RVM}}(Y_{j+1}) - \mathcal{N}_{\mathrm{RVM}}(Y_j) + 1 \leq Y_j\ell(Y_{j+1})/2\pi + 1$. The second-derivative test for exponential sums [21, Thm. 5.9] gives, for every run $a < k \leq b$ of consecutive indices of $\mathcal{B}_j$,

$$\Big|\sum_{a<k\leq b} e^{iu\tau_k}\Big| \leq A\Big(16\Big(\frac{Y_j\ell(Y_{j+1})}{2\pi} + 1\Big)\kappa_j^{1/2} + \kappa_j^{-1/2}\Big) =: P_j, \tag{5.14}$$

with an absolute constant A; a run of length less than 1 has at most one term, and $1 \le \kappa_j^{-1/2}$ because $\kappa_j \le u/(4(1+u)\log^3 2) < 1$. The weights $1/\tau_k$ are positive and decreasing, and $1/\tau_k < 1/Y_j$ on $\mathcal{B}_j$, so Abel's inequality bounds the contribution of any run in $\mathcal{B}_j$ by P_j/Y_j:

$$\frac{P_j}{Y_j} \le A\left(\frac{16}{2\pi}\Big(\frac{\pi u}{Y_j \ell(Y_{j+1})}\Big)^{1/2} + \frac{16}{Y_j} + \Big(\frac{\ell(Y_{j+1})^3}{\pi u Y_j}\Big)^{1/2}\right). \tag{5.15}$$

With $Y_j = 2^j \cdot 4\pi(1+u)$ and $\ell(Y_{j+1}) = \log(2^{j+2}(1+u)) \le j + 2 + \log(1+u)$, each of the three terms is at most an absolute constant times $(j+2)^{3/2}2^{-j/2}$, uniformly in $u \ge \log 2$, because $\sup_{u \ge \log 2}(1+\log(1+u))^3/(u(1+u)) < \infty$. Hence $\sum_{j \ge 0} P_j/Y_j \le C_{\mathcal{A}}$ with $C_{\mathcal{A}}$ absolute.

Convergence and bound. If $\tau_K \in \mathcal{B}_J$, the terms with $k > K$ form a run of $\mathcal{B}_J$ followed by the full blocks $\mathcal{B}_j$, $j > J$, so $|\sum_{k>K'}^{K''} e^{iu\tau_k}/\tau_k| \le \sum_{j \ge J} P_j/Y_j \to 0$ for $K'' > K' \ge K$, $K \to \infty$. The series converges, and (5.12) together with $\sum_j P_j/Y_j \le C_{\mathcal{A}}$ gives (5.11). □

Theorem 5.5 (Square-root estimate for J_{RVM} and π_{RVM})**.** *For $x \ge 8$,*

$$J_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.16}$$

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O\left(\frac{\sqrt{x}(\log\log x)^2}{\log x}\right), \tag{5.17}$$

with absolute implied constants. No hypothesis on ζ is used.

Proof. Let $u = \log x \ge \log 8$ and $\rho_k = \frac{1}{2} + i\tau_k$. Lemma 5.1 with $w = \rho_k u$, $a = u/2$, $b = \tau_k u$ gives

$$\mathrm{Ei}(\rho_k u) = \frac{e^{\rho_k u}}{\rho_k u} + \varrho_k, \qquad |\varrho_k| \le \frac{e^{u/2}}{\tau_k^2 u^2}. \tag{5.18}$$

Moreover $\frac{1}{\rho_k} = -\frac{i}{\tau_k} + r_k$ with $r_k = \frac{i/2}{\tau_k \rho_k}$, so $|r_k| \le \frac{1}{2\tau_k^2}$. Since $\sum_k \tau_k^{-2} < \infty$, Lemma 5.4 shows that the series in (5.2) converges in the order of k and

$$\sum_{k \ge 1} \mathrm{Ei}(\rho_k u) = \frac{e^{u/2}}{u}\Big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + \sum_k r_k e^{iu\tau_k}\Big) + \sum_k \varrho_k = \frac{e^{u/2}}{u}\big(-i\,\mathcal{A}_{\mathrm{RVM}}(u) + O(1)\big). \tag{5.19}$$

For $u \ge 2$ one has $\log(2+2u) \le \log(2u) + \log 2 \le 3\log u$, so (5.11) gives $\mathcal{A}_{\mathrm{RVM}}(u) = O(\log^2 u)$. The remaining terms $-\log 2 + \int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) are bounded for $x \ge 2$. With $e^{u/2} = \sqrt{x}$ and $\log u = \log\log x$ this proves (5.16).

For (5.17), (5.7) gives

$$\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = \big(J_{\mathrm{RVM}}(x) - \mathrm{li}(x)\big) + \sum_{2 \le m \le \log_2 x} \frac{\mu(m)}{m}\, J^*_{\mathrm{RVM}}(x^{1/m}). \tag{5.20}$$

By (5.16), $\mathrm{li}(y) \ll y/\log y$, and the continuity of J_{RVM} on $[2,8]$ (Theorem 5.2), $|J^*_{\mathrm{RVM}}(y)| \le C\,y/\log y$ for all $y \ge 2$. The term $m = 2$ is therefore $O(\sqrt{x}/\log x)$, and the terms $3 \le m \le \log_2 x$ add up to at most $C\,x^{1/3}\log_2 x/\log x = O(x^{1/3})$. Both are absorbed in (5.17). □

Corollary 5.6 (The three limits)**.** *As $x \to \infty$,*

$$\frac{\mathrm{li}(x)}{\pi_{\mathrm{RVM}}(x)} \to 1, \qquad \frac{\pi(x)}{\mathrm{li}(x)} \to 1, \qquad \frac{\pi(x)}{\pi_{\mathrm{RVM}}(x)} \to 1. \tag{5.21}$$

More precisely, for some absolute $c > 0$,

$$\pi(x) - \pi_{\mathrm{RVM}}(x) = O\Big(x\,e^{-c\sqrt{\log x}}\Big). \tag{5.22}$$

Proof. Since $\mathrm{li}(x) \sim x/\log x$, (5.17) gives $\pi_{\mathrm{RVM}}(x) = \mathrm{li}(x)(1 + O(x^{-1/2}(\log\log x)^2))$; in particular $\pi_{\mathrm{RVM}}(x) > 0$ for large x, and the first limit follows. The second is the prime number theorem [15, 8], in the form $\pi(x) = \mathrm{li}(x) + O(x\, e^{-c\sqrt{\log x}})$ [7, Ch. 18]. Subtracting (5.17) gives (5.22), and dividing by $\pi_{\mathrm{RVM}}(x) \sim x/\log x$ gives the third limit. □

The first limit and (5.17) concern only the synthetic system; the second and third compare it with the primes and use the prime number theorem. Figures 8 and 10 show them numerically.

6. Integer weights and the continuous-to-atomic problem

Define

$$c_n^{\mathrm{RVM}} := \pi_{\mathrm{RVM}}(n) - \pi_{\mathrm{RVM}}(n-1), \qquad n \geq 2, \tag{6.1}$$

and, initially for $\Re s > 1$,

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) := \prod_{n\geq 2} (1 - n^{-s})^{-c_n^{\mathrm{RVM}}}. \tag{6.2}$$

No positivity or 0/1 property is built into (6.1). Define the integer-step prime-power count and the right-continuous extension

$$J_{\mathbb{Z},\mathrm{RVM}}(x) := \sum_{1\leq m\leq \log_2 x} \frac{1}{m}\, \pi_{\mathrm{RVM}}(\lfloor x^{1/m}\rfloor), \tag{6.3}$$

$$J^{*,\mathbb{Z}}_{\mathrm{RVM}}(x) := J^{*}_{\mathrm{RVM}}(\lfloor x\rfloor). \tag{6.4}$$

Theorem 6.1 (Lipschitz control of J_{RVM})**.** *There is an absolute constant C_J such that*

$$|J_{\mathrm{RVM}}(x+\theta) - J_{\mathrm{RVM}}(x)| \leq C_J\, \theta \qquad (x \geq 2,\ 0 < \theta \leq x). \tag{6.5}$$

Proof. Put $u_1 = \log x$, $u_2 = \log(x+\theta)$, so $0 < u_2 - u_1 = \log(1+\theta/x) \leq \theta/x$ and $u_2 - u_1 \leq \log 2$. The terms li and $\int_x^\infty dy/(y(y^2-1)\log y)$ of (5.2) have derivatives bounded by $1/\log 2$ and $3/\log 2$ on $[2,\infty)$, so they change by $O(\theta)$. It remains to bound $\sum_k D_k$, where, by (5.1) and $\frac{d}{dv}\,\mathrm{Ei}(\rho v) = e^{\rho v}/v$,

$$D_k := \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_2) - \mathrm{Ei}(\rho_k^{\mathrm{RVM}} u_1) = \int_{u_1}^{u_2} \frac{e^{v/2} e^{iv\tau_k}}{v}\, dv. \tag{6.6}$$

The series $\sum_k D_k$ converges (Theorem 5.2 and its proof). By Abel's theorem for the increasing exponents τ_k, $\sum_k D_k = \lim_{\varepsilon\downarrow 0} \sum_k e^{-\varepsilon\tau_k} D_k$, and for $\varepsilon > 0$ Fubini gives

$$\sum_k e^{-\varepsilon\tau_k} D_k = \int_{u_1}^{u_2} \frac{e^{v/2}}{v}\, S_\varepsilon(v)\, dv, \qquad S_\varepsilon(v) := \sum_{k\geq 1} g(\tau_k), \quad g(\tau) := e^{(iv-\varepsilon)\tau}. \tag{6.7}$$

Clock decomposition. With $\tau_0 = 4\pi$ and the sawtooth count (8.8), Stieltjes integration exactly as in (8.9) ($g(\infty) = 0$, $g' \in L^1$) gives

$$S_\varepsilon(v) = \int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}} + g(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}) g'(\tau)\, d\tau. \tag{6.8}$$

The middle term is at most $\frac{1}{2}$. For the first, two integrations by parts with $\mathcal{N}'_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi}\log\frac{\tau}{2\pi}$, $\mathcal{N}''_{\mathrm{RVM}}(\tau) = \frac{1}{2\pi\tau}$ and $v \geq \log 2$ give $|\int_{\tau_0}^\infty g\, d\mathcal{N}_{\mathrm{RVM}}| \leq C/v$, uniformly in $\varepsilon \in (0,1]$.

Harmonics. The partial sums of $\psi(y+\frac{1}{2}) = -\sum_{m\geq 1}(-1)^m \sin(2\pi m y)/(\pi m)$ are uniformly bounded and converge off a null set, and $g' \in L^1$ for $\varepsilon > 0$, so termwise integration is allowed:

$$\int_{\tau_0}^\infty \psi(\mathcal{N}_{\mathrm{RVM}} + \tfrac{1}{2}) g'\, d\tau = -\sum_{m\geq 1} \frac{(-1)^m}{2\pi i\, m}(K_m^+ - K_m^-), \qquad K_m^\pm := \int_{\tau_0}^\infty (iv-\varepsilon) e^{-\varepsilon\tau} e^{i\Phi_m^\pm(\tau)}\, d\tau, \tag{6.9}$$

with $\Phi_m^\pm(\tau) = v\tau \pm 2\pi m \mathcal{N}_{\mathrm{RVM}}(\tau)$. The weight $e^{-\varepsilon\tau}$ is positive and decreasing, so by the second mean-value theorem each integral is at most $2(v+1)\sup_\xi |\int_a^\xi e^{i\Phi} d\tau|$ over the relevant interval $[a,b]$.

We use the first- and second-derivative tests [21, Lemmas 4.2 and 4.4]: $|\int_a^b e^{i\Phi}| \le 4/\mu$ if Φ' is monotone with $|\Phi'| \ge \mu$, and $\le 8/\sqrt{\lambda}$ if $|\Phi''| \ge \lambda$.

For K_m^+: $(\Phi_m^+)' = v + m\log\frac{\tau}{2\pi} \ge v + m\log 2$ is increasing, so $|K_m^+| \le 8(v+1)/(v+m\log 2)$.

For K_m^-: $(\Phi_m^-)' = v - m\log\frac{\tau}{2\pi}$ is decreasing and vanishes at $\tau_m := 2\pi e^{v/m} > 2\pi$, and $(\Phi_m^-)'' = -m/\tau$. On $[\tau_0, \tau_m/2]$ (if nonempty) $(\Phi_m^-)' \ge m\log 2$; on $[2\tau_m, \infty)$, $(\Phi_m^-)' \le -m\log 2$; on $[\max(\tau_0, \tau_m/2), 2\tau_m]$, $|(\Phi_m^-)''| \ge m/(2\tau_m)$. Hence

$$|K_m^-| \le 2(v+1)\Big(\frac{8}{m\log 2} + 8\sqrt{\frac{2\tau_m}{m}}\Big) = 2(v+1)\Big(\frac{8}{m\log 2} + 16\sqrt{\pi}\,\frac{e^{v/(2m)}}{\sqrt{m}}\Big). \tag{6.10}$$

Summing with the weights $1/(2\pi m)$ and using $\sum_m m^{-3/2}e^{v/(2m)} \le e^{v/2} + \zeta(\frac{3}{2})e^{v/4}$,

$$|S_\varepsilon(v)| \le C\,(v+1)\,e^{v/2} \qquad (v \ge \log 2,\ 0 < \varepsilon \le 1), \tag{6.11}$$

with an absolute constant C.

Conclusion. Inserting (6.11) into (6.7), for $v \in [u_1, u_2]$ we have $e^{v/2} \le \sqrt{2x}$ and $(v+1)/v \le 1 + 1/\log 2$, so

$$\Big|\sum_k e^{-\varepsilon\tau_k} D_k\Big| \le C'(u_2 - u_1)\,x \le C'\theta, \tag{6.12}$$

uniformly in ε. Letting $\varepsilon \downarrow 0$ and taking $2\Re$ proves (6.5). □

The $m = 1$ harmonic, with stationary point $\tau_1 = 2\pi x$, produces the unit-periodic ripple of J_{RVM} (Figure 8); the harmonics $m \ge 2$ have stationary points $2\pi x^{1/m}$ and contribute $O(x^{-1/2+1/(2m)}m^{-3/2})$ to a unit increment.

Corollary 6.2 (Bounded cell masses)**.**

$$\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y\rfloor) = O(1), \qquad c_n^{\mathrm{RVM}} = O(1). \tag{6.13}$$

Proof. By (5.7), $c_n^{\mathrm{RVM}} = \sum_{m \le \log_2 n} \frac{\mu(m)}{m}\big(J^*_{\mathrm{RVM}}(n^{1/m}) - J^*_{\mathrm{RVM}}((n-1)^{1/m})\big)$. The term $m = 1$ is $O(1)$ by (6.5). For $m \ge 2$ and $(n-1)^{1/m} \ge 2$, (6.5) with $\theta = n^{1/m} - (n-1)^{1/m} \le n^{1/m-1}$ bounds the term by $C_J n^{1/m-1} \le C_J n^{-1/2}$, and there are at most $\log_2 n$ such terms. The jump of J^*_{RVM} at 2 enters only for $n = 2^m$, once, with size $|J_{\mathrm{RVM}}(2)|/m$. The same argument applies to $\pi_{\mathrm{RVM}}(y) - \pi_{\mathrm{RVM}}(\lfloor y\rfloor)$. □

Corollary 6.3 (The integer-weighted zeta is a Dirichlet series)**.** *The product* (6.2) *converges absolutely for* $\Re s > 1$ *and*

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{N\ge 1} \frac{a_N^{\mathbb{Z}}}{N^s}, \qquad a_1^{\mathbb{Z}} = 1, \qquad \sum_{N\ge 1} \frac{|a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \tag{6.14}$$

The inverse is also an absolutely convergent Dirichlet series on the whole half-plane:

$$\frac{1}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)} = \sum_{N\ge 1} \frac{\bar a_N^{\mathbb{Z}}}{N^s}, \qquad \bar a_1^{\mathbb{Z}} = 1, \qquad \sum_{N\ge 1} \frac{|\bar a_N^{\mathbb{Z}}|}{N^\sigma} < \infty \quad (\sigma > 1). \tag{6.15}$$

Proof. By Corollary 6.2,

$$\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \sum_{n\ge 2} c_n^{\mathrm{RVM}} \sum_{m\ge 1} \frac{n^{-ms}}{m} = \sum_{N\ge 2} \frac{b_N}{N^s}, \qquad b_N := \sum_{n^m = N} \frac{c_n^{\mathrm{RVM}}}{m}, \tag{6.16}$$

with $\sum_N |b_N| N^{-\sigma} \le C \sum_{n\ge 2}\sum_m n^{-m\sigma}/m < \infty$ for $\sigma > 1$. The exponential of an absolutely convergent Dirichlet series without constant term is an absolutely convergent Dirichlet series with $a_1^{\mathbb{Z}} = 1$, and $\sum_N |a_N^{\mathbb{Z}}| N^{-\sigma} \le \exp(\sum_N |b_N| N^{-\sigma})$. The same applies to $1/\zeta_{\mathbb{Z},\mathrm{RVM}} = \exp(-\sum_N b_N N^{-s})$, which gives (6.15). □

Numerically the cell masses lie in $[-0.38, 1.04]$ for $n \le 1000$ (Experiment 13.6).

Theorem 6.4 (Mellin–Stieltjes identification)**.** *There is a function $J^{\mathrm{M}}_{\mathrm{RVM}}$ on $(1,\infty)$ with $J^{\mathrm{M}}_{\mathrm{RVM}}(x)x^{-c} \in L^2((1,\infty), dx/x)$ for every $c > 1$ such that*

$$\log \zeta_{\mathrm{RVM}}(s) = s \int_1^\infty J^{\mathrm{M}}_{\mathrm{RVM}}(x)\, x^{-s-1}\, dx \qquad (\Re s > 1), \tag{6.17}$$

and $J^{\mathrm{M}}_{\mathrm{RVM}} - J_{\mathrm{RVM}}$ is constant on $(1,\infty)$. In particular $J^{\mathrm{M}}_{\mathrm{RVM}} = J^{*}_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ on $[2,\infty)$ for a constant C_{RVM}.

The proof is given in Section 12 (Theorems 12.5 and 12.6); it uses the half-plane bound of Theorem 8.10.

Corollary 6.5 (Weighted factorization)**.**

$$J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{*}_{\mathrm{RVM}}(x) = O(\log\log x), \tag{6.18}$$

and

$$\zeta_{\mathbb{Z},\mathrm{RVM}}(s) = \zeta_{\mathrm{RVM}}(s)\, e^{h_{\mathrm{RVM}}(s)}, \tag{6.19}$$

where

$$h_{\mathrm{RVM}}(s) := s \int_1^\infty (J_{\mathbb{Z},\mathrm{RVM}}(x) - J^{\mathrm{M}}_{\mathrm{RVM}}(x)) x^{-s-1}\, dx \tag{6.20}$$

is holomorphic for $\Re s > 0$. The two functions have the same zeros in $\Re s > 0$.

Proof. By (6.13) replacing $x^{1/m}$ by its integer part costs $O(1/m)$ in the mth summand; summing m^{-1} up to $\log_2 x$ gives (6.18). The product (6.2) gives $\log \zeta_{\mathbb{Z},\mathrm{RVM}}(s) = s\int_1^\infty J_{\mathbb{Z},\mathrm{RVM}}(x)x^{-s-1}dx$ for $\Re s > 1$, because $\sum_{n\le y} c_n^{\mathrm{RVM}} = \pi_{\mathrm{RVM}}(\lfloor y \rfloor)$. Subtracting (6.17) gives (6.19) on $\Re s > 1$. On $[2,\infty)$, $J_{\mathbb{Z},\mathrm{RVM}} - J^{\mathrm{M}}_{\mathrm{RVM}} = (J_{\mathbb{Z},\mathrm{RVM}} - J^{*}_{\mathrm{RVM}}) - C_{\mathrm{RVM}} = O(\log\log x)$ by (6.18); on $(1,2)$, $J_{\mathbb{Z},\mathrm{RVM}} = 0$ and $J^{\mathrm{M}}_{\mathrm{RVM}}$ is square-integrable. Hence (6.20) converges locally normally for $\Re s > 0$, and exponentiation introduces no zeros. □

Lemma 6.6 (Exact first-harmonic cell cancellation)**.** *For every integer n,*

$$\int_{n-1}^{n} 2\cos(2\pi x)\, dx = 0. \tag{6.21}$$

Hypothesis 6.7 (Uniform cell remainder)**.**

$$c_n^{\mathrm{RVM}} - \int_{n-1}^{n} \left(\frac{1}{\log x} + 2\cos(2\pi x)\right) dx = o\Big(\frac{1}{\log n}\Big). \tag{6.22}$$

Proposition 6.8 (Conditional eventual positivity)**.** *Hypothesis 6.7 implies*

$$c_n^{\mathrm{RVM}} \sim \frac{1}{\log n}, \tag{6.23}$$

hence $0 < c_n^{\mathrm{RVM}} < 1$ for all large n.

Proof. The main-density integral is $1/\log n + O(1/(n\log^2 n))$; the first harmonic vanishes by Lemma 6.6. □

7. The critical RVM Mertens transform

Define

$$F_{\mathrm{RVM}}(s) = \widehat{M}^{\mathrm{crit}}_{\mathrm{RVM}}(s) := \frac{1}{\zeta_{\mathrm{RVM}}(s)} = \frac{G(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{7.1}$$

Formally the associated summatory object is the inverse Mellin transform

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = \frac{1}{2\pi i}\int_{(c)} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\, ds; \tag{7.2}$$

its existence as a symmetric limit, for every $c > \frac{1}{2}$, is proved in Theorem 8.12.

Proposition 7.1 (Synthetic pole abscissa)**.** *F_{RVM} is holomorphic in $\Re s > \frac{1}{2}$. Its poles are the simple poles*

$$\frac{1}{2} \pm i\tau_k \quad (k \geq 1) \qquad \text{and} \qquad s = -2, -4, -6, \dots \tag{7.3}$$

and $s = 1$ is a simple zero of F_{RVM}.

Proof. This is Theorem 3.1 applied to $F_{\mathrm{RVM}} = G/P^{\mathrm{crit}}_{\mathrm{RVM}}$: the zeros of $P^{\mathrm{crit}}_{\mathrm{RVM}}$ are (3.6), and the poles of G are those of $\Gamma(s/2)$ except $s = 0$, where $s\Gamma(s/2) \to 2$. □

Proposition 7.2 (Gaussian square-root bound under vertical growth)**.** *Fix $c > 1$ and $0 < \varepsilon < c - \frac{1}{2}$. Assume, uniformly for $\frac{1}{2} + \varepsilon \leq \sigma \leq c$,*

$$F_{\mathrm{RVM}}(\sigma + it) = \exp(o(t^2)) \qquad (|t| \to \infty). \tag{7.4}$$

Define

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) := \frac{1}{2\pi i} \int_{(c)} e^{s^2} F_{\mathrm{RVM}}(s) \frac{x^s}{s}\, ds. \tag{7.5}$$

Then

$$M^{\mathrm{crit}}_{\mathrm{RVM},G}(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{7.6}$$

Proof. On $s = \sigma + it$, $|e^{s^2}| = e^{\sigma^2 - t^2}$, so (7.4) gives absolute convergence on vertical lines and makes the horizontal integrals tend to 0. The strip $\frac{1}{2} + \varepsilon \leq \Re s \leq c$ contains no pole of F_{RVM} (Proposition 7.1) and does not contain $s = 0$, so Cauchy's theorem moves the line to $\Re s = \frac{1}{2} + \varepsilon$, where $|x^s| = x^{1/2+\varepsilon}$. □

Theorem 8.12 below removes both the assumption (7.4) and the smoothing.

8. The transition theorem and the unsmoothed synthetic Mertens bound

Let

$$\chi(s) := 2^s \pi^{s-1} \sin\Big(\frac{\pi s}{2}\Big) \Gamma(1 - s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \tag{8.1}$$

the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1 - s)$ [21, §2.1]. Write

$$s = \tfrac{1}{2} + z, \quad z = \sigma' + it, \qquad \tau_* := -iz = t - i\sigma', \qquad w(s) := \exp\{-2\pi i\, \mathcal{N}_{\mathrm{RVM}}(\tau_*)\}. \tag{8.2}$$

Stirling's formula gives, uniformly on compact σ-sets,

$$|\chi(\sigma + it)| = \Big(\frac{|t|}{2\pi}\Big)^{1/2-\sigma} (1 + O(|t|^{-1})). \tag{8.3}$$

Theorem 8.1 (Transition evaluation of the midpoint product)**.** *There is a constant $\varpi_{\mathrm{RVM}} > 0$ with the following properties.*

(a) *For every $B > 0$ there are $t_0(B)$ and C_B such that, for $0 < \sigma' \leq B$ and $t \geq t_0(B)$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + w(s))(1 + \mathcal{E}_0(s)), \qquad |\mathcal{E}_0(s)| \leq \frac{C_B}{t}. \tag{8.4}$$

(b) *For every $0 < \sigma_1' < \sigma_2' < \frac{1}{2}$ there are t_0 and C such that, for $\sigma_1' \leq \sigma' \leq \sigma_2'$ and $t \geq t_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}(1 + \chi(s))(1 + \mathcal{E}(s)), \qquad |\mathcal{E}(s)| \leq \frac{C}{t}. \tag{8.5}$$

For $t < 0$ the same statements hold by $\zeta_{\mathrm{RVM}}(\bar{s}) = \overline{\zeta_{\mathrm{RVM}}(s)}$.

The phase identity $w = \chi(s)(1+O(1/t))$ alone does not give (8.4); the product has to be evaluated across the transition region $\tau_k \asymp t$. Log denotes the principal logarithm and $\operatorname{Log}\Gamma$ the branch of $\log\Gamma$ on $\Re v > 0$ that is real on $(0, \infty)$. Nothing about ζ is used.

8.1. **Proof of Theorem 8.1.** Throughout, $t > 0$, $0 < \sigma' \le B$, $\tau_0 := 4\pi$, $t^2 \ge 2(\tau_0^2 + B^2)$, and implied constants depend only on B (and on σ_1', σ_2' in part (b)).

Step 0: reduction to a sum over the clock. Since $q(s) = z^2 - \frac{1}{4}$, each factor of (3.2) equals $(\tau_k^2 + z^2)/(\tau_k^2 + \frac{1}{4})$, so

$$\frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})} = \prod_{k\ge1}\Big(1 + \frac{z^2}{\tau_k^2}\Big) = \exp\Lambda(z), \qquad \Lambda(z) := \sum_{k\ge1} f(\tau_k), \quad f(\tau) := \mathrm{Log}\Big(1 + \frac{z^2}{\tau^2}\Big). \tag{8.6}$$

For $\tau > 0$, $\Im(1 + z^2/\tau^2) = 2\sigma' t/\tau^2 > 0$, so f is smooth on $(0,\infty)$, $f(\tau) = O(|z|^2\tau^{-2})$, the series converges absolutely, and

$$f'(\tau) = -\frac{2z^2}{\tau(\tau^2 + z^2)} = -\frac{2}{\tau} + \frac{1}{\tau - iz} + \frac{1}{\tau + iz}. \tag{8.7}$$

Step 1: the midpoint sawtooth. The clock increases on $[2\pi,\infty)$, $\mathcal{N}_{\mathrm{RVM}}(\tau_0) = 2\log\frac{2}{e} + \frac{7}{8} \in (0, \frac{1}{2})$, and $\tau_1 > \tau_0$. With $\psi(y) := y - \lfloor y\rfloor - \frac{1}{2}$, the count (4.12) satisfies, for $\tau \ge \tau_0$,

$$n_{\mathrm{RVM}}(\tau) = \lfloor \mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}\rfloor = \mathcal{N}_{\mathrm{RVM}}(\tau) - \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2}), \qquad n_{\mathrm{RVM}}(\tau_0) = 0. \tag{8.8}$$

Stieltjes integration of f against (8.8), followed by integration by parts in the ψ-part ($f(\infty) = 0$, $|\psi| \le \frac{1}{2}$, $\psi(\mathcal{N}_{\mathrm{RVM}}(\tau_0) + \frac{1}{2}) = \mathcal{N}_{\mathrm{RVM}}(\tau_0)$), gives

$$\begin{aligned} &\Lambda(z) = \Lambda_{\mathrm{sm}}(z) + \Lambda_{\mathrm{saw}}(z), \qquad \Lambda_{\mathrm{sm}}(z) := \int_{\tau_0}^{\infty} f\, d\mathcal{N}_{\mathrm{RVM}} + f(\tau_0)\mathcal{N}_{\mathrm{RVM}}(\tau_0), \\ &\Lambda_{\mathrm{saw}}(z) := \int_{\tau_0}^{\infty} \psi(\mathcal{N}_{\mathrm{RVM}}(\tau) + \tfrac{1}{2})\, f'(\tau)\, d\tau. \end{aligned} \tag{8.9}$$

Lemma 8.2 (Smooth part)**.** *With* $\log G(s) := \mathrm{Log}(z^2 - \frac{1}{4}) + \mathrm{Log}\,\Gamma(\frac{s}{2}) - \frac{s}{2}\log\pi$,

$$\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(t^{-1}), \qquad \nu_3 \in \mathbb{R} \text{ independent of } z. \tag{8.10}$$

Proof. A second integration by parts ($f\mathcal{N}_{\mathrm{RVM}} \to 0$ at ∞) cancels the boundary term:

$$\Lambda_{\mathrm{sm}}(z) = -\int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}} f'\, d\tau = \int_{\tau_0}^{\infty} \mathcal{N}_{\mathrm{RVM}}(\tau)\Big(\frac{2}{\tau} - \frac{2\tau}{\tau^2 + z^2}\Big) d\tau. \tag{8.11}$$

The constant $\frac{7}{8}$ of the clock contributes $\frac{7}{8}\mathrm{Log}(1 + z^2/\tau_0^2)$, because $\mathrm{Log}\,\frac{\tau^2}{\tau^2+z^2}$ is an antiderivative of the bracket on $(0,\infty)$ tending to 0 at ∞. As $0 < \arg z < \frac{\pi}{2}$, $\arg(1 + \tau_0^2/z^2) = O(t^{-2})$ and

$$\tfrac{7}{8}\,\mathrm{Log}\Big(1 + \frac{z^2}{\tau_0^2}\Big) = \tfrac{7}{4}\,\mathrm{Log}\, z - \tfrac{7}{4}\log\tau_0 + O(t^{-2}). \tag{8.12}$$

The term $\frac{\tau}{2\pi}\log\frac{\tau}{2\pi e}$ contributes $\frac{1}{\pi}\int_{\tau_0}^{\infty}\log\frac{\tau}{2\pi e}\frac{z^2}{\tau^2+z^2}\,d\tau$. Put

$$\mathcal{I}(z) := \frac{1}{\pi}\int_0^{\infty} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2 + z^2}\, d\tau. \tag{8.13}$$

The integrand is holomorphic in z on $\Re z > 0$ (its singularities are at $z = \pm i\tau$), $O(|z|^2\tau^{-2}\log\tau)$ at ∞ and $O(|\log\tau|)$ at 0, locally uniformly; so $\mathcal{I}$ is holomorphic on $\Re z > 0$. For real $z = x > 0$ the substitution $\tau = xv$ and $\int_0^\infty \frac{dv}{1+v^2} = \frac{\pi}{2}$, $\int_0^\infty \frac{\log v}{1+v^2}\,dv = 0$ give $\mathcal{I}(x) = \frac{x}{2}\log\frac{x}{2\pi e}$; by the identity theorem $\mathcal{I}(z) = \frac{z}{2}\,\mathrm{Log}\,\frac{z}{2\pi e}$ on $\Re z > 0$. On $[0,\tau_0]$, $|\tau^2 + z^2| \ge t^2/2$, so $\frac{z^2}{\tau^2+z^2} = 1 + O(t^{-2})$ there. Hence

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\,\mathrm{Log}\,\frac{z}{2\pi e} + \frac{7}{4}\,\mathrm{Log}\, z + \nu_1 + O(t^{-2}), \qquad \nu_1 = -\tfrac{7}{4}\log\tau_0 - \tfrac{1}{\pi}\int_0^{\tau_0}\log\tfrac{\tau}{2\pi e}\,d\tau. \tag{8.14}$$

Stirling's formula $\operatorname{Log}\Gamma(v) = (v-\frac{1}{2})\operatorname{Log} v - v + \frac{1}{2}\log 2\pi + O(|v|^{-1})$ [11, §5.11] with $v = \frac{s}{2} = \frac{z+1/2}{2}$ and $\operatorname{Log} v = \operatorname{Log} z - \log 2 + \frac{1}{2z} + O(t^{-2})$ gives

$$\operatorname{Log}\Gamma(\tfrac{s}{2}) = \tfrac{z}{2}\operatorname{Log}\tfrac{z}{2} - \tfrac{1}{4}\operatorname{Log} z + \tfrac{1}{4}\log 2 - \tfrac{z}{2} + \tfrac{1}{2}\log 2\pi + O(t^{-1}), \tag{8.15}$$

and with $\operatorname{Log}(z^2-\frac{1}{4}) = 2\operatorname{Log} z + O(t^{-2})$, $-\frac{s}{2}\log\pi = -\frac{z}{2}\log\pi - \frac{1}{4}\log\pi$,

$$\log G(s) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{4}\operatorname{Log} z + \nu_2 + O(t^{-1}), \qquad \nu_2 = \tfrac{1}{4}\log 2 + \tfrac{1}{2}\log 2\pi - \tfrac{1}{4}\log\pi. \tag{8.16}$$

Subtracting gives (8.10) with $\nu_3 = \nu_1 - \nu_2$. □

Lemma 8.3 (The clock as a holomorphic phase)**.** *$\mathcal{N}_{\mathrm{RVM}}(\tau) = \frac{\tau}{2\pi}(\operatorname{Log}\frac{\tau}{2\pi} - 1) + \frac{7}{8}$ is holomorphic on $\Re\tau > 0$, $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$, and for $\Re\tau \ge \tau_0$,*

$$2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \ge (\log 2)\,\Im\tau \quad (\Im\tau \ge 0), \qquad 2\pi\,\Im\mathcal{N}_{\mathrm{RVM}}(\tau) \le (\log 2)\,\Im\tau \quad (\Im\tau \le 0). \tag{8.17}$$

Proof. For $\tau = X + iY$, $X \ge \tau_0$, $Y > 0$, put $v = Y/X$. Then

$$2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau) = Y\log\frac{|\tau|}{2\pi e} + X\arg\tau = Y\Big[\log\frac{X}{2\pi e} + \tfrac{1}{2}\log(1+v^2) + \frac{\arctan v}{v}\Big] \ge Y\big[\log 2 + \varphi(v)\big], \tag{8.18}$$

using $X \ge 4\pi$, where $\varphi(v) = \frac{1}{2}\log(1+v^2) + \frac{\arctan v}{v} - 1$. Since $\varphi(0^+) = 0$ and $\varphi'(v) = (v - \arctan v)/v^2 \ge 0$, $\varphi \ge 0$. The case $Y < 0$ follows by conjugation. □

In particular $|w(s)| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} \le 2^{-\sigma'} < 1$, and on the vertical rays $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 \pm iy)}| \le 2^{-my}$.

Lemma 8.4 (Abel-regularized harmonic sum)**.**

$$\Lambda_{\mathrm{saw}}(z) = \operatorname{Log}\big(1 + w(s)\big) + \nu_4 + O(t^{-2}), \qquad \nu_4 \in \mathbb{R} \text{ independent of } z. \tag{8.19}$$

Proof. Regularization. For $0 < r < 1$ let

$$\psi_r(y) := -\sum_{m\ge 1} r^m \frac{\sin 2\pi m y}{\pi m} = \frac{1}{\pi}\Im\operatorname{Log}\big(1 - re^{2\pi i y}\big). \tag{8.20}$$

Since $\Re(1 - re^{2\pi iy}) > 0$, $|\psi_r| \le \frac{1}{2}$, and $\psi_r(y) \to \psi(y)$ as $r \uparrow 1$ for $y \notin \mathbb{Z}$ (Abel summation of the convergent Fourier series of ψ). The set $\{\tau : \mathcal{N}_{\mathrm{RVM}}(\tau) + \frac{1}{2} \in \mathbb{Z}\}$ is discrete and $f' \in L^1(\tau_0, \infty)$, so by dominated convergence

$$\Lambda_{\mathrm{saw}}(z) = \lim_{r\uparrow 1}\Lambda_r(z), \qquad \Lambda_r(z) := \int_{\tau_0}^{\infty} \psi_r\big(\mathcal{N}_{\mathrm{RVM}} + \tfrac{1}{2}\big) f'\,d\tau. \tag{8.21}$$

Termwise integration. For fixed $r < 1$ the series for $\psi_r(\mathcal{N}_{\mathrm{RVM}} + \frac{1}{2})$, with $\sin 2\pi m(y + \frac{1}{2}) = (-1)^m \sin 2\pi m y$, is dominated by $\sum_m r^m/m$; by Fubini,

$$\Lambda_r(z) = -\sum_{m\ge 1} \frac{(-r)^m}{2\pi i\, m}\big(I_m^+ - I_m^-\big), \qquad I_m^{\pm} := \int_{\tau_0}^{\infty} e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau)} f'(\tau)\,d\tau. \tag{8.22}$$

Contours. Let $U^{\pm} = \{\Re\tau \ge \tau_0,\ \pm\Im\tau \ge 0\}$. On $U^{\pm}$, $|e^{\pm 2\pi i m\mathcal{N}_{\mathrm{RVM}}}| \le 1$ by Lemma 8.3, and $|f'(\tau)| \le 4|z|^2|\tau|^{-3}$ for $|\tau| \ge 2|z|$, so the quarter-circle arcs of radius R contribute $O(|z|^2R^{-2}) \to 0$. The poles of f' are 0, $iz = -t + i\sigma' \notin U^{\pm}$, and $\tau_* = -iz = t - i\sigma' \in U^-$, the last with residue 1 (from $\frac{1}{\tau + iz}$). Define the ray integrals

$$R_m^+ := i\int_0^{\infty} e^{2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 + iy)} f'(\tau_0 + iy)\,dy, \qquad R_m^- := -i\int_0^{\infty} e^{-2\pi i m\mathcal{N}_{\mathrm{RVM}}(\tau_0 - iy)} f'(\tau_0 - iy)\,dy. \tag{8.23}$$

On U^+ Cauchy's theorem gives $I_m^+ = R_m^+$. On U^- the positively oriented boundary of the quarter disc runs down the ray, along the arc, and back along $[\tau_0, \tau_0 + R]$ from right to left; letting $R \to \infty$,

$$R_m^- - I_m^- = 2\pi i\, e^{-2\pi i m \mathcal{N}_{\mathrm{RVM}}(\tau_*)} = 2\pi i\, w^m. \tag{8.24}$$

Therefore, using $|rw| < 1$,

$$\Lambda_r(z) = -\sum_{m\ge 1} \frac{(-rw)^m}{m} + \Phi_r(z) = \mathrm{Log}(1 + rw) + \Phi_r(z), \qquad \Phi_r(z) := -\sum_{m\ge 1} \frac{(-r)^m}{2\pi i\, m}(R_m^+ - R_m^-). \tag{8.25}$$

The rays. Split $f'(\tau) = -\frac{2}{\tau} + g_z(\tau)$, $g_z(\tau) = \frac{2\tau}{\tau^2+z^2}$, and correspondingly $R_m^\pm = R_m^{\pm,(0)} + R_m^{\pm,(z)}$, $\Phi_r = \Phi_r^{(0)} + \Phi_r^{(z)}$. The ray integrands carry the factor 2^{-my}. Since $|2/\tau| \le 2/\tau_0$ on the rays, $|R_m^{\pm,(0)}| \le 2/(\tau_0 m \log 2)$, so the series for $\Phi_r^{(0)}$ is dominated by $\sum_m C/m^2$ uniformly in r, and $\Phi_r^{(0)} \to \nu_4 := \Phi_1^{(0)}$, a constant depending only on τ_0. It is real: $\mathcal{N}_{\mathrm{RVM}}(\bar\tau) = \overline{\mathcal{N}_{\mathrm{RVM}}(\tau)}$ gives $R_m^{-,(0)} = \overline{R_m^{+,(0)}}$, so each term equals $-(-r)^m \Im R_m^{+,(0)}/(\pi m)$. For $\tau = \tau_0 \pm iy$, $\Re(\tau^2 + z^2) = \tau_0^2 + \sigma'^2 - y^2 - t^2 \le -\frac12(y^2 + t^2)$ because $t^2 \ge 2(\tau_0^2 + B^2)$; hence $|g_z(\tau)| \le 4(\tau_0 + y)t^{-2}$ and

$$|R_m^{\pm,(z)}| \le \frac{4}{t^2}\int_0^\infty 2^{-my}(\tau_0 + y)\, dy = \frac{4}{t^2}\Big(\frac{\tau_0}{m\log 2} + \frac{1}{(m\log 2)^2}\Big). \tag{8.26}$$

So $|\Phi_r^{(z)}(z)| \le C\, t^{-2}\sum_m m^{-2}$, uniformly in $0 < r < 1$ and $0 < \sigma' \le B$, and $\Phi_r^{(z)}$ converges as $r \uparrow 1$ by dominated convergence of the series. Since $\mathrm{Log}(1 + rw) \to \mathrm{Log}(1 + w)$, letting $r \uparrow 1$ in (8.25) and (8.21) gives (8.19). □

Step 2: assembly. By (8.6), (8.9) and Lemmas 8.2 and 8.4,

$$\begin{aligned}\zeta_{\mathrm{RVM}}(s) &= \frac{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}{G(s)} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\exp\big(\Lambda_{\mathrm{sm}}(z) - \log G(s) + \Lambda_{\mathrm{saw}}(z)\big)\\ &= P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4}(1 + w)\big(1 + O(t^{-1})\big),\end{aligned} \tag{8.27}$$

uniformly for $0 < \sigma' \le B$. This is (8.4) with

$$\varpi_{\mathrm{RVM}} := P^{\mathrm{crit}}_{\mathrm{RVM}}(\tfrac12)\, e^{\nu_3+\nu_4} > 0, \tag{8.28}$$

which depends only on the clock; its numerical value $0.9619415776\ldots$ plays no role in the proof.

Step 3: from the clock phase to χ. Let

$$\vartheta(v) := \frac{1}{2i}\Big(\mathrm{Log}\,\Gamma\big(\tfrac14 + \tfrac{iv}{2}\big) - \mathrm{Log}\,\Gamma\big(\tfrac14 - \tfrac{iv}{2}\big)\Big) - \frac{v}{2}\log\pi, \tag{8.29}$$

holomorphic on $|\Im v| < \frac12$. The identity $\chi(\frac12 + iv) = e^{-2i\vartheta(v)}$ holds for real v, hence on the strip, and $s = \frac12 + i\tau_*$ with $|\Im\tau_*| = \sigma' < \frac12$. Stirling's formula in the strip gives $\vartheta(v) = \frac{v}{2}\,\mathrm{Log}\,\frac{v}{2\pi} - \frac{v}{2} - \frac{\pi}{8} + O(|v|^{-1})$, i.e.

$$\mathcal{N}_{\mathrm{RVM}}(v) = \frac{\vartheta(v)}{\pi} + 1 + O(|v|^{-1}). \tag{8.30}$$

Therefore

$$w = e^{-2i\vartheta(\tau_*) - 2\pi i + O(1/t)} = \chi(s)\big(1 + O(t^{-1})\big). \tag{8.31}$$

For $\sigma' \ge \sigma_1'$, (8.3) gives $|\chi(s)| \le \frac12$ when $t \ge t_0(\sigma_1')$, so $1 + w = (1 + \chi)(1 + (w - \chi)/(1 + \chi)) = (1 + \chi)(1 + O(t^{-1}))$, which with (8.4) gives (8.5). This completes the proof of Theorem 8.1. □

Lemma 8.5 (Extension to the critical line)**.** *The function $\mathcal{E}_0 = \zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1 + w)) - 1$ extends holomorphically to $\{|\sigma'| < \frac12,\ t > t_0\}$, and $|\mathcal{E}_0| \le C/t$ for $0 \le \sigma' \le \frac14$.*

Proof. By Lemma 8.3, $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) \neq 0$ for $\sigma' \neq 0$, so $1+w$ vanishes in the strip only on $\sigma' = 0$, exactly at $t = \tau_k$; these zeros are simple because $\frac{d}{dt}w = -\log(\tau_*/2\pi)\, w \neq 0$. The zeros of ζ_{RVM} in the strip are the same simple points. Hence $\mathcal{E}_0$ has only removable singularities, and the bound extends to $\sigma' = 0$ by continuity. □

Corollary 8.6 (Derivative at the zeros)**.** *For k with $\tau_k > t_0$,*

$$\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} \log \frac{\tau_k}{2\pi}\Big(1 + O\Big(\frac{1}{\tau_k}\Big)\Big). \tag{8.32}$$

Proof. By Lemma 8.5, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ holds holomorphically near $\sigma' = 0$. At $s = \rho_k^{\mathrm{RVM}}$, $1 + w = 0$, so $\zeta'_{\mathrm{RVM}}(\rho_k^{\mathrm{RVM}}) = \varpi_{\mathrm{RVM}} w'(\rho_k^{\mathrm{RVM}})(1 + \mathcal{E}_0(\rho_k^{\mathrm{RVM}}))$. From (8.2), $w'(s) = -2\pi i\, \mathcal{N}'_{\mathrm{RVM}}(\tau_*) \cdot (-i)\, w(s) = -\log(\tau_*/2\pi)\, w(s)$, and at the zero $w = -1$, $\tau_* = \tau_k$. Finally $|\mathcal{E}_0| \le C/\tau_k$. □

Remark 8.7 (Structure of the proof). The delicate points are handled as follows: the smooth part is computed with principal branches by the identity theorem (Lemma 8.2); the clock estimate is quantified (Lemma 8.3); the sawtooth series is replaced by its Abel means, termwise integration is justified for $r < 1$, and $r \uparrow 1$ is taken after the contour argument uniformly in r and z (Lemma 8.4); the orientation of the lower contour and the sign of the residue are written out in (8.24); the z-dependence of the rays is isolated in g_z and is $O(t^{-2})$, the rest is the explicit real constant ν_4; Stirling is used uniformly for $0 < \sigma' \le B$, and the only place where $\sigma' < \frac{1}{2}$ enters is Step 3.

Corollary 8.8 (Vertical constant limit)**.** *For every fixed $\sigma > \frac{1}{2}$,*

$$\zeta_{\mathrm{RVM}}(\sigma + it) \longrightarrow \varpi_{\mathrm{RVM}}, \qquad F_{\mathrm{RVM}}(\sigma + it) \longrightarrow L := \frac{1}{\varpi_{\mathrm{RVM}}} \qquad (|t| \to \infty). \tag{8.33}$$

More precisely, with $\sigma' = \sigma - \frac{1}{2}$,

$$|w(\sigma + it)| = O_\sigma(|t|^{-\sigma'}), \tag{8.34}$$

and, with $\alpha_c := \min(c - \frac{1}{2}, 1)$ for $c > \frac{1}{2}$,

$$F_{\mathrm{RVM}}(c + it) - L = O_c(|t|^{-\alpha_c}). \tag{8.35}$$

Proof. Expanding the clock at $\tau_* = t - i\sigma'$ gives $\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*) = -\frac{\sigma'}{2\pi}\log\frac{t}{2\pi} + O_\sigma(t^{-1})$, hence $|w| = e^{2\pi\Im\mathcal{N}_{\mathrm{RVM}}(\tau_*)} = O_\sigma(t^{-\sigma'})$. Inverting (8.4) for large t gives $F_{\mathrm{RVM}} = L(1 + O(|w|) + O(t^{-1}))$. □

8.2. **Bounds on the whole half-plane.** The constants of Theorem 8.1 depend on the width B of the band. The following estimate holds on the whole half-plane and does not use contours. Put

$$\mathfrak{s}_1(\tau) := \int_{\tau_0}^{\tau} \psi(\mathcal{N}_{\mathrm{RVM}}(v) + \tfrac{1}{2})\, dv \qquad (\tau \ge \tau_0). \tag{8.36}$$

Lemma 8.9 (Bounded primitive of the midpoint sawtooth)**.** $|\mathfrak{s}_1(\tau)| \le \dfrac{\pi}{4\log 2}$ *for all $\tau \ge \tau_0$.*

Proof. Substitute $y = \mathcal{N}_{\mathrm{RVM}}(v)$, $dv = \mathcal{T}'(y)\, dy$ with $\mathcal{T}'(y) = 2\pi/\log(\mathcal{T}(y)/2\pi)$ from (4.4). The weight $\mathcal{T}'$ is positive and decreasing, with $\mathcal{T}' \le 2\pi/\log 2$ on $y \ge \mathcal{N}_{\mathrm{RVM}}(\tau_0)$ because $\log(\tau_0/2\pi) = \log 2$. The primitive of $y \mapsto \psi(y + \frac{1}{2})$ is periodic with values in $[-\frac{1}{8}, 0]$, so every integral of $\psi(y + \frac{1}{2})$ over an interval has modulus at most $\frac{1}{8}$. The second mean-value theorem gives $|\mathfrak{s}_1(\tau)| \le \frac{2\pi}{\log 2} \cdot \frac{1}{8}$. □

Theorem 8.10 (Two-sided bound and limit at $+\infty$)**.** *There is an absolute constant R_0 such that, for $s = \frac{1}{2} + z$, $z = \sigma' + it$, $\sigma' > \frac{1}{2}$ and $|z| \ge R_0$,*

$$\zeta_{\mathrm{RVM}}(s) = \varpi_{\mathrm{RVM}}\, e^{\theta_{\mathrm{RVM}}(s)}, \qquad |\theta_{\mathrm{RVM}}(s)| \le \frac{\pi^2}{2\log 2} \cdot \frac{1}{\sigma'} + \frac{C}{|z|}. \tag{8.37}$$

Consequently, for every $\sigma_0 > 1$ there is $C(\sigma_0)$ with

$$C(\sigma_0)^{-1} \le |\zeta_{\mathrm{RVM}}(s)| \le C(\sigma_0) \qquad (\Re s \ge \sigma_0), \tag{8.38}$$

and $\zeta_{\mathrm{RVM}}(s) \to \varpi_{\mathrm{RVM}}$ *uniformly in* t *as* $\Re s \to +\infty$. *Moreover*

$$\nu_4 = -2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau. \tag{8.39}$$

Proof. Smooth part. For $\Re z > 0$ the computation in the proof of Lemma 8.2 is exact:

$$\Lambda_{\mathrm{sm}}(z) = \frac{z}{2}\operatorname{Log}\frac{z}{2\pi e} + \frac{7}{8}\operatorname{Log}\Big(1+\frac{z^2}{\tau_0^2}\Big) - \frac{1}{\pi}\int_0^{\tau_0} \log\frac{\tau}{2\pi e}\,\frac{z^2}{\tau^2+z^2}\, d\tau. \tag{8.40}$$

For $|z| \ge 2\tau_0$ one has $|\tau^2+z^2| \ge \frac{3}{4}|z|^2$ on $[0,\tau_0]$, and Stirling's formula is uniform in $|\arg(s/2)| < \frac{\pi}{2}$; the expansions (8.14) and (8.16) therefore hold with $O(|z|^{-1})$ in place of $O(t^{-1})$, uniformly on $\Re z > \frac{1}{2}$, and $\Lambda_{\mathrm{sm}}(z) - \log G(s) = \nu_3 + O(|z|^{-1})$.

Sawtooth part. Integrating (8.9) by parts with (8.36) ($\mathfrak{s}_1(\tau_0) = 0$, $f'(\tau) = O(|z|^2\tau^{-3})$),

$$\Lambda_{\mathrm{saw}}(z) = -\int_{\tau_0}^{\infty} \mathfrak{s}_1 f''\, d\tau, \qquad f''(\tau) = \frac{2}{\tau^2} - \frac{1}{(\tau - iz)^2} - \frac{1}{(\tau+iz)^2}. \tag{8.41}$$

Since $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$ and $\int_{\mathbb{R}} \frac{d\tau}{(\tau-a)^2+\sigma'^2} = \frac{\pi}{\sigma'}$, Lemma 8.9 gives

$$\Big|\Lambda_{\mathrm{saw}}(z) + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{s}_1(\tau)}{\tau^2}\, d\tau\Big| \le \frac{\pi}{4\log 2}\cdot\frac{2\pi}{\sigma'}. \tag{8.42}$$

Assembly. By (8.27), $\log\zeta_{\mathrm{RVM}}(s) = \log\varpi_\infty + \theta_{\mathrm{RVM}}(s)$ with $\varpi_\infty := P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac{1}{2})\exp(\nu_3 - 2\int_{\tau_0}^{\infty}\mathfrak{s}_1\tau^{-2}d\tau) > 0$ and θ_{RVM} bounded as in (8.37). For fixed σ' and $t \to \infty$, Corollary 8.8 gives $|\log\varpi_{\mathrm{RVM}} - \log\varpi_\infty| \le \pi^2/(2\sigma'\log 2)$; letting $\sigma' \to \infty$, $\varpi_\infty = \varpi_{\mathrm{RVM}}$, which is (8.39) by (8.28). On the compact set $\{\Re s \ge \sigma_0,\ |z| \le R_0\}$ the function ζ_{RVM} is continuous and has no zeros or poles, which gives (8.38). □

Theorem 8.10 extends the two-sided bound (9.9) of Corollary 9.3 from bands to the whole half-plane $\Re s \ge \sigma_0 > 1$, and identifies ϖ_{RVM} also as the limit of $\zeta_{\mathrm{RVM}}(\sigma)$ as $\sigma \to +\infty$.

Theorem 8.11 (No absolutely convergent Dirichlet expansion)**.** *Neither* ζ_{RVM} *nor* $F_{\mathrm{RVM}} = 1/\zeta_{\mathrm{RVM}}$ *is represented by a Dirichlet series* $\sum_n a_n n^{-s}$ *converging absolutely in some half-plane* $\Re s > \sigma_0$.

Proof. Suppose $F_{\mathrm{RVM}}(s) = \sum_{n\ge1} a_n n^{-s}$ absolutely for $\Re s > \sigma_0$. Fix $\sigma > \max(\sigma_0, \frac{1}{2})$ and put $\tilde a_1 = a_1 - L$, $\tilde a_n = a_n$ $(n \ge 2)$, so $F_{\mathrm{RVM}}(\sigma+it) - L = \sum_n \tilde a_n n^{-\sigma-it}$ with $\sum_n |\tilde a_n| n^{-\sigma} < \infty$. Then

$$\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{m,n} \tilde a_m\overline{\tilde a_n}(mn)^{-\sigma}\,\frac{1}{2T}\int_{-T}^{T}\Big(\frac{n}{m}\Big)^{it} dt, \tag{8.43}$$

the interchange being justified by absolute convergence of the double series, uniformly in t. The inner average is 1 for $m = n$ and is bounded by $\min(1, 1/(T|\log(n/m)|)) \to 0$ for $m \ne n$; each term is bounded by $|\tilde a_m\tilde a_n|(mn)^{-\sigma}$, which is summable, so dominated convergence gives

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\big|F_{\mathrm{RVM}}(\sigma+it) - L\big|^2 dt = \sum_{n\ge1}|\tilde a_n|^2 n^{-2\sigma}. \tag{8.44}$$

By (8.33) the integrand tends to 0 and is bounded on the line (continuous, with a finite limit), so the left side of (8.44) is 0. Hence every $\tilde a_n = 0$, $F_{\mathrm{RVM}} \equiv L$ in the half-plane and, by analytic continuation, everywhere, which contradicts $F_{\mathrm{RVM}}(1) = 0$. The same argument applied to $\zeta_{\mathrm{RVM}} - \varpi_{\mathrm{RVM}}$ forces an absolutely convergent Dirichlet series for ζ_{RVM} to be the constant ϖ_{RVM}, which contradicts its pole at $s = 1$. □

Theorem 8.12 (Unsmoothed synthetic Mertens bound)**.** *For every* $x > 1$ *and every* $c > \frac{1}{2}$ *the symmetric Perron integral*

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) := \lim_{U\to\infty}\frac{1}{2\pi i}\int_{c-iU}^{c+iU} F_{\mathrm{RVM}}(s)\frac{x^s}{s}\, ds \tag{8.45}$$

exists, is independent of $c > \frac{1}{2}$, and equals the absolutely convergent expression

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + \frac{1}{2\pi i}\int_{(c)} (F_{\mathrm{RVM}}(s) - L)\frac{x^s}{s}\,ds. \tag{8.46}$$

For every $\varepsilon \in (0, \frac{1}{2})$,

$$M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon}). \tag{8.47}$$

Proof. Fix $c > \frac{1}{2}$. By (8.35) and continuity of F_{RVM} on $\Re s = c$ (no poles there by Proposition 7.1), the integral in (8.46) converges absolutely. The Perron kernel gives, for $x > 1$ and $c > 0$ [7, Ch. 17],

$$\lim_{U\to\infty} \frac{1}{2\pi i}\int_{c-iU}^{c+iU} \frac{x^s}{s}\,ds = 1. \tag{8.48}$$

Splitting $F_{\mathrm{RVM}} = L + (F_{\mathrm{RVM}} - L)$ proves existence of (8.45) and (8.46). If $\frac{1}{2} < c < c'$, the integrand of (8.45) is holomorphic on the closed strip $c \le \Re s \le c'$ (Proposition 7.1); by Theorem 8.1(a) with $B = c' - \frac{1}{2}$, F_{RVM} is bounded on the horizontal segments $\Im s = \pm U$, $U \ge t_0(B)$, so these contribute $O(x^{c'}U^{-1}) \to 0$, and Cauchy's theorem gives independence of c. Take $c = \frac{1}{2} + \varepsilon$ in (8.46):

$$|M^{\mathrm{crit}}_{\mathrm{RVM}}(x) - L| \le \frac{x^{1/2+\varepsilon}}{2\pi}\int_{-\infty}^{\infty} \frac{|F_{\mathrm{RVM}}(\frac{1}{2}+\varepsilon+it) - L|}{|\frac{1}{2}+\varepsilon+it|}\,dt, \tag{8.49}$$

and the integral is finite by (8.35), the integrand being $O(|t|^{-1-\alpha_c})$. □

Remark 8.13 (Scope). Theorem 8.12 concerns $1/\zeta_{\mathrm{RVM}}$. The transfer to M goes through the endpoint of the λ-family and is Theorem 11.8.

9. The exact Hadamard product and the comparison with ζ

Put

$$P_\zeta(s) := G(s)\zeta(s) = 2\xi(s), \qquad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{9.1}$$

Theorem 9.1 (Exact Hadamard product)**.** *Let ρ run over the nontrivial zeros of ζ with $\Im\rho > 0$, with multiplicity, ordered by $\Im\rho$. The product*

$$P_\zeta(s) = \prod_{\Im\rho>0}\Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big) \tag{9.2}$$

converges absolutely and locally uniformly in $s \in \mathbb{C}$, and the identity holds with no exponential or constant factor.

Proof. ξ is entire of order one with $\xi(0) = \frac{1}{2}$, so Hadamard's factorization [15], [21, §2.12], [7, Ch. 12] gives

$$2\xi(s) = e^{Bs}\prod_\rho \Big(1 - \frac{s}{\rho}\Big)e^{s/\rho}, \tag{9.3}$$

the product running over all nontrivial zeros and converging absolutely because $\sum_\rho |\rho|^{-2} < \infty$. The functional equation $\xi(s) = \xi(1-s)$ makes $\rho \mapsto 1-\rho$ an involution of the zero multiset; no nontrivial zero is real, since $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\ge1}(-1)^{n-1}n^{-\sigma} > 0$ for $0 < \sigma < 1$; so this involution exchanges $\Im\rho > 0$ and $\Im\rho < 0$. Grouping ρ with $1-\rho$, which absolute convergence permits,

$$\Big(1 - \frac{s}{\rho}\Big)\Big(1 - \frac{s}{1-\rho}\Big)e^{s/\rho + s/(1-\rho)} = \Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big)e^{s/(\rho(1-\rho))}. \tag{9.4}$$

Since $|\rho(1-\rho)| \ge (\Im\rho)^2$ and $\sum_{\Im\rho>0}(\Im\rho)^{-2} < \infty$, the numbers $1/(\rho(1-\rho))$ are absolutely summable, so the exponentials collect into a single factor:

$$2\xi(s) = e^{B's}\,\Pi_\zeta(s), \qquad \Pi_\zeta(s) := \prod_{\Im\rho>0}\Big(1 + \frac{q(s)}{\rho(1-\rho)}\Big), \qquad B' = B + \sum_{\Im\rho>0}\frac{1}{\rho(1-\rho)}. \tag{9.5}$$

Π_ζ depends on s only through $q(s)$, hence $\Pi_\zeta(1-s) = \Pi_\zeta(s)$. From $\xi(1-s) = \xi(s)$ we get $e^{B's} = e^{B'(1-s)}$ on the open set where $\Pi_\zeta \neq 0$, so $B' = 0$. □

Define the exact quotient

$$\mathcal{Q}_{\mathrm{RVM}}(s) := \frac{\zeta(s)}{\zeta_{\mathrm{RVM}}(s)} = \frac{P_\zeta(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}. \tag{9.6}$$

Proposition 9.2 (Exact transform relation)**.** *For $\Re s > 1$, and by meromorphic continuation,*

$$F_{\mathrm{RVM}}(s) = \mathcal{Q}_{\mathrm{RVM}}(s)\,\frac{1}{\zeta(s)}. \tag{9.7}$$

If the inverse Mellin transform $\mathcal{Q}^\vee_{\mathrm{RVM}}$ of $\mathcal{Q}_{\mathrm{RVM}}$ and the inverse defining $M^{\mathrm{crit}}_{\mathrm{RVM}}$ exist in a common distribution space with Mellin convolution, then

$$M^{\mathrm{crit}}_{\mathrm{RVM}} = \mathcal{Q}^\vee_{\mathrm{RVM}} *_{\mathrm{Mel}} M. \tag{9.8}$$

Proof. (9.7) is the identity $1/\zeta_{\mathrm{RVM}} = (\zeta/\zeta_{\mathrm{RVM}})(1/\zeta)$; (9.8) is the Mellin convolution theorem under the stated assumptions. □

Corollary 9.3 (All growth of ζ to the right of $\frac12$ lies in $\mathcal{Q}_{\mathrm{RVM}}$)**.** *Let $0 < \varepsilon < c' - \frac12$. There are t_0 and $0 < c_- \le c_+$ such that*

$$c_- \le |\zeta_{\mathrm{RVM}}(\sigma + it)| \le c_+ \qquad (\tfrac12 + \varepsilon \le \sigma \le c',\ |t| \ge t_0), \tag{9.9}$$

and therefore, in the same region,

$$c_-|\mathcal{Q}_{\mathrm{RVM}}(s)| \le |\zeta(s)| \le c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|, \qquad \frac{1}{c_+|\mathcal{Q}_{\mathrm{RVM}}(s)|} \le \Big|\frac{1}{\zeta(s)}\Big| \le \frac{1}{c_-|\mathcal{Q}_{\mathrm{RVM}}(s)|}. \tag{9.10}$$

Consequently:

(i) *the Lindelöf hypothesis holds if and only if $\mathcal{Q}_{\mathrm{RVM}}(\sigma + it) = O_{\sigma,\varepsilon}(|t|^\varepsilon)$ for every $\sigma > \frac12$;*
(ii) *every bound for ζ or $1/\zeta$ on $\frac12 + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$, is equivalent, up to the constants $c_\pm$, to the same bound for $\mathcal{Q}_{\mathrm{RVM}}$ or $1/\mathcal{Q}_{\mathrm{RVM}}$.*

Proof. By Theorem 8.1(a) with $B = c' - \frac12$, $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|\mathcal{E}_0| \le C_B/|t|$ and, by (8.34), $|w| \le C(|t|/2\pi)^{-\varepsilon}$; for $|t| \ge t_0$ this gives $\frac12 \le |1+w| \le 2$ and $\frac12 \le |1+\mathcal{E}_0| \le 2$, hence (9.9) with $c_- = \varpi_{\mathrm{RVM}}/4$, $c_+ = 4\varpi_{\mathrm{RVM}}$. Then (9.10) follows from $\zeta = \mathcal{Q}_{\mathrm{RVM}}\zeta_{\mathrm{RVM}}$. For (i): the Lindelöf hypothesis is equivalent to $\zeta(\sigma + it) = O(|t|^\varepsilon)$ for every $\sigma > \frac12$ [21, Ch. XIII], and (9.10) transfers this to $\mathcal{Q}_{\mathrm{RVM}}$. (ii) is (9.10). □

On the prime side define

$$\mathfrak{d}^{\mathrm{RVM}}_n := \mathbf{1}_{\{n \text{ prime}\}} - c^{\mathrm{RVM}}_n. \tag{9.11}$$

Since $\pi_{\mathrm{RVM}}(1) = 0$, for every integer $N \ge 2$

$$\sum_{2 \le n \le N} \mathfrak{d}^{\mathrm{RVM}}_n = \pi(N) - \pi_{\mathrm{RVM}}(N). \tag{9.12}$$

10. The λ-family from ζ_{RVM} to ζ_{Z1}

For $\lambda \in [0,1]$ define the moving ordinates, the family product and the family zeta function

$$\gamma_k(\lambda) := (1-\lambda)\gamma_k + \lambda\tau_k = \gamma_k + \lambda\eta_k, \tag{10.1}$$

$$P_\lambda(s) := \prod_{k \ge 1} \left(1 + \frac{q(s)}{\frac14 + \gamma_k(\lambda)^2}\right), \tag{10.2}$$

$$\zeta_\lambda(s) := \frac{P_\lambda(s)}{G(s)}, \qquad F_\lambda(s) := \frac{1}{\zeta_\lambda(s)}, \tag{10.3}$$

and the endpoint

$$P_{Z1} := P_0, \qquad \zeta_{Z1} := \zeta_0, \qquad \zeta_1 = \zeta_{\mathrm{RVM}}. \tag{10.4}$$

Theorem 10.1 (The λ-family)**.** *The product* (10.2) *converges absolutely and locally uniformly jointly on* $[0,1]\times\mathbb{C}$*. For every* λ*,* P_λ *is entire of order exactly one,* $P_\lambda(1-s)=P_\lambda(s)$*,* $P_\lambda(\bar s)=\overline{P_\lambda(s)}$*,* $P_\lambda(0)=P_\lambda(1)=1$*, and its zeros, with multiplicity, are exactly*

$$\frac12 \pm i\gamma_k(\lambda), \qquad k\ge 1. \tag{10.5}$$

For $0<\lambda\le 1$ *the ordinates* $\gamma_k(\lambda)$ *are strictly increasing in* k *and all zeros of* P_λ *are simple; at* $\lambda=0$ *the multiplicities are those of the ordinate list* (γ_k)*. For every* $j\ge 0$*,* $P_\lambda^{(j)}\to P_{\lambda_0}^{(j)}$ *locally uniformly as* $\lambda\to\lambda_0$*. In particular no member* ζ_λ *of the family,* $\zeta_{Z1}=\zeta_0$ *included, can have a zero off the critical line other than* $-2,-4,\dots$*.*

Proof. By Lemma 2.2, $|\gamma_k(\lambda)-\gamma_k|\le C_\eta$, and $\gamma_k\asymp k/\log k$; so $\gamma_k(\lambda)\ge\frac12\gamma_k$ for $k\ge k_0$, uniformly in λ. On a compact $K\subset\mathbb{C}$,

$$\sum_{k\ge1}\sup_{s\in K,\ 0\le\lambda\le1}\left|\frac{q(s)}{\frac14+\gamma_k(\lambda)^2}\right|<\infty, \tag{10.6}$$

which gives joint normal convergence and, by Cauchy's formula, convergence of all s-derivatives. For $|s|\le R_0$, uniformly in λ,

$$\log\max_{|s|\le R_0}|P_\lambda(s)|\le\sum_{k\ge1}\log\Big(1+\frac{R_0^2+R_0}{\frac14+\gamma_k(\lambda)^2}\Big)=O(R_0\log R_0), \tag{10.7}$$

so the order is at most one; the zero count is $\asymp T\log T$, so the exponent of convergence is one and the order is exactly one. Symmetries, normalization and divisor follow factor by factor as in Theorem 3.1. For $0<\lambda\le1$,

$$\gamma_{k+1}(\lambda)-\gamma_k(\lambda)=(1-\lambda)(\gamma_{k+1}-\gamma_k)+\lambda(\tau_{k+1}-\tau_k)\ge\lambda(\tau_{k+1}-\tau_k)>0, \tag{10.8}$$

so distinct factors vanish at distinct points and each zero is simple; at $\lambda=0$ equal ordinates give equal factors. □

Remark 10.2 (All differences lie in $\gamma_k-\tau_k$)*.* For $\lambda,\lambda'\in[0,1]$ and s off the divisors,

$$\frac{\zeta_\lambda(s)}{\zeta_{\lambda'}(s)}=\prod_{k\ge1}\frac{(\frac14+(\gamma_k+\lambda'\eta_k)^2)(\frac14+(\gamma_k+\lambda\eta_k)^2+q(s))}{(\frac14+(\gamma_k+\lambda\eta_k)^2)(\frac14+(\gamma_k+\lambda'\eta_k)^2+q(s))}. \tag{10.9}$$

The gamma factor cancels; the members differ only through $\eta_k=\tau_k-\gamma_k$, and all of them coincide if every η_k vanishes.

Theorem 10.3 (Infinitesimal generator)**.** *Put* $A_{k,\lambda}:=\frac14+\gamma_k(\lambda)^2$*. Off the divisor of* P_λ*,*

$$\mathcal{V}_\lambda(s):=\partial_\lambda\log P_\lambda(s)=-2q(s)\sum_{k\ge1}\frac{\gamma_k(\lambda)\,\eta_k}{A_{k,\lambda}\big(A_{k,\lambda}+q(s)\big)}, \tag{10.10}$$

the series converging locally uniformly on compact sets separated from the divisor. Since G *does not depend on* λ*,*

$$\partial_\lambda\log\zeta_\lambda=\mathcal{V}_\lambda, \qquad \partial_\lambda\log F_\lambda=-\mathcal{V}_\lambda. \tag{10.11}$$

Proof. Since $\partial_\lambda A_{k,\lambda}=2\gamma_k(\lambda)\eta_k$,

$$\partial_\lambda\log\Big(1+\frac{q}{A_{k,\lambda}}\Big)=-\frac{2q\,\gamma_k(\lambda)\eta_k}{A_{k,\lambda}(A_{k,\lambda}+q)}. \tag{10.12}$$

On compact sets separated from the divisor the summand is $O_K(|\eta_k|\gamma_k^{-3})$, and $\sum_k\gamma_k^{-3}<\infty$, so termwise differentiation is justified. □

Proposition 10.4 (Integrated multiplier)**.** *On every component avoiding the moving divisor,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\exp\Big(\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.13}$$

$$F_\lambda(s) = F_{Z1}(s)\exp\Big(-\int_0^\lambda \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big), \tag{10.14}$$

and the total multiplier is

$$\Pi_{\mathrm{tr}}(s) := \exp\Big(-\int_0^1 \mathcal{V}_{\lambda'}(s)\,d\lambda'\Big) = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}, \qquad \zeta_{Z1} = \zeta_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}. \tag{10.15}$$

The identities extend meromorphically to $\mathbb{C}$ *through the product form.*

Proof. (10.13) and (10.14) are the fundamental theorem of calculus applied to (10.11). For (10.15), integrate each mode with $r = \gamma_k(\lambda')$, $dr = \eta_k\,d\lambda'$:

$$\int_0^1 \partial_{\lambda'}\log\Big(1+\frac{q}{A_{k,\lambda'}}\Big)d\lambda' \equiv \log\frac{1+q/(\frac14+\tau_k^2)}{1+q/(\frac14+\gamma_k^2)} \pmod{2\pi i}. \tag{10.16}$$

Normal convergence permits summation, and exponentiation removes the branch ambiguity. □

Corollary 10.5 (Expansion at the endpoint)**.** *Let* $K \subset \mathbb{C}$ *be compact and disjoint from the zeros of* P_{Z1}*. Uniformly for* $s \in K$*, as* $\lambda \downarrow 0$*,*

$$\zeta_\lambda(s) = \zeta_{Z1}(s)\big(1 + \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big), \tag{10.17}$$

$$F_\lambda(s) = F_{Z1}(s)\big(1 - \lambda\mathcal{V}_0(s) + O_K(\lambda^2)\big). \tag{10.18}$$

Proof. By Theorem 10.1 the zeros of P_λ converge to those of P_{Z1}, so K avoids the moving divisor for small λ. The series for $\mathcal{V}_\lambda$ and its λ-derivative converge normally on K (the differentiated summands are $O_K(\gamma_k^{-3})$), so $\mathcal{V}_\lambda = \mathcal{V}_0 + O_K(\lambda)$, and (10.13) gives $\log(\zeta_\lambda/\zeta_{Z1}) = \lambda\mathcal{V}_0 + O_K(\lambda^2)$. □

Proposition 10.6 (Residues along the family)**.** *Suppose* γ_k *is a simple ordinate (it occurs once in the list). Put* $\rho_{k,\lambda} = \frac12 + i\gamma_k(\lambda)$ *and, for* $0 < \lambda \le 1$*,*

$$\mathfrak{r}_{k,\lambda} := \operatorname*{Res}_{s=\rho_{k,\lambda}} \frac{1}{s\,\zeta_\lambda(s)} = \frac{G(\rho_{k,\lambda})}{\rho_{k,\lambda}P'_\lambda(\rho_{k,\lambda})}. \tag{10.19}$$

Then

$$\mathfrak{r}_{k,\lambda} \longrightarrow \frac{G(\frac12+i\gamma_k)}{(\frac12+i\gamma_k)P'_{Z1}(\frac12+i\gamma_k)} \qquad (\lambda\downarrow 0). \tag{10.20}$$

Proof. Theorem 10.1 gives convergence of the zero locations and local uniform convergence of P'_λ; the limit denominator is nonzero because the endpoint zero is simple, and G is continuous. □

Remark 10.7 (Multiplicity). For $0 < \lambda \le 1$ every zero of ζ_λ is simple. At $\lambda = 0$ a repeated ordinate of ζ appears in ζ_{Z1} with the same multiplicity, and the residues of Proposition 10.6 are replaced by the full principal part. Multiplicity plays no role in Step 3: Theorem 11.8 uses only that ζ does not vanish in $\Re s > \frac12$.

11. The endpoint ζ_{Z1}, Hamburger's theorem, and Step 3

11.1. The analytic class of the endpoint.

Proposition 11.1 (Analytic class of ζ_{Z1})**.** *The endpoint*

$$P_{Z1}(s) = \prod_{k\ge1}\left(1+\frac{q(s)}{\frac14+\gamma_k^2}\right), \qquad \zeta_{Z1}(s) = \frac{P_{Z1}(s)}{G(s)}, \tag{11.1}$$

has the following properties.

(i) *P_{Z1} is entire of order exactly one, $P_{Z1}(1-s)=P_{Z1}(s)$, $P_{Z1}(0)=P_{Z1}(1)=1$;*
(ii) *the nontrivial zeros of ζ_{Z1} are exactly*

$$\frac{1}{2}\pm i\gamma_k, \qquad k\ge 1, \tag{11.2}$$

with the multiplicities of the ordinate list; its other zeros are $s=-2,-4,\dots$. It is impossible for ζ_{Z1} to vanish at any point off the critical line other than $-2,-4,\dots$: every factor of P_{Z1} vanishes only at $\frac{1}{2}\pm i\gamma_k$;
(iii) *ζ_{Z1} has a simple pole at $s=1$ with residue 1;*
(iv) *ζ_{Z1} satisfies the functional equation of ζ:*

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s)=\pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta_{Z1}(1-s); \tag{11.3}$$

(v) *$(s-1)\zeta_{Z1}(s)$ is entire of order exactly one.*

Proof. (i) and the divisor in (ii) are the case $\lambda=0$ of Theorem 10.1; the trivial zeros and the absence of other zeros come from $1/G$ as in Theorem 3.1. For (iii), $P_{Z1}(1)=1$ and

$$G(s)=(s-1)+O\big((s-1)^2\big) \qquad (s\to 1). \tag{11.4}$$

For (iv),

$$\pi^{-s/2}\Gamma(s/2)\zeta_{Z1}(s)=\frac{P_{Z1}(s)}{q(s)}, \tag{11.5}$$

and both P_{Z1} and q are invariant under $s\mapsto 1-s$. For (v),

$$(s-1)\zeta_{Z1}(s)=P_{Z1}(s)\,\frac{\pi^{s/2}}{s\,\Gamma(s/2)}, \tag{11.6}$$

where $s\Gamma(s/2)\to 2$ at $s=0$ and $1/\Gamma$ is entire of order one; the product has order at most one, and the zero count $\asymp T\log T$ gives order at least one. □

Define the ratio

$$\mathscr{R}_{Z1}(s):=\frac{\zeta_{Z1}(s)}{\zeta(s)}=\frac{P_{Z1}(s)}{P_\zeta(s)}, \tag{11.7}$$

a meromorphic function; $\mathscr{R}_{Z1}\equiv 1$ is the identity $\zeta_{Z1}=\zeta$.

11.2. **Hamburger's theorem at the endpoint.**

Theorem 11.2 (Hamburger closure)**.** *Consider the condition*

$$\textit{(D)} \qquad \zeta_{Z1}(s)=\sum_{n\ge 1}\frac{a_n}{n^s} \qquad \textit{absolutely for } \Re s>1. \tag{11.8}$$

Then

$$\text{(D)} \iff \mathscr{R}_{Z1}\equiv 1 \iff \zeta_{Z1}=\zeta \iff \text{RH}. \tag{11.9}$$

Proof. Assume (D). Then $\zeta_{Z1}(1-s)=\sum_n a_n n^{-(1-s)}$ converges absolutely for $\Re s<0$, $(s-1)\zeta_{Z1}$ is entire of finite order (Proposition 11.1(v)), and (11.3) holds. These are the hypotheses of Hamburger's theorem [16], [21, §2.13], which gives

$$\zeta_{Z1}=C_H\,\zeta \tag{11.10}$$

for a constant C_H. Both functions have residue 1 at $s=1$, so $C_H=1$. If $\zeta_{Z1}=\zeta$, the nontrivial zeros of ζ are those of ζ_{Z1}, namely (11.2), which is RH. If RH holds, every ρ with $\Im\rho>0$ is $\frac{1}{2}+i\gamma_k$, so $\rho(1-\rho)=\frac{1}{4}+\gamma_k^2$, and Theorem 9.1 gives

$$P_\zeta(s)=\prod_{k\ge 1}\Big(1+\frac{q(s)}{\frac{1}{4}+\gamma_k^2}\Big)=P_{Z1}(s) \tag{11.11}$$

exactly, hence $\zeta = \zeta_{Z1}$. Finally $\zeta = \sum_n n^{-s}$ converges absolutely for $\Re s > 1$, so $\zeta_{Z1} = \zeta$ gives (D). □

11.3. **Condition (D) and the integer-weighted zeta.** By Corollary 6.3, $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ with first coefficient 1, unconditionally. Define the multiplier

$$\mathcal{K}(s) := \frac{\zeta_{Z1}(s)}{\zeta_{\mathbb{Z},\mathrm{RVM}}(s)}. \tag{11.12}$$

Theorem 11.3 (Reduction of (D) to the multiplier)**.** (i) *If $\mathcal{K}$ has an expansion $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ that converges absolutely in $\Re s > 1$, then* (11.8) *holds.*

(ii) *If* (11.8) *holds, then $\mathcal{K}(s) = \sum_{n\geq 1} k_n n^{-s}$ absolutely on the whole half-plane $\Re s > 1$.*

Thus (D) holds if and only if $\mathcal{K}$ is an absolutely convergent Dirichlet series in $\Re s > 1$. Hence, by Theorem 11.2, an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ gives $\zeta_{Z1} = \zeta$, RH, and, by Theorem 11.8, $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$.

Proof. (i) $\zeta_{Z1} = \zeta_{\mathbb{Z},\mathrm{RVM}}\,\mathcal{K}$ is the Dirichlet convolution of two absolutely convergent series in $\Re s > 1$, and $\sum_N |(a^{\mathbb{Z}} * k)_N| N^{-\sigma} \leq \sum_N |a^{\mathbb{Z}}_N| N^{-\sigma} \sum_n |k_n| n^{-\sigma} < \infty$. (ii) By (6.15), $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, and so is its product $\mathcal{K}$ with (11.8). □

By (10.15) and (6.19) (Theorem 6.4),

$$\mathcal{K}(s) = \Pi_{\mathrm{tr}}(s)\, e^{-h_{\mathrm{RVM}}(s)} = \frac{P_{Z1}(s)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(s)}\, e^{-h_{\mathrm{RVM}}(s)}, \tag{11.13}$$

an explicit function of the displacements $\gamma_k - \tau_k$ and of the cell masses c_n^{RVM}. Condition (D) is thereby located in a single object built from the two constructions of this paper: the λ-family and the integer weights.

Remark 11.4 (Continuous-to-atomic meaning of (D))*.* The prime-power count J^{M}_{Z1} of ζ_{Z1} (Theorem 12.5) is a continuous object built from the zeros, and (D) makes its measure the atomic measure $\sum_{p,m} m^{-1}\delta_{p^m}$. The final step of the method is therefore the passage of ζ_{Z1} into the discrete Dirichlet category. Theorem 12.7 states it exactly: (D) holds if and only if J^{M}_{Z1} is constant between consecutive integers, and Corollary 12.3 reduces it to the single identity (12.12) between the displacements $\gamma_k - \tau_k$ and ϖ_{RVM}.

11.4. **Step 3: the Mertens bound from $\zeta_{Z1} = \zeta$.**

Lemma 11.5 (Vertical growth of ζ)**.** *For $\frac{1}{4} \leq \sigma \leq 4$ and $|t| \geq 2$,*

$$|\zeta(\sigma + it)| \leq 15|t|. \tag{11.14}$$

Proof. For $\sigma > 0$ [21, (2.1.5)],

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{y\} y^{-s-1}\, dy. \tag{11.15}$$

For $|t| \geq 2$ and $\sigma \leq 4$ one has $|s| \leq 3|t|$, $|s/(s-1)| \leq |s|/|t| \leq 3$, and the integral is at most $1/\sigma \leq 4$; hence $|\zeta(s)| \leq 3 + 12|t| \leq 15|t|$. □

Lemma 11.6 (Logarithm of ζ off the critical line)**.** *Assume $\zeta(s) \neq 0$ for $\Re s > \frac{1}{2}$. Let $0 < \delta \leq \frac{1}{4}$ and*

$$\omega(\delta) := \frac{\log(3 - 2\delta)}{\log(3 - \delta)} < 1. \tag{11.16}$$

Then, with the branch of $\log\zeta$ equal to $\sum_{n\geq 2} \Lambda(n) n^{-s}/\log n$ on $\Re s > 1$,

$$|\log\zeta(\sigma + it)| \leq \Big(\frac{48}{\delta}\log|t|\Big)^{\omega(\delta)} \qquad (\sigma \geq \tfrac{1}{2} + \delta,\ |t| \geq 4). \tag{11.17}$$

Consequently, for every $\varepsilon > 0$ there is $t_2 = t_2(\delta, \varepsilon)$ with

$$\Big|\frac{1}{\zeta(\sigma + it)}\Big| \le |t|^{\varepsilon} \qquad (\sigma \ge \tfrac{1}{2} + \delta,\ |t| \ge t_2). \tag{11.18}$$

Proof. Fix t with $|t| \ge 4$ and put $s_0 = 2 + it$, $R = \frac{3}{2} - \frac{\delta}{4}$, $r_3 = \frac{3}{2} - \frac{\delta}{2}$. The closed disc $|s - s_0| \le R$ lies in $\frac{1}{2} + \frac{\delta}{4} \le \Re s \le \frac{7}{2}$, $|\Im s - t| \le \frac{3}{2}$; it avoids $s = 1$ and, by hypothesis, every zero of ζ. Hence $\log \zeta$ has a holomorphic branch on a neighbourhood of it, equal to the Dirichlet series on its intersection with $\Re s > 1$, which is convex. On the disc $|\Im s| \ge |t|/2 \ge 2$ and $|\Im s| \le 2|t|$, so Lemma 11.5 gives $\Re \log \zeta = \log |\zeta| \le \log(30|t|)$ there, while $|\log \zeta(s_0)| \le \log \zeta(2)$. The Borel–Carathéodory lemma [20, §5.5], applied with radii $r_3 < R$,

$$\max_{|s-s_0| \le r_3} |\log \zeta(s)| \le \frac{2r_3}{R - r_3} \max_{|s-s_0| = R} \Re \log \zeta(s) + \frac{R + r_3}{R - r_3} |\log \zeta(s_0)|, \tag{11.19}$$

together with $R - r_3 = \delta/4$, $2r_3 \le 3$, $R + r_3 \le 3$ and $\log 30 + \log \zeta(2) < 4 \le 3 \log |t|$, gives

$$\max_{|s-s_0| \le r_3} |\log \zeta(s)| \le \frac{12}{\delta}\big(\log(30|t|) + \log \zeta(2)\big) \le \frac{48}{\delta} \log |t|. \tag{11.20}$$

Now let $\frac{1}{2} + \delta \le \sigma < \frac{3}{2}$ and set $r_1 = \frac{1}{2}$, $r_2 = 2 - \sigma \in (\frac{1}{2}, \frac{3}{2} - \delta]$. On $|s - s_0| = r_1$ one has $\Re s \ge \frac{3}{2}$, so $|\log \zeta(s)| \le \log \zeta(\frac{3}{2}) < 1$. Hadamard's three-circle theorem [20, §5.3] gives, with $M_j = \max_{|s-s_0|=r_j} |\log \zeta|$,

$$M_2 \le M_1^{1-\omega} M_3^{\omega}, \qquad \omega = \frac{\log(r_2/r_1)}{\log(r_3/r_1)} \le \frac{\log(3 - 2\delta)}{\log(3 - \delta)} = \omega(\delta). \tag{11.21}$$

Since $M_1 < 1$ and $M_3 \le \frac{48}{\delta} \log |t|$ with $\frac{48}{\delta} \log |t| > 1$, (11.21) gives (11.17) at $s = \sigma + it$, which lies on the circle of radius r_2. For $\sigma \ge \frac{3}{2}$, $|\log \zeta| \le \log \zeta(\frac{3}{2}) < 1$. Finally $|1/\zeta| = \exp(-\Re \log \zeta) \le \exp |\log \zeta|$, and $(\frac{48}{\delta} \log |t|)^{\omega(\delta)} \le \varepsilon \log |t|$ for $|t| \ge t_2(\delta, \varepsilon)$ because $\omega(\delta) < 1$. □

Lemma 11.7 (Perron kernel)**.** *For $y > 0$, $y \ne 1$, $c > 0$, $T > 0$ [7, Ch. 17],*

$$\left| \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{y^s}{s}\, ds - \mathbf{1}_{\{y>1\}} \right| \le y^c \min\Big(1, \frac{1}{T|\log y|}\Big). \tag{11.22}$$

Theorem 11.8 (Step 3)**.** *If $\zeta_{Z1} = \zeta$, then for every $\varepsilon > 0$*

$$M(x) = O_{\varepsilon}(x^{1/2+\varepsilon}) \qquad (x \ge 1). \tag{11.23}$$

Proof. Non-vanishing. By Proposition 11.1(ii), ζ_{Z1} has no zeros in $\Re s > \frac{1}{2}$; since $\zeta = \zeta_{Z1}$, neither has ζ, and Lemma 11.6 applies.

Truncation. Let $x = N + \frac{1}{2}$ with $N \ge 3$ an integer, $c = 1 + 1/\log x$ and $T = x$. Applying (11.22) with $y = x/n$ to each term of $\sum_n \mu(n) n^{-s} = 1/\zeta(s)$, which converges absolutely on $\Re s = c$, gives

$$M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\, ds + O\Big(\sum_{n \ge 1} \Big(\frac{x}{n}\Big)^c \min\Big(1, \frac{1}{T|\log(x/n)|}\Big)\Big). \tag{11.24}$$

For $n \le x/2$ or $n \ge 2x$, $|\log(x/n)| \ge \log 2$, and these terms contribute at most $x^c \zeta(c)/(T \log 2) \le e\, x(1 + \log x)/(T \log 2)$. For $x/2 < n < 2x$, $(x/n)^c \le 4$ and $|\log(x/n)| \ge |n - x|/(2x)$; as $|n - x| \in \{j - \frac{1}{2} : j \ge 1\}$, each value taken at most twice, these terms contribute at most $8 \sum_{1 \le j \le 2x} \min(1, 4x/(T(2j-1))) \ll 1 + (x/T) \log x$. Hence

$$M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s\,\zeta(s)}\, ds + O(\log x). \tag{11.25}$$

Contour shift. Fix $0 < \delta \le \frac{1}{4}$. The integrand is holomorphic on the closed rectangle with vertices $c \pm iT$, $\frac{1}{2} + \delta \pm iT$: ζ has no zeros there, $s = 1$ is a zero of $1/\zeta$, and $s = 0$ lies outside. By Cauchy's theorem

$$\int_{c-iT}^{c+iT} = \int_{\frac{1}{2}+\delta-iT}^{\frac{1}{2}+\delta+iT} + \int_{\frac{1}{2}+\delta+iT}^{c+iT} - \int_{\frac{1}{2}+\delta-iT}^{c-iT}. \tag{11.26}$$

For $x \ge t_2(\delta, \varepsilon)$, (11.18) on $|t| = T = x$ bounds the horizontal integrals by

$$\frac{1}{2\pi} \int_{\frac{1}{2}+\delta}^{c} \frac{x^\sigma T^\varepsilon}{T}\, d\sigma \le x^c T^{\varepsilon-1} = e\, x^\varepsilon. \tag{11.27}$$

On the left side, $1/\zeta$ is continuous on the compact segment $\{\frac{1}{2} + \delta + it : |t| \le t_2\}$, bounded by a constant $C(\delta, \varepsilon)$, and (11.18) holds for $|t| \ge t_2$; hence

$$\frac{1}{2\pi} \int_{-T}^{T} \frac{x^{1/2+\delta}}{|\frac{1}{2} + \delta + it|\, |\zeta(\frac{1}{2} + \delta + it)|}\, dt \le x^{1/2+\delta} \Big(C'(\delta, \varepsilon) + \int_{t_2}^{T} t^{\varepsilon-1} dt \Big) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}. \tag{11.28}$$

Conclusion. (11.25)–(11.28) give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$ for $x = N + \frac{1}{2}$; taking $\delta = \varepsilon = \varepsilon_0/2$ (with $\varepsilon_0 \le \frac{1}{2}$) gives $O_{\varepsilon_0}(x^{1/2+\varepsilon_0})$. For arbitrary $x \ge 3$, $M(x) = M(\lfloor x \rfloor + \frac{1}{2})$ and $\lfloor x \rfloor + \frac{1}{2} \le 2x$; for $1 \le x < 3$, $|M(x)| \le 2$. □

Theorem 11.9 (Transfer of the synthetic Mertens bound to M)**.** *Assume $\zeta_{Z1} = \zeta$ and let $0 < \delta \le \frac{1}{4}$, $\varepsilon > 0$. Then:*

(i) *on $\Re s > \frac{1}{2}$,*

$$\frac{1}{\zeta(s)} = F_{\mathrm{RVM}}(s)\, \Pi_{\mathrm{tr}}(s)^{-1}, \qquad \Pi_{\mathrm{tr}} = \frac{P_{Z1}}{P_{\mathrm{RVM}}^{\mathrm{crit}}} = \frac{\zeta}{\zeta_{\mathrm{RVM}}}, \tag{11.29}$$

where Π_{tr} is holomorphic and zero-free in $\Re s > \frac{1}{2}$ and $\Pi_{\mathrm{tr}}(1) = 1$;

(ii) *there is $t_3 = t_3(\delta, \varepsilon)$ such that, for $\frac{1}{2} + \delta \le \sigma \le 2$ and $|t| \ge t_3$,*

$$|F_{\mathrm{RVM}}(\sigma + it)| \le 2L, \qquad |\Pi_{\mathrm{tr}}(\sigma + it)^{\pm 1}| \le |t|^\varepsilon; \tag{11.30}$$

(iii) *the synthetic bound of Theorem 8.12, $M_{\mathrm{RVM}}^{\mathrm{crit}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$, carries over to the Mertens function:*

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}). \tag{11.31}$$

Proof. (i) By (10.15), $\zeta_{Z1} = \zeta_{\mathrm{RVM}} \Pi_{\mathrm{tr}}$; with $\zeta_{Z1} = \zeta$ this is (11.29). In $\Re s > \frac{1}{2}$, $\zeta = \zeta_{Z1}$ has no zeros (Proposition 11.1(ii)) and ζ_{RVM} has none (Theorem 3.1); both have a simple pole of residue 1 at $s = 1$, so Π_{tr} is holomorphic there with $\Pi_{\mathrm{tr}}(1) = 1$, and it is zero-free in $\Re s > \frac{1}{2}$.

(ii) By Theorem 8.1(a) with $B = \frac{3}{2}$, for $\delta \le \sigma' \le \frac{3}{2}$ and $|t| \ge t_0$ we have $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+\mathcal{E}_0)$ with $|w| \le C|t|^{-\delta} \le \frac{1}{2}$ and $|\mathcal{E}_0| \le \frac{1}{2}$ for $|t|$ large; hence $|F_{\mathrm{RVM}}| \le 2L$ and

$$\log \zeta_{\mathrm{RVM}} = \log \varpi_{\mathrm{RVM}} + \mathrm{Log}(1 + w) + \mathrm{Log}(1 + \mathcal{E}_0) = O(1). \tag{11.32}$$

By Lemma 11.6, which applies because $\zeta = \zeta_{Z1}$ has no zeros in $\Re s > \frac{1}{2}$, $|\log \zeta(\sigma + it)| \le (\frac{48}{\delta} \log |t|)^{\omega(\delta)}$ with $\omega(\delta) < 1$. Therefore

$$|\log \Pi_{\mathrm{tr}}(\sigma + it)| = |\log \zeta - \log \zeta_{\mathrm{RVM}}| \le \Big(\frac{48}{\delta} \log |t| \Big)^{\omega(\delta)} + O(1) \le \varepsilon \log |t| \tag{11.33}$$

for $|t| \ge t_3(\delta, \varepsilon)$, which gives (11.30).

(iii) The proof of Theorem 8.12 moves the Perron line of $F_{\mathrm{RVM}} x^s/s$ to $\Re s = \frac{1}{2} + \delta$ using only the holomorphy of F_{RVM} in $\Re s > \frac{1}{2}$ and the vertical control of F_{RVM}. By (11.29) the integrand for M is the same integrand multiplied by Π_{tr}^{-1}, which is holomorphic in $\Re s > \frac{1}{2}$ by (i) and at most $|t|^\varepsilon$ by (ii). Hence $|1/\zeta(\sigma + it)| \le 2L|t|^\varepsilon$ for $\frac{1}{2} + \delta \le \sigma \le 2$, $|t| \ge t_3$, and the truncated Perron formula, contour shift and estimates (11.25)–(11.28) of Theorem 11.8, applied with this bound, give $M(x) \ll_{\delta,\varepsilon} x^{1/2+\delta+\varepsilon}$. Choosing $\delta = \varepsilon = \varepsilon_0/2$ gives (11.31). □

Remark 11.10 (The transfer in Mellin form). Formally, (11.29) reads $M = (\Pi_{\mathrm{tr}}^{-1})^{\vee} *_{\mathrm{Mel}} M_{\mathrm{RVM}}^{\mathrm{crit}}$: the Mertens function is the synthetic Mertens function convolved with the inverse Mellin transform of the inverse transport multiplier. The synthetic factor F_{RVM} contributes the exponent $\frac{1}{2}$ and is bounded (Theorem 8.1); the multiplier contributes at most $|t|^{\varepsilon}$. All the information of the λ-family enters through $\Pi_{\mathrm{tr}} = P_{Z1}/P_{\mathrm{RVM}}^{\mathrm{crit}}$, that is, through the displacements $\gamma_k - \tau_k$.

Corollary 11.11 (Step 3 through (D))**.** *Condition* (11.8) *implies* $M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ *for every* $\varepsilon > 0$.

Proof. Theorem 11.2 and Theorem 11.8. □

Remark 11.12 (Status of Step 3). The implication $\zeta_{Z1} = \zeta \Rightarrow M(x) = O_{\varepsilon}(x^{1/2+\varepsilon})$ is proved completely above, and it requires nothing beyond the identity: no separate statement about the zeros of ζ enters, because $\zeta = \zeta_{Z1}$ already carries the divisor (11.2). The identity $\zeta_{Z1} = \zeta$ itself, equivalently (D), equivalently an absolutely convergent Dirichlet expansion of the multiplier $\mathcal{K}$ of (11.12) in $\Re s > 1$ (Theorem 11.3), is not proved in this paper. It is the open step of the method.

11.5. **Reciprocal endpoint and finite-height Perron integrals.** Put

$$F_{Z1}(s) := \frac{1}{\zeta_{Z1}(s)} = F_{\mathrm{RVM}}(s)\,\Pi_{\mathrm{tr}}(s)^{-1}, \tag{11.34}$$

the second identity by (10.15). For $c > 1$ and finite U define

$$M_{\lambda,U}(x) := \frac{1}{2\pi i}\int_{c-iU}^{c+iU} F_{\lambda}(s)\frac{x^s}{s}\,ds, \qquad M_{Z1,U} := M_{0,U}. \tag{11.35}$$

Proposition 11.13 (Finite-height Perron integrals along the family)**.** *For fixed* $c > 1$, $U < \infty$ *and* $x > 0$,

$$M_{\lambda,U}(x) \longrightarrow M_{Z1,U}(x) \qquad (\lambda \downarrow 0). \tag{11.36}$$

Proof. The segment $\{c + it : |t| \le U\}$ is compact and disjoint from every divisor (10.5), and $P_{\lambda} \to P_{Z1}$ uniformly on it, so $F_{\lambda} \to F_{Z1}$ uniformly there. □

Theorem 11.14 (Reciprocal-coefficient criterion)**.** *Assume that in* $\Re s > 1$

$$F_{Z1}(s) = \sum_{n\ge 1} \frac{\mu_{Z1}(n)}{n^s} \tag{11.37}$$

absolutely, and let $M_{Z1}(x) = \sum_{n\le x} \mu_{Z1}(n)$, *so that*

$$\mu_{Z1}(n) = M_{Z1}(n) - M_{Z1}(n-1). \tag{11.38}$$

Then the following are equivalent: (i) $M_{Z1}(n) = M(n)$ *for every* $n \ge 1$*; (ii)* $\mu_{Z1} = \mu$*; (iii)* $F_{Z1} = 1/\zeta$ *in* $\Re s > 1$*; (iv)* $\zeta_{Z1} = \zeta$*; (v) RH.*

Proof. (i)⇔(ii) by (11.38); (ii)⇔(iii) by uniqueness of absolutely convergent Dirichlet coefficients; (iii)⇔(iv) by analytic continuation, both functions being nonzero in $\Re s > 1$; (iv)⇔(v) by Theorem 11.2. □

Remark 11.15 (A route that does not close Step 3). The following approach does not work. One may try to expand both factors of $F_{Z1} = F_{\mathrm{RVM}}\,\Pi_{\mathrm{tr}}^{-1}$ in (11.34) as absolutely convergent Dirichlet series and identify their Dirichlet convolution with μ. By Theorem 8.11 the expansion of F_{RVM} does not exist in any half-plane, so there are no coefficients to convolve. The endpoint criterion concerns ζ_{Z1} itself, as in Theorem 11.2.

12. The endpoint at infinity and the Euler products

This section applies the clock decomposition of Theorem 8.10 to the ordinates γ_k of the endpoint, and constructs the prime-power counts of ζ_{RVM} and ζ_{Z1} as Mellin transforms of their logarithms. No hypothesis is used.

12.1. **The value of the endpoint at infinity.** With the zero count N_ζ of (2.2) put

$$\mathfrak{e}(\tau) := N_\zeta(\tau) - \mathcal{N}_{\mathrm{RVM}}(\tau), \qquad \mathfrak{e}_1(\tau) := \int_{\tau_0}^{\tau} \mathfrak{e}(v)\,dv \qquad (\tau \ge \tau_0 = 4\pi). \tag{12.1}$$

By (2.2), $\mathfrak{e}(\tau) = S(\tau) + O(1/\tau)$, and Littlewood's bound $\int_0^T S(t)\,dt = O(\log T)$ [21, Thm. 9.9] gives an absolute constant $C_{\mathfrak{e}}$ with

$$|\mathfrak{e}_1(\tau)| \le C_{\mathfrak{e}} \log \tau \qquad (\tau \ge \tau_0). \tag{12.2}$$

For the clock the corresponding primitive is $-\mathfrak{s}_1$, bounded by Lemma 8.9.

Theorem 12.1 (Clock decomposition of the endpoint)**.** *Let $s = \frac12 + z$, $z = \sigma' + it$, $\sigma' > 0$. Then*

$$\log \zeta_{Z1}(s) = \log \varpi_{Z1} - \int_{\tau_0}^{\infty} \mathfrak{e}_1(\tau)\Big(\frac{1}{(\tau - iz)^2} + \frac{1}{(\tau + iz)^2}\Big)d\tau + O\Big(\frac{1}{|z|}\Big) \qquad (|z| \ge R_0), \tag{12.3}$$

where

$$\varpi_{Z1} := P_{Z1}(\tfrac12)\exp\Big(\nu_3 + 2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau)}{\tau^2}\,d\tau\Big) > 0. \tag{12.4}$$

Consequently:

(i) *for $\sigma' > 0$ and $|z| \ge R_0$,*

$$\big|\log \zeta_{Z1}(s) - \log \varpi_{Z1}\big| \le \frac{2(\pi+2)C_{\mathfrak{e}}}{\sigma'}\Big(1 + \log(2|t| + 2\sigma' + \tau_0)\Big) + \frac{C}{|z|}; \tag{12.5}$$

(ii) $\zeta_{Z1}(\sigma) \to \varpi_{Z1}$ *as $\sigma \to +\infty$;*

(iii) *the value at infinity is the ordered product*

$$\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_{k\ge1} \frac{\frac14 + \tau_k^2}{\frac14 + \gamma_k^2} := \varpi_{\mathrm{RVM}} \lim_{K\to\infty} \prod_{k\le K} \frac{\frac14 + \tau_k^2}{\frac14 + \gamma_k^2}. \tag{12.6}$$

Proof. As in (8.6), $\log P_{Z1}(s) - \log P_{Z1}(\frac12) = \sum_k f(\gamma_k)$ with $f(\tau) = \mathrm{Log}(1 + z^2/\tau^2)$. Since $\gamma_1 > \tau_0$, $N_\zeta(\tau_0) = 0$ and $\mathfrak{e}(\tau_0) = -\mathcal{N}_{\mathrm{RVM}}(\tau_0)$. Stieltjes integration against $N_\zeta = \mathcal{N}_{\mathrm{RVM}} + \mathfrak{e}$ and two integrations by parts ($f\mathfrak{e} \to 0$ and $\mathfrak{e}_1 f' \to 0$ at ∞, $\mathfrak{e}_1(\tau_0) = 0$) give

$$\sum_k f(\gamma_k) = \Lambda_{\mathrm{sm}}(z) + \int_{\tau_0}^{\infty} \mathfrak{e}_1 f''\,d\tau, \tag{12.7}$$

with the same Λ_{sm} as in (8.9). The proof of Theorem 8.10 gives $\Lambda_{\mathrm{sm}} - \log G = \nu_3 + O(|z|^{-1})$, and inserting f'' from (8.41) gives (12.3).

For (i), $|\tau \mp iz|^2 = (\tau \pm t)^2 + \sigma'^2$. For real a, splitting at $X = 2|a| + 2\sigma' + \tau_0$ and using $(\tau - a)^2 \ge \tau^2/4$ for $\tau \ge X$,

$$\int_{\tau_0}^{\infty} \frac{\log \tau}{(\tau - a)^2 + \sigma'^2}\,d\tau \le \frac{\pi}{\sigma'}\log X + 4\,\frac{1 + \log X}{X} \le \frac{\pi+2}{\sigma'}(1 + \log X). \tag{12.8}$$

With (12.2) this gives (12.5). For (ii) take $t = 0$: the right side of (12.5) tends to 0 as $\sigma' \to \infty$.

For (iii), Theorem 8.10 gives $\varpi_{\mathrm{RVM}} = P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)\exp(\nu_3 - 2\int \mathfrak{s}_1\tau^{-2}d\tau)$, so

$$\frac{\varpi_{Z1}}{\varpi_{\mathrm{RVM}}} = \frac{P_{Z1}(\frac12)}{P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)}\exp\Big(2\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1(\tau) + \mathfrak{s}_1(\tau)}{\tau^2}\,d\tau\Big). \tag{12.9}$$

Since $\mathfrak{e} + \psi(\mathcal{N}_{\mathrm{RVM}} + \frac12) = N_\zeta - n_{\mathrm{RVM}}$ and $(\mathfrak{e}_1 + \mathfrak{s}_1)(X)/X \to 0$,

$$\int_{\tau_0}^{\infty} \frac{\mathfrak{e}_1 + \mathfrak{s}_1}{\tau^2}\,d\tau = \lim_{X\to\infty}\int_{\tau_0}^{X} \frac{N_\zeta(\tau) - n_{\mathrm{RVM}}(\tau)}{\tau}\,d\tau = \lim_{K\to\infty}\sum_{k\le K} \log\frac{\tau_k}{\gamma_k}. \tag{12.10}$$

In the last step, $\int_{\tau_0}^{X}(N_\zeta - n_{\mathrm{RVM}})\tau^{-1}d\tau = \sum_{\gamma_k\le X}\log\frac{X}{\gamma_k} - \sum_{\tau_k\le X}\log\frac{X}{\tau_k}$; the indices counted by only one of the two sums are $O(\log X)$ in number and, by Lemma 2.2, have ordinates in $[X - C_\eta, X]$, so

they contribute $O(\log X/X)$. Finally $P_{Z1}(\frac12)/P^{\mathrm{crit}}_{\mathrm{RVM}}(\frac12)=\prod_k \frac{\gamma_k^2}{\frac14+\gamma_k^2}\cdot\frac{\frac14+\tau_k^2}{\tau_k^2}$, an absolutely convergent product, and multiplying by $\prod_{k\le K}\tau_k^2/\gamma_k^2$ gives (12.6). □

The ordinates enter ϖ_{Z1} only through the displacements $\gamma_k-\tau_k$, in agreement with Remark 10.2.

Theorem 12.2 (The value at infinity decides the endpoint identity)**.** *$\varpi_{Z1}\ge 1$, and $\varpi_{Z1}=1$ if and only if $\zeta_{Z1}=\zeta$.*

Proof. Write $\zeta_{Z1}=\zeta\mathscr{R}_{Z1}$ with $\mathscr{R}_{Z1}=P_{Z1}/P_\zeta$ as in (11.7), and compare the products factor by factor, with the zeros $\rho_k=\beta_k+i\gamma_k$ listed with the ordinates (Theorem 9.1). Factors with $\rho_k(1-\rho_k)=\frac14+\gamma_k^2$ cancel. The remaining factors of P_ζ come in pairs $\{\rho,1-\bar\rho\}$ with the same ordinate γ; with $\delta=\Re\rho-\frac12$ and $W=(s-\frac12)^2+\gamma^2$, the corresponding factor of $\mathscr{R}_{Z1}$ is

$$\frac{W^2}{(W-\delta^2)^2+4\delta^2\gamma^2}\cdot c_\gamma,\qquad c_\gamma=\frac{|\rho(1-\rho)|^2}{(\frac14+\gamma^2)^2}=1+\frac{2\delta^2(\gamma^2-\frac14)+\delta^4}{(\frac14+\gamma^2)^2}>1. \tag{12.11}$$

For real $\sigma>1$, $W\ge\gamma^2\ge\gamma_1^2$, and the logarithm of the first factor is at most $14\delta^2/W\le 14\delta^2/\gamma^2$ in modulus; since $\delta\le\frac12$ and $\sum\gamma^{-2}<\infty$, dominated convergence and $\zeta(\sigma)\to1$ give $\varpi_{Z1}=\lim_{\sigma\to\infty}\zeta_{Z1}(\sigma)=\prod c_\gamma\ge1$, with equality exactly when there are no such pairs, that is, when $\mathscr{R}_{Z1}\equiv1$. □

Corollary 12.3 (The endpoint identity as an identity of the displacements)**.** *The following are equivalent: (D); $\zeta_{Z1}=\zeta$; $\varpi_{Z1}=1$;*

$$\prod_{k\ge1}\frac{\frac14+\gamma_k^2}{\frac14+\tau_k^2}=\varpi_{\mathrm{RVM}} \tag{12.12}$$

(ordered product); and $\lim_{\sigma\to+\infty}\Pi_{\mathrm{tr}}(\sigma)=L=1/\varpi_{\mathrm{RVM}}$.

Proof. Theorems 11.2, 12.1 and 12.2, and $\Pi_{\mathrm{tr}}=\zeta_{Z1}/\zeta_{\mathrm{RVM}}\to\varpi_{Z1}/\varpi_{\mathrm{RVM}}$ by Theorems 8.10 and 12.1(ii). □

Proposition 12.4 (Boundedness of the endpoint)**.** *Let $\sigma_0>1$. Then $|\zeta_{Z1}|$ is bounded above and below on $\Re s\ge\sigma_0$ if and only if*

$$\sup_{\sigma'\ge\sigma_0-\frac12,\ t\ge0}\Big|\Re\int_{\tau_0}^{\infty}\frac{\mathfrak{e}_1(\tau)+\mathfrak{s}_1(\tau)}{(\tau-t+i\sigma')^2}\,d\tau\Big|<\infty. \tag{12.13}$$

Proof. $\tau+iz=\tau-t+i\sigma'$. For $t\ge0$ the term with $(\tau-iz)^{-2}$ in (12.3) is bounded by (12.2), since $|\tau-iz|\ge\tau+t$; the term with $\mathfrak{s}_1$ is bounded by Lemma 8.9. By (12.3), $\log|\zeta_{Z1}|$ is bounded on $\Re s\ge\sigma_0$, $t\ge0$, exactly when (12.13) holds, and $\zeta_{Z1}(\bar s)=\overline{\zeta_{Z1}(s)}$ covers $t<0$. □

The function in (12.13) is, after one integration by parts, the conjugate Poisson integral at distance σ' of the count difference $N_\zeta-n_{\mathrm{RVM}}$; it depends only on the displacements $\gamma_k-\tau_k$.

12.2. The Euler products of ζ_{RVM} and ζ_{Z1}.

Theorem 12.5 (Prime-power counts as Mellin transforms)**.** *Let $X\in\{\mathrm{RVM},Z1\}$. There is a unique function J_X^{M} on $(1,\infty)$ with $J_X^{\mathrm{M}}(x)x^{-c}\in L^2((1,\infty),dx/x)$ for every $c>1$ and*

$$\log\zeta_X(s)=s\int_1^\infty J_X^{\mathrm{M}}(x)\,x^{-s-1}\,dx\qquad(\Re s>1), \tag{12.14}$$

the branch of $\log\zeta_X$ being real on $(1,\infty)$. Equivalently $\zeta_X(s)=\exp\big(\int_{[1,\infty)}x^{-s}\,dJ_X^{\mathrm{M}}(x)\big)$.

Proof. ζ_X has no zeros or poles in $\Re s > 1$, so $\log \zeta_X$ is holomorphic there. For $c > 1$, Theorem 8.10 ($X = \mathrm{RVM}$) and Theorem 12.1 ($X = Z1$), together with continuity on the compact part, give $|\log \zeta_X(s)| \le C_c(1 + \log(|t| + 2))$ on $\Re s \ge c$. Hence $F_\sigma(t) := \log \zeta_X(\sigma + it)/(\sigma + it)$ lies in $L^2(\mathbb{R})$ with $\|F_\sigma\|_2 \le B_c$ for all $\sigma \ge c$. In the variable $u = \log x$, define $J_X^{\mathrm{M}}(e^u)e^{-\sigma u}$ as the inverse Fourier–Plancherel transform of F_σ; Cauchy's theorem and the decay of F on horizontal segments show that J_X^{M} does not depend on σ. Plancherel gives $\int_{\mathbb{R}} |J_X^{\mathrm{M}}(e^u)|^2 e^{-2\sigma u} du = \|F_\sigma\|_2^2/2\pi \le B_c^2/2\pi$ for all $\sigma \ge c$; letting $\sigma \to \infty$ forces $J_X^{\mathrm{M}}(e^u) = 0$ for almost every $u < 0$. Plancherel again gives (12.14), and uniqueness follows from the injectivity of the Fourier transform. □

Theorem 12.6 (Distributional explicit formula)**.** *Let $X \in \{\mathrm{RVM}, Z1\}$, with ordinates $\kappa_k = \tau_k$ for $X = \mathrm{RVM}$ and $\kappa_k = \gamma_k$ for $X = Z1$. In $\mathcal{D}'(1, \infty)$,*

$$(\log x)\,\frac{dJ_X^{\mathrm{M}}}{dx} = 1 - \frac{2}{\sqrt{x}}\,\Re \sum_{k \ge 1} x^{i\kappa_k} - \frac{1}{x(x^2 - 1)}, \tag{12.15}$$

the series converging in $\mathcal{D}'(1, \infty)$. The explicit-formula count

$$J_X(x) := \mathrm{li}(x) - 2\Re \sum_{k \ge 1} \mathrm{Ei}\Big(\big(\tfrac{1}{2} + i\kappa_k\big)\log x\Big) - \log 2 + \int_x^\infty \frac{dy}{y(y^2 - 1)\log y} \tag{12.16}$$

converges in $\mathcal{D}'(1, \infty)$ (J_{RVM} is the function (5.2)), and $J_X^{\mathrm{M}} - J_X$ is a constant C_X on $(1, \infty)$.

Proof. With $\rho_k = \frac{1}{2} + i\kappa_k$, the factorization $\zeta_X = P_X/G$, $P_X'/P_X = \sum_k (\frac{1}{s - \rho_k} + \frac{1}{s - \bar{\rho}_k})$, and the digamma expansion $\frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2}) + \frac{1}{s} = -\frac{\gamma_E}{2} + \sum_{n \ge 1}(\frac{1}{2n} - \frac{1}{s + 2n})$ give, for $\Re s > 1$,

$$-\frac{\zeta_X'}{\zeta_X}(s) = \frac{1}{s - 1} - \sum_{k \ge 1}\Big(\frac{1}{s - \rho_k} + \frac{1}{s - \bar{\rho}_k}\Big) + \sum_{n \ge 1}\Big(\frac{1}{2n} - \frac{1}{s + 2n}\Big) - \frac{\gamma_E}{2} - \frac{1}{2}\log \pi. \tag{12.17}$$

Each term is a Mellin–Stieltjes transform $\int_{[1,\infty)} x^{-s} d\mu$: $dx \mapsto \frac{1}{s-1}$, $x^{\rho - 1}dx \mapsto \frac{1}{s - \rho}$, $x^{-2n-1}dx \mapsto \frac{1}{s+2n}$, $\delta_1 \mapsto 1$. After multiplication by x^{-c} and the change $u = \log x$, the Fourier transforms of the partial sums over k and over n converge locally uniformly with polynomial bounds, so the series converge in $\mathcal{S}'$, and on $(1, \infty)$ the limit measure is the right side of (12.15) times dx. Differentiating (12.14) in s shows that $-\zeta_X'/\zeta_X$ is the Mellin–Stieltjes transform of $(\log x)\, dJ_X^{\mathrm{M}}$; injectivity gives (12.15). The series over k converges in $\mathcal{D}'(1, \infty)$ because $\int \varphi(x) x^{-1/2 + i\kappa} dx$ is rapidly decreasing in κ for $\varphi \in C_c^\infty(1, \infty)$ and $\kappa_k \asymp k/\log k$. By Lemma 5.1, $\mathrm{Ei}(\rho_k u) = e^{\rho_k u}/(\rho_k u) + \varrho_k$ with $\sum |\varrho_k|$ locally uniformly convergent, so (12.16) converges in $\mathcal{D}'(1, \infty)$ for the same reason; differentiating termwise with $\frac{d}{dx}\mathrm{Ei}(\rho \log x) = x^{\rho - 1}/\log x$, $(\log x)J_X'$ equals the right side of (12.15). A distribution on an interval with zero derivative is constant. □

Proof of Theorem 6.4. Take $X = \mathrm{RVM}$ in Theorems 12.5 and 12.6. By Theorem 5.2, J_{RVM} is continuous on $(1, \infty)$, so $J_{\mathrm{RVM}}^{\mathrm{M}} = J_{\mathrm{RVM}} + C_{\mathrm{RVM}}$ pointwise, and $J_{\mathrm{RVM}}^{*} = J_{\mathrm{RVM}}$ on $[2, \infty)$. □

Theorem 12.7 (Cell form of the endpoint identity)**.** *The following are equivalent.*

(i) *(D), that is, $\zeta_{Z1} = \zeta$.*
(ii) *J_{Z1}^{M} is constant on every interval $(n, n + 1)$, $n \ge 1$, and its jumps $\mathsf{b}_n := v_n - v_{n-1}$, where v_n is its value on $(n, n + 1)$ and $v_0 := 0$, satisfy $\sum_{n \le X} |\mathsf{b}_n| = O_\varepsilon(X^{1+\varepsilon})$ for every $\varepsilon > 0$.*
(iii) *For every $n \ge 1$, in $\mathcal{D}'(n, n + 1)$,*

$$2\,\Re \sum_{k \ge 1}\big(x^{i\gamma_k} - x^{i\tau_k}\big) = \sqrt{x}\,\log x\; J_{\mathrm{RVM}}'(x), \tag{12.18}$$

and the jumps of J_{Z1}^{M} satisfy the bound in (ii).

In this case J_{Z1}^{M} is Riemann's prime-power count $J(x) = \sum_{p^m \le x} 1/m$.

Proof. (ii)⇒(i): $|v_n| \le \sum_{m\le n} |\mathsf{b}_m| \ll n^{1+\varepsilon}$, so for $\Re s > 1$ summation by parts in (12.14) gives $\log \zeta_{Z1}(s) = \sum_n v_n(n^{-s} - (n+1)^{-s}) = \sum_{n\ge1} \mathsf{b}_n n^{-s}$, absolutely convergent by partial summation. The exponential of an absolutely convergent Dirichlet series is one, so ζ_{Z1} satisfies (11.8).

(i)⇒(ii): by Theorem 11.2, $\zeta_{Z1} = \zeta$, and $\log\zeta(s) = s\int_1^\infty J(x)x^{-s-1}dx$ for $\Re s > 1$; by uniqueness in Theorem 12.5, $J^{\mathrm{M}}_{Z1} = J$, which is constant between consecutive integers, with jumps $1/m$ at p^m and $\sum_{n\le X} |\mathsf{b}_n| = J(X) \ll X$.

(ii)⇔(iii): Theorem 12.6 for $X = Z1$ and $X = \mathrm{RVM}$, with $dJ^{\mathrm{M}}_{\mathrm{RVM}}/dx = J'_{\mathrm{RVM}}$, gives in $\mathcal{D}'(1,\infty)$

$$(\log x)\Big(\frac{dJ^{\mathrm{M}}_{Z1}}{dx} - J'_{\mathrm{RVM}}(x)\Big) = -\frac{2}{\sqrt{x}}\,\Re\sum_{k\ge1}\big(x^{i\gamma_k} - x^{i\tau_k}\big), \tag{12.19}$$

so $dJ^{\mathrm{M}}_{Z1}/dx = 0$ on $(n, n+1)$ is (12.18). □

Remark 12.8 (The final step in cell form). Theorem 12.7 is the exact form of the continuous-to-atomic passage of Remark 11.4. Both sides of (12.18) are built from the midpoint construction: the right side from the clock alone (J'_{RVM} is bounded on $[2,\infty)$ by Theorem 6.1, equal to $1/\log x$ plus the unit-periodic first harmonic and the harmonics with stationary points $2\pi x^{1/m}$, Section 6), the left side from the displacements $\gamma_k - \tau_k$. The final step of the method is the identity (12.18) on every cell, equivalently (12.12). Numerically, the explicit formula (12.16) with the γ_k is constant between consecutive integers up to the truncation of the list (Experiment 13.6, Figure 8), and the partial products of (12.12) approach ϖ_{RVM} (Experiment 13.3).

13. Numerical illustration with 2 001 052 zeros

This section shows what is being constructed and checks that the theorems are consistent with computation. It uses the ordinates $\gamma_1, \dots, \gamma_N$ of the first $N = 2\,001\,052$ zeros of ζ (file `zeross.bin`, $\gamma_N = 1\,132\,490.6587$) and the midpoint ordinates $\tau_1, \dots, \tau_N$ computed from (2.5) (file `zLamb.bin`, $\tau_N = 1\,132\,490.7226$). The programs are listed in Appendix A and their complete console output in Appendix B; all figures have linear axes. Each item is an *experiment*: it reports a computation and does not replace a proof.

Products over the ordinates are evaluated as follows: the first 10^5 factors exactly, the factors $10^5 < k \le N$ through power sums of $(\frac14 + o_k^2)^{-1}$, and the factors $k > N$ by the midpoint rule for the clock, which gives the tail $\frac{1}{2\pi}(q\frac{L_U+1}{U} - \frac{q^2}{2}\frac{3L_U+1}{9U^3} + \frac{q^3}{3}\frac{5L_U+1}{25U^5})$ with $\mathcal{N}_{\mathrm{RVM}}(U) = N$, $L_U = \log(U/2\pi)$.

13.1. The clock and the zeros.

Experiment 13.1 (Clock, midpoints and displacements). *Object:* $\mathcal{N}_{\mathrm{RVM}}$, N_ζ, τ_k, $\eta_k = \tau_k - \gamma_k$. *Range:* $k \le N$. *Expected:* (2.4) to rounding, and $|\eta_k| \le C_\eta$ (Lemma 2.2). *Observed:* $\max_k |\mathcal{N}_{\mathrm{RVM}}(\tau_k) - (k - \frac12)| = 6.985\cdot10^{-10}$; $\max_k|\eta_k| = 0.9937$ (at $k = 34$), mean $|\eta_k| = 0.1512$, root mean square 0.1881, mean $\eta_k = 1.7\cdot10^{-7}$. By blocks of k, $\max|\eta_k|$ is $0.840, 0.994, 0.915, 0.834, 0.786, 0.763, 0.717$ on $[1,10]$, $[11,10^2]$, $[10^2,10^3]$, …, $[10^6, N]$.

Figure 1 shows how the clock places the midpoint ordinates. The staircase N_ζ jumps by one at each γ_k; the smooth clock $\mathcal{N}_{\mathrm{RVM}}$ crosses the half-integer level $k - \frac12$ at τ_k. Since $N_\zeta - \mathcal{N}_{\mathrm{RVM}} = S(T) + O(1/T)$ (Figure 2), the horizontal distance between the open square $(\gamma_k, k-\frac12)$ and the red point $(\tau_k, k-\frac12)$ is the remainder S converted into ordinate units by the slope $\mathcal{N}'_{\mathrm{RVM}} = \frac{1}{2\pi}\log\frac{T}{2\pi}$. The right panel repeats the picture at height $1.13\cdot10^6$, where the mean spacing is 0.519: every τ_k still lies next to its γ_k.

13.2. The λ-family and the endpoint.

Figure 3 shows the core of the method. The lower left panel draws the moving ordinates $\gamma_k(\lambda)$ in clock units, $\mathcal{N}_{\mathrm{RVM}}(\gamma_k(\lambda)) - (k-\frac12)$: at $\lambda = 1$ all lines meet at 0 (the τ_k), at $\lambda = 0$ they sit at the ordinates of ζ; for $\lambda > 0$ the ordinates $\gamma_k(\lambda)$ remain strictly increasing in k (Theorem 10.1). The lower right panel draws the real functions $Z_\lambda(t) = e^{i\vartheta(t)}\zeta_\lambda(\frac12 + it)$

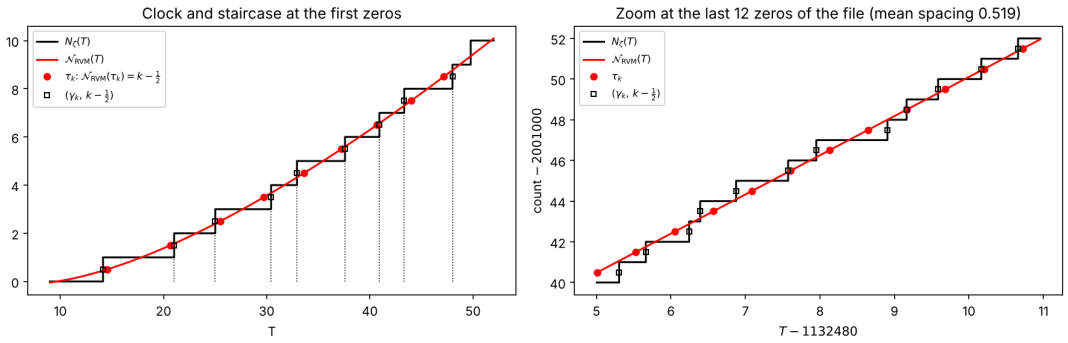


FIGURE 1. The midpoint clock $\mathcal{N}_{\mathrm{RVM}}$ (red), the zero staircase N_ζ (black), the midpoint ordinates τ_k (red points) and the points $(\gamma_k, k-\frac{1}{2})$ (squares). Left: the first nine zeros. Right: the last twelve zeros of the file.

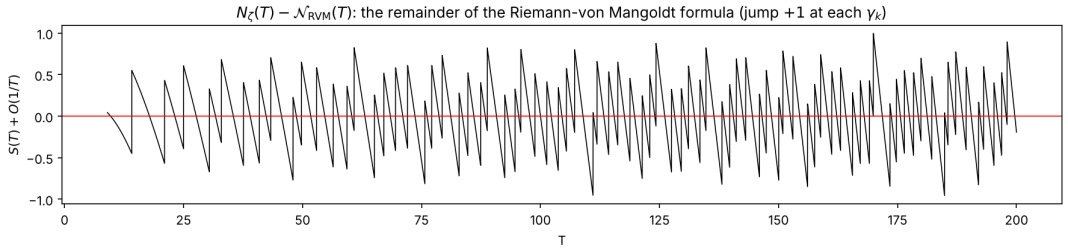


FIGURE 2. The remainder $N_\zeta(T) - \mathcal{N}_{\mathrm{RVM}}(T) = S(T) + O(1/T)$ of (2.2) for $T \le 200$.

for $\lambda = 0, 0.25, 0.5, 0.75, 1$ together with Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$: the zeros of every Z_λ are real, they move continuously with λ, and the curve for $\lambda = 0$ lies on top of Z.

Experiment 13.2 (The endpoint ζ_{Z1} against ζ)**.** *Object:* $\zeta_{Z1} = P_{Z1}/G$ computed from the N ordinates γ_k, and ζ computed by mpmath. *Range:* $\sigma \in \{2, 1.25, 0.75\}$, $1 \le t \le 2000$ (499 points per line), and the critical line $0.5 \le t \le 60$. *Expected:* if $\zeta_{Z1} = \zeta$, the ratio is 1 up to the truncation of the product. *Observed:* $\max|\zeta_{Z1}/\zeta - 1| = 8.7\cdot10^{-10}$, $1.1\cdot10^{-9}$, $2.6\cdot10^{-9}$ on the three lines, and $\max|Z_{Z1} - Z|/\max(1,|Z|) = 8.4\cdot10^{-10}$ on the critical line (Figure 4).

Experiment 13.2 is consistent with the identity $\zeta_{Z1} = \zeta$ to nine digits in the range computed; it is a finite computation and not a proof of the identity.

Experiment 13.3 (The value at infinity: the product identity)**.** *Object:* the ordered partial products $P_K = \prod_{k\le K}(\frac{1}{4}+\gamma_k^2)/(\frac{1}{4}+\tau_k^2)$ of (12.12). *Range:* $K \le N$. *Expected:* by Corollary 12.3, $\zeta_{Z1} = \zeta$ is the statement $P_K \to \varpi_{\mathrm{RVM}}$. *Observed:* $P_K = 0.96256$, 0.96248, 0.96229, 0.961951, 0.961930, 0.9619418189, 0.9619415390 for $K = 10, 10^2, \dots, 10^6, N$, against $\varpi_{\mathrm{RVM}} = 0.9619415807$ from Experiment 13.4; for $K \ge 10^6$, P_K stays in $[0.9619389, 0.9619440]$ with mean 0.9619416633 (Figure 5).

13.3. The transition theorem and ζ_{RVM} against ζ.

Experiment 13.4 (Transition theorem)**.** *Object:* ζ_{RVM}, $w(s)$, ϖ_{RVM}, L. *Range:* $\sigma' \in \{0.25, 0.5, 1.5\}$, $10 \le t \le 10^4$. *Expected:* Theorem 8.1 and Corollary 8.8: $t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1|$ bounded, $|w| = (t/2\pi)^{-\sigma'}(1+O(1/t))$. *Observed:* Richardson extrapolation on $\Re s = 2$ gives $\varpi_{\mathrm{RVM}} = 0.9619415807$ and $L = 1.0395641690$; $\max_{t>100} t\,|\zeta_{\mathrm{RVM}}/(\varpi_{\mathrm{RVM}}(1+w)) - 1| = 0.0617, 0.0618, 0.0620$ for the three values of σ', and 0.0790 for $\varpi_{\mathrm{RVM}}(1+\chi)$ at $\sigma' = 0.25$; $|w|\,t^{\sigma'} = 1.5832$, 2.5066, 15.66–15.75, i.e. $(2\pi)^{\sigma'}$.

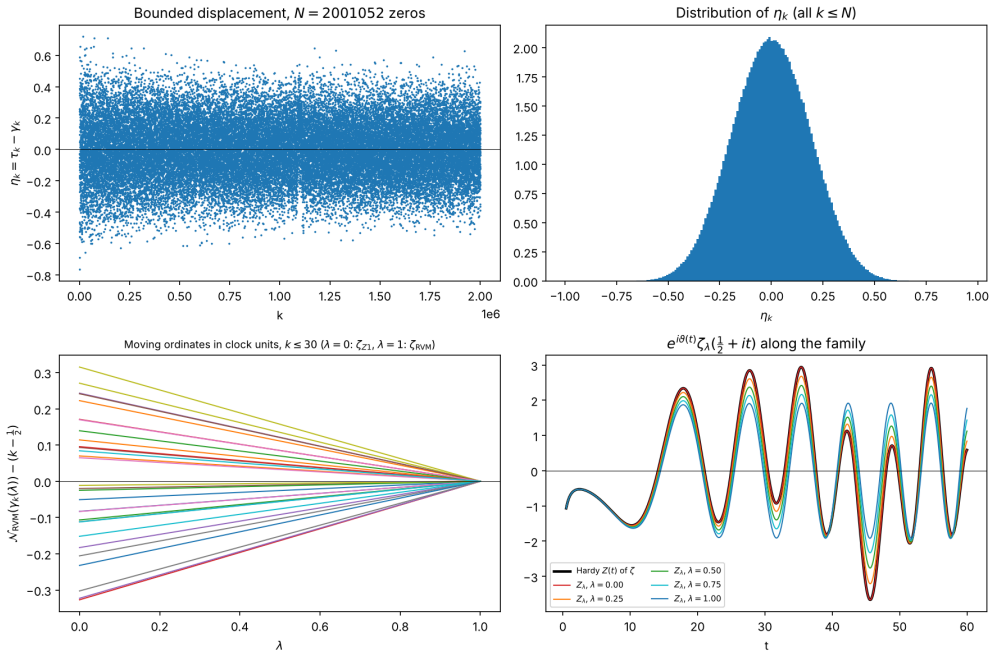


FIGURE 3. Top: η_k for every 50th $k \le N$ and its distribution. Bottom: moving ordinates in clock units, $k \le 30$, and the functions $Z_\lambda(t)$ on the critical line.

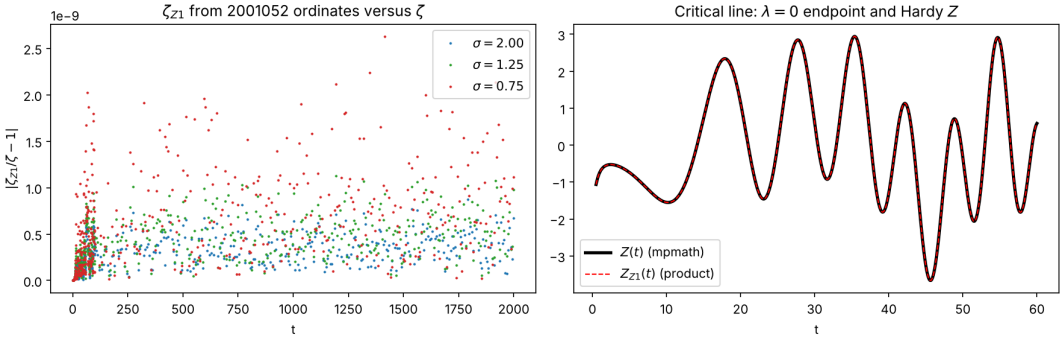


FIGURE 4. Left: $|\zeta_{Z1}/\zeta - 1|$ on three vertical lines. Right: Z_{Z1} and Hardy's Z on the critical line.

The bottom panels of Figure 6 compare $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on two vertical lines: ζ carries the Euler product and oscillates with the primes, whereas ζ_{RVM} approaches the constant ϖ_{RVM} on $\sigma = 2$ and $\varpi_{\mathrm{RVM}}|1+\chi|$ on $\sigma = 0.75$. This is the analytic content of Theorem 8.11.

13.4. **Exponential sums.**

Experiment 13.5 (Exponential sums over the two spectra)**.** *Object:* $S(N;\log x) = \sum_{k\le N} x^{io_k}$ for $o_k = \tau_k$ and $o_k = \gamma_k$. *Range:* $N \le 2\,001\,052$, $x \in \{2,3,4,6,10\}$. *Expected:* $|S_{\mathrm{RVM}}| \ll 1 + \log N$ (Theorem 4.1); for γ_k, Landau's term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$. *Observed:* $\max_N |S_{\mathrm{RVM}}(N;\log x)| = 3.57, 3.98, 3.95, 4.53, 9.70$; at $N = 2\,001\,052$, $|\sum x^{i\gamma_k}| = 88\,301$, $114\,268$, $62\,435$, 0.26, 2.42, against Landau's $88\,342$, $114\,324$, $62\,467$, 0, 0.

13.5. **Prime counts: π and π_{RVM}.**

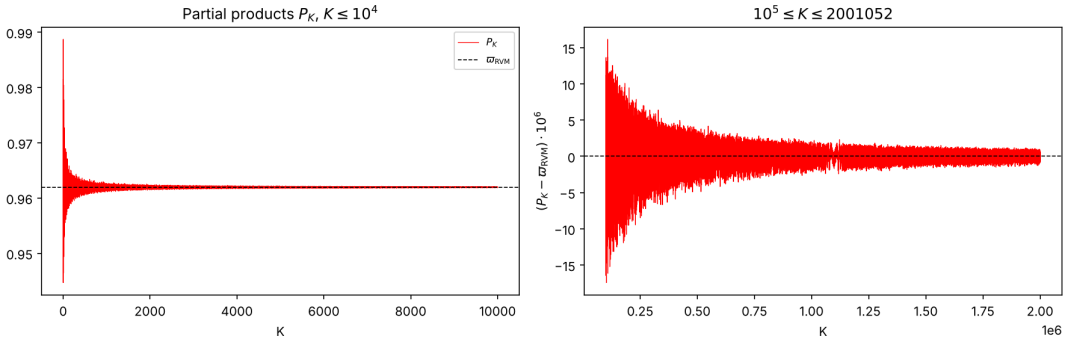


FIGURE 5. The partial products P_K of (12.12) against ϖ_{RVM} (dashed). Left: $K \le 10^4$. Right: $(P_K - \varpi_{\mathrm{RVM}}) \cdot 10^6$ for $10^5 \le K \le N$. The value of the endpoint at $+\infty$ is $\varpi_{Z1} = \varpi_{\mathrm{RVM}} / \lim_K P_K$ (Theorem 12.1).

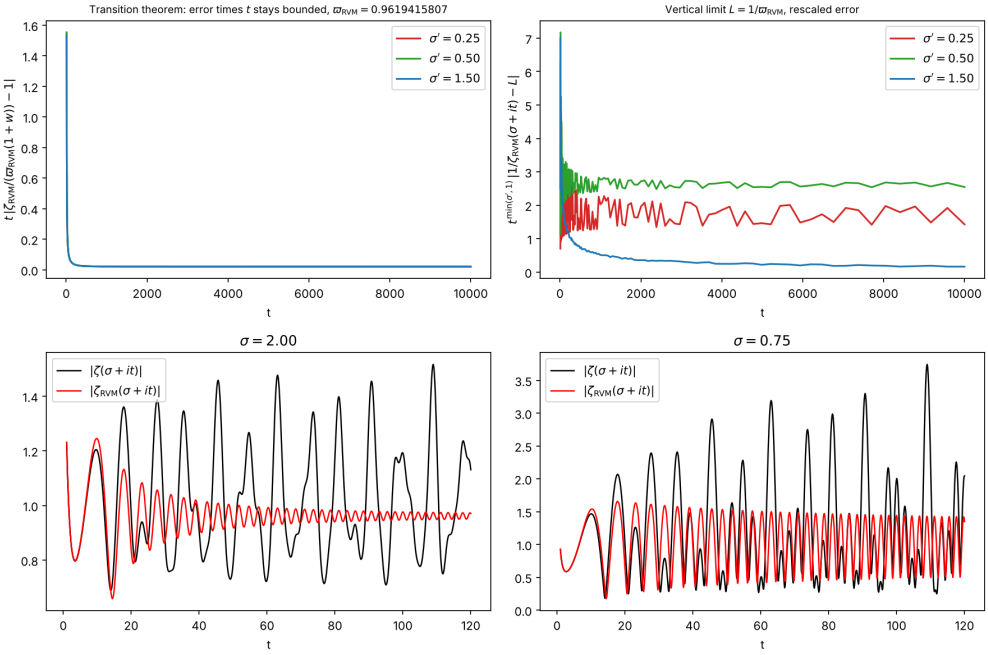


FIGURE 6. Top: the transition error multiplied by t, and the rescaled vertical limit $t^{\min(\sigma',1)}|1/\zeta_{\mathrm{RVM}} - L|$. Bottom: $|\zeta|$ and $|\zeta_{\mathrm{RVM}}|$ on $\sigma = 2$ and $\sigma = 0.75$.

Experiment 13.6 (Prime counts and RVM primes)**.** *Object:* J_{RVM}, π_{RVM}, c_n^{RVM}, p_j^{RVM} against J, π, the primes. *Range:* $2 \le x \le 100$ with all N ordinates of each list (3000 points); $2 \le x \le 1000$ with the first $K = 17\,061$ ordinates and a smooth cut (40 000 points). *Expected:* the explicit formula with the γ_k reproduces $\pi(x)$; π_{RVM} is continuous and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x) = O(\sqrt{x}(\log\log x)^2/\log x)$ (Theorem 5.5). *Observed:* away from the primes, $|\pi_\gamma - \pi| \le 0.40$ on $[2, 100]$; $\pi_{\mathrm{RVM}}(100) = 26.08$, $\pi(100) = 25$, $\mathrm{li}(100) = 30.13$; $\pi_{\mathrm{RVM}}(1000) = 169.98$, $\pi(1000) = 168$, $\mathrm{li}(1000) = 177.61$. π_{RVM} is not monotone: it carries a ripple of unit scale, with minimum grid increment -0.25 on $[2, 100]$. The first crossings of the levels $j = 1, \ldots, 8$ are $p_j^{\mathrm{RVM}} = 2.21, 3.16, 5.19, 7.16, 11.14, 14.02, 17.26, 19.18$ against $p_j = 2, 3, 5, 7, 11, 13, 17, 19$.

Figure 8 shows π and π_{RVM} in the usual format. The RVM count follows the prime staircase much more closely than li: replacing each γ_k by τ_k moves every zero by less than one unit (Experiment 13.1),

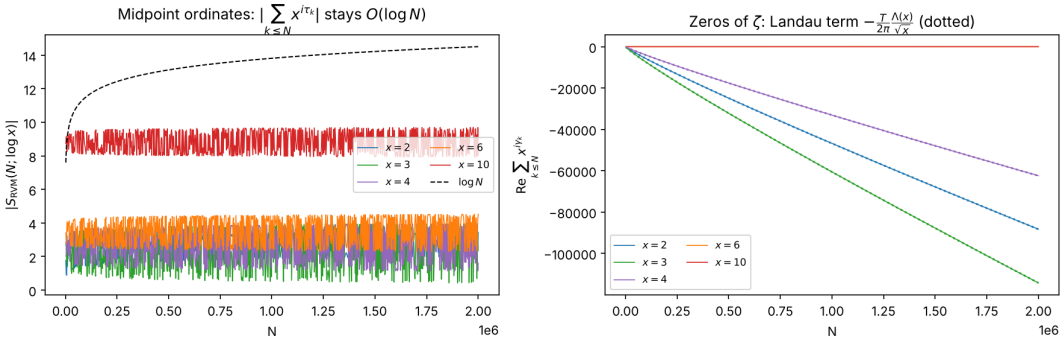


FIGURE 7. The midpoint spectrum carries no prime frequencies (left); the spectrum of ζ resonates at prime powers (right).

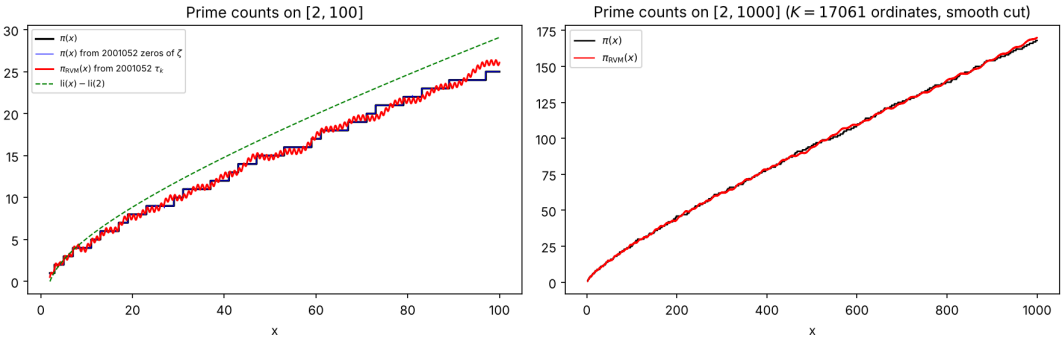


FIGURE 8. $\pi(x)$ (black staircase), the explicit formula with the zeros of ζ (blue), $\pi_{\mathrm{RVM}}(x)$ (red) and $\mathrm{li}(x)-\mathrm{li}(2)$ (green). The distance between the red and green curves is the quantity bounded in Theorem 5.5; the ratios are in Figure 10.

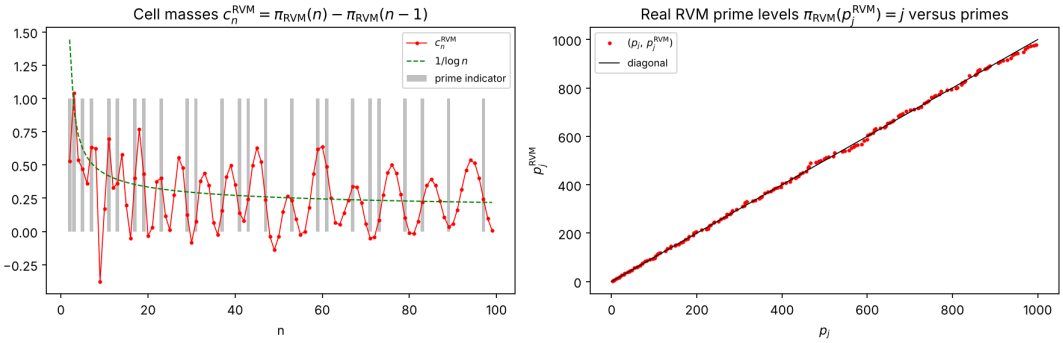


FIGURE 9. Left: cell masses c_n^{RVM}, the prime indicator and $1/\log n$. Right: real RVM prime levels against the primes up to 1000.

and the low ordinates, which shape $\pi(x)$ for small x, are moved least in relative terms. The steps are replaced by a continuous ripple, so the cell masses c_n^{RVM} of Figure 9 are real numbers, occasionally negative, and are not a prime indicator.

Experiment 13.7 (The three ratios and the square-root scale)**.** *Object:* $\mathrm{li}/\pi_{\mathrm{RVM}}$, π/li, π/π_{RVM}, and $(\pi_{\mathrm{RVM}}-\mathrm{li})/B$, $(\pi-\mathrm{li})/B$ with $B(x)=\sqrt{x}(\log\log x)^2/\log x$. *Range:* the grid of Experiment 13.6 on $[8,1000]$ ($K=17\,061$ ordinates). *Expected:* the three ratios tend to 1 (Corollary 5.6) and $(\pi_{\mathrm{RVM}}-$

li)/B stays bounded (Theorem 5.5). *Observed:* at $x = 100, 500, 1000$, $\mathrm{li}/\pi_{\mathrm{RVM}} = 1.155, 1.086, 1.045$, $\pi/\mathrm{li} = 0.830, 0.933, 0.946$, $\pi/\pi_{\mathrm{RVM}} = 0.959, 1.013, 0.988$; $\max|\pi_{\mathrm{RVM}} - \mathrm{li}|/B = 2.72$ on $[8, 1000]$, 1.35 on $[50, 1000]$ and 0.72 on $[500, 1000]$, while $\max|\pi - \mathrm{li}|/B = 2.45$ on $[8, 1000]$ (Figure 10).

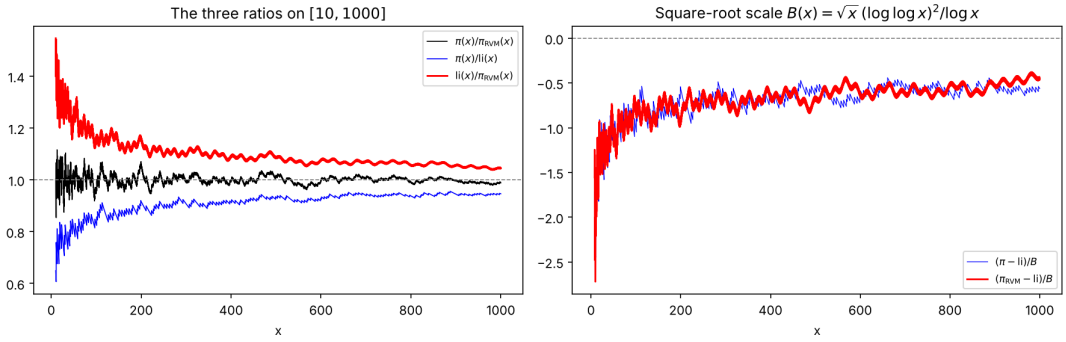


FIGURE 10. Left: the three ratios of Corollary 5.6. Right: $\pi_{\mathrm{RVM}} - \mathrm{li}$ and $\pi - \mathrm{li}$ in units of the scale $B(x)$ of Theorem 5.5. The ratio π/π_{RVM} (black) is the closest of the three to 1; the ratios with li approach 1 at the rate $\frac{1}{2}\,\mathrm{li}(\sqrt{x})/\,\mathrm{li}(x)$ of the bias $J - \pi$.

On $[2, 1000]$ the two normalized errors in Figure 10 have the same size and sign: both counts lie below li by about $\frac{1}{2}\,\mathrm{li}(\sqrt{x})$, the $m = 2$ term of the Möbius inversion (5.20). This is why π/π_{RVM} is much closer to 1 than either ratio with li.

13.6. **The Mertens functions and the ternary division.** The Mertens function is the sum of the ternary sequence $\mu(n) \in \{-1, 0, +1\}$. Its values split into two independent pieces of information:

(a) *the zero set* $\{n : \mu(n) = 0\}$, the non-squarefree integers, of density $1 - 6/\pi^2 = 0.392073$;
(b) *the signs* on the squarefree integers, $\mu(n) = (-1)^{\omega(n)}$, each sign of density $3/\pi^2 = 0.303964$.

The ternary walk μ_h of (1.9)–(1.10) keeps (a) exactly, $\mu_h(n) = 0 \Leftrightarrow \mu(n) = 0$, and replaces (b): at a squarefree n it steps up when $M_h(n-1)$ lies below the target $\mathcal{M}(n) = M^{\mathrm{crit}}_{\mathrm{RVM}}(n)$ and down otherwise. Thus

$$M_h(n) = M_h(n-1) + \mu(n)^2\,\varsigma_n, \qquad |M_h(n) - \mathcal{M}(n)| \text{ small}, \tag{13.1}$$

so that M_h is an integer walk with the zero set of μ that follows $M^{\mathrm{crit}}_{\mathrm{RVM}}$, in the same way as M is the integer walk with the zero set of μ whose signs are $(-1)^{\omega(n)}$.

Experiment 13.8 (Mertens functions and the ternary walk)**.** *Object:* M, $M^{\mathrm{crit}}_{\mathrm{RVM}}$, M_h; the explicit formulas. *Range:* $n \le 10^6$; explicit formulas on $[2, 2000]$ with all N ordinates of each list. *Expected:* the explicit formula with the zeros of ζ reproduces M; $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = O(x^{1/2+\varepsilon})$ (Theorem 8.12); M_h tracks $M^{\mathrm{crit}}_{\mathrm{RVM}}$ with the densities of μ. *Observed:*

- explicit formula for M with the γ_k: root mean square error 0.0005 on $[2, 2000]$ off the integers;
- $\zeta'_{\mathrm{RVM}}(\rho^{\mathrm{RVM}}_k)/\log(\tau_k/2\pi) = 0.96194$ at $k = 1000$, against $\varpi_{\mathrm{RVM}} = 0.96194$ (Corollary 8.6); the values $|\zeta'(\rho_k)|/\log(\gamma_k/2\pi)$ are spread over $(0, 3)$ (Figure 13);
- spectral sums $2\sum|a|^2$: 0.029034 for ζ, 0.027203 for the midpoint system;
- the integer target agrees with the full explicit formula for $M^{\mathrm{crit}}_{\mathrm{RVM}}$ to 0.019 at 100 integers $\le 10^5$;
- $\max|M_h - \mathcal{M}| = 2.62$ on $[1, 10^6]$ and 1.65 on $[10^4, 10^6]$;
- densities of $(-1, 0, +1)$ for μ: 0.303857, 0.392074, 0.304069;
- densities of $(-1, 0, +1)$ for μ_h: 0.304003, 0.392074, 0.303923;

n	$\mu(n)$	$\mu(n)^2$	$\mathcal{M}(n)$	$M_h(n-1)$	ς_n	$\mu_h(n)$	$M_h(n)$	$M(n)$
1	1	1	0.2925	0	+1	1	1	1
2	−1	1	0.4563	1	−1	−1	0	0
3	−1	1	−0.6523	0	−1	−1	−1	−1
4	0	0	−0.8720	−1	–	0	−1	−1
5	−1	1	−1.7255	−1	−1	−1	−2	−2
6	1	1	−1.3063	−2	+1	1	−1	−1
7	−1	1	−1.5931	−1	−1	−1	−2	−2
8	0	0	−2.4029	−2	–	0	−2	−2
9	0	0	−1.3292	−2	–	0	−2	−2
10	1	1	−1.3280	−2	+1	1	−1	−1
11	−1	1	−2.3272	−1	−1	−1	−2	−2
12	0	0	−2.2067	−2	–	0	−2	−2
13	−1	1	−1.8468	−2	+1	1	−1	−3
14	1	1	−1.9324	−1	−1	−1	−2	−2
15	1	1	−1.5435	−2	+1	1	−1	−1

TABLE 2. The ternary division for $n \le 15$ (the first 40 values are in Appendix B.6). The zero columns of μ and μ_h coincide; the signs of μ_h are chosen by comparing $M_h(n-1)$ with $\mathcal{M}(n)$. For $n \le 12$ the two walks agree; at $n = 13$ the target -1.85 lies above $M_h(12) = -2$, so $\mu_h(13) = +1$ while $\mu(13) = -1$.

- $\mu_h(n) = \mu(n)$ at 50.03% of the squarefree $n \le 10^6$, and $\sum_{n\le 10^6} \mu(n)\mu_h(n)/\#\{n \le 10^6 : \mu(n) \ne 0\} = 0.0007$: the sign sequences are uncorrelated although both walks have the same zero set and comparable size;
- $\max_{10\le n\le 10^6} |W(n)|/\sqrt{n} = 0.832, 0.754, 0.688$ for $W = M, M^{\mathrm{crit}}_{\mathrm{RVM}}, M_h$;
- energies $\frac{1}{N}\sum_{n\le N} W(n)^2$ at $N = 10^6$: 14 851, 12 538, 12 538.

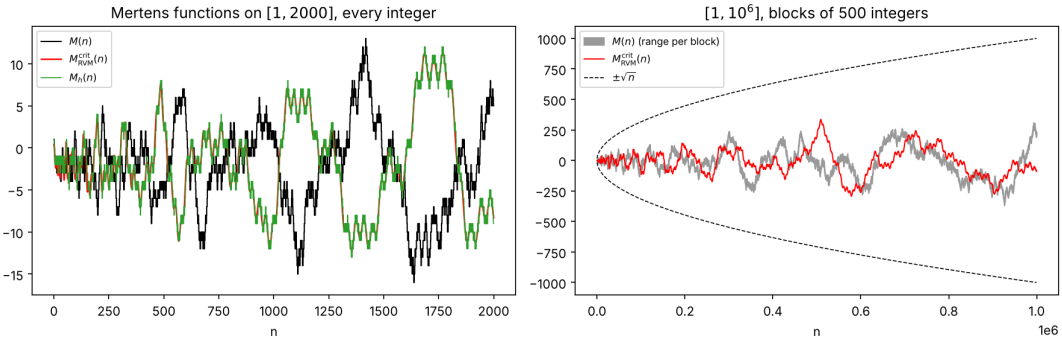


FIGURE 11. Left: M (black), $M^{\mathrm{crit}}_{\mathrm{RVM}}$ (red) and M_h (green) at every integer $n \le 2000$. Right: range of M and mean of $M^{\mathrm{crit}}_{\mathrm{RVM}}$ in blocks of 500 integers up to 10^6, with $\pm\sqrt{n}$.

Figures 11–13 summarize the Mertens side. $M^{\mathrm{crit}}_{\mathrm{RVM}}$ has the size and the visual texture of M, it stays inside $\pm\sqrt{n}$ as Theorem 8.12 requires, and its residues are deterministic; M_h is its ternary integer version with the zero set of μ. The passage from these synthetic objects to M itself is not numerical: it is the identity $\zeta_{Z1} = \zeta$ of Step 3, which Experiment 13.2 is consistent with.

14. Conclusions and open problems

The following statements are proved:

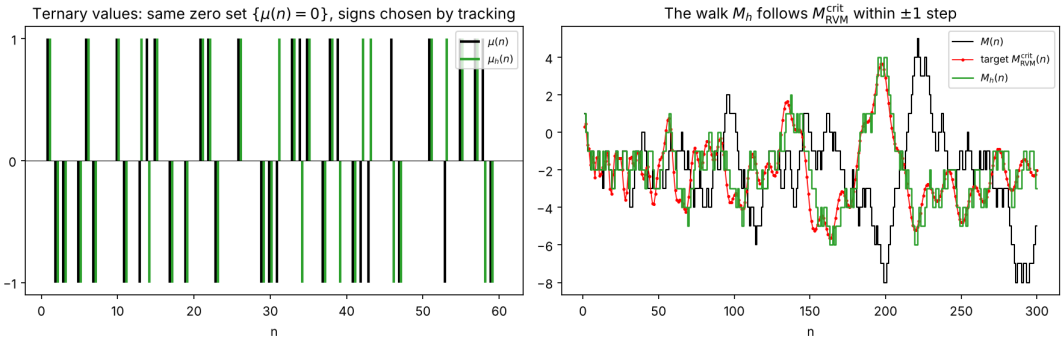


FIGURE 12. Left: $\mu(n)$ and $\mu_h(n)$ for $n \le 60$: same zeros, different signs. Right: M, the target $M^{\mathrm{crit}}_{\mathrm{RVM}}$ and M_h for $n \le 300$.

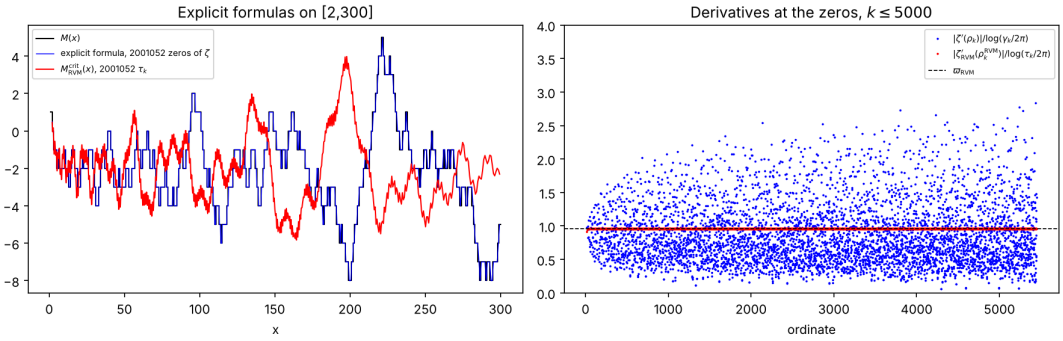


FIGURE 13. Left: $M(x)$, its explicit formula from the zeros of ζ and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x)$ from the τ_k on $[2, 300]$. Right: normalized derivatives at the zeros: erratic for ζ, constant ϖ_{RVM} for ζ_{RVM}.

(1) the midpoint ordinates τ_k are given by the Lambert formula (2.5) and stay within a bounded distance of the ordinates γ_k of ζ (Lemma 2.2);
(2) $P^{\mathrm{crit}}_{\mathrm{RVM}}$ is entire and symmetric, and ζ_{RVM} has exactly the simple zeros $\frac{1}{2} \pm i\tau_k$, the trivial zeros and a residue-one pole at 1 (Theorem 3.1);
(3) $S_{\mathrm{RVM}}(N; u) \ll 1 + \log N$ uniformly on compact positive frequency intervals, so the point centroid and $M^{\mathrm{RVM}}_H(x)$ tend to 0 (Section 4);
(4) J_{RVM} exists compact-uniformly and is related to π_{RVM} by exact finite Möbius inversion (Section 5);
(5) $J_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ and $\pi_{\mathrm{RVM}}(x) - \mathrm{li}(x)$ are $O(\sqrt{x}(\log\log x)^2/\log x)$, so $\mathrm{li}/\pi_{\mathrm{RVM}} \to 1$, and with the prime number theorem $\pi/\mathrm{li} \to 1$ and $\pi/\pi_{\mathrm{RVM}} \to 1$ (Theorem 5.5, Corollary 5.6);
(6) J_{RVM} is Lipschitz on $[2, \infty)$, $c^{\mathrm{RVM}}_n = O(1)$, and $\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$ (Section 6);
(7) $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+w)(1+O(1/|t|))$ on every band to the right of $\frac{1}{2}$ and $\zeta_{\mathrm{RVM}} = \varpi_{\mathrm{RVM}}(1+\chi)(1+O(1/|t|))$ on closed substrips of $\frac{1}{2} < \Re s < 1$ (Theorem 8.1);
(8) $1/\zeta_{\mathrm{RVM}} \to L$ on every vertical line to the right of $\frac{1}{2}$, $\zeta^{\pm 1}_{\mathrm{RVM}}$ have no absolutely convergent Dirichlet expansion, and $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ (Section 8);
(9) 2ξ equals its Hadamard product over $\rho(1-\rho)$ exactly (Theorem 9.1), and $|\zeta| \asymp |\mathcal{Q}_{\mathrm{RVM}}|$ on $\frac{1}{2} + \varepsilon \le \sigma \le c'$, $|t| \ge t_0$ (Corollary 9.3);
(10) the λ-family from ζ_{RVM} to ζ_{Z1} converges jointly, has simple zeros for $\lambda > 0$, an explicit generator and total multiplier, and its members differ only through $\gamma_k - \tau_k$ (Section 10);

(11) ζ_{Z1} has the functional equation, the pole, the residue and the order of ζ, and $\zeta_{Z1} = \zeta \Leftrightarrow$(D)$\Leftrightarrow$RH (Theorem 11.2);
(12) ζ_{RVM} is bounded above and below on every half-plane $\Re s \geq \sigma_0 > 1$ and tends to ϖ_{RVM} as $\Re s \to +\infty$ (Theorem 8.10); $1/\zeta_{\mathbb{Z},\mathrm{RVM}}$ is an absolutely convergent Dirichlet series in $\Re s > 1$, so (D) is equivalent to an absolutely convergent Dirichlet expansion of $\mathcal{K}$ in $\Re s > 1$ (Theorem 11.3);
(13) no member of the λ-family, ζ_{Z1} included, can have a zero off the critical line other than the trivial zeros (Theorem 10.1);
(14) the endpoint has the value $\varpi_{Z1} = \varpi_{\mathrm{RVM}} \prod_k (\frac{1}{4} + \tau_k^2)/(\frac{1}{4} + \gamma_k^2) \geq 1$ at $+\infty$, and $\zeta_{Z1} = \zeta$ if and only if $\varpi_{Z1} = 1$ (Theorems 12.1, 12.2, Corollary 12.3);
(15) the prime-power counts of ζ_{RVM} and ζ_{Z1} are the Mellin transforms of their logarithms and satisfy a distributional explicit formula; this proves the Mellin–Stieltjes identification and the weighted factorization $\zeta_{\mathbb{Z},\mathrm{RVM}} = \zeta_{\mathrm{RVM}} e^{h_{\mathrm{RVM}}}$, and (D) holds if and only if the prime-power count of ζ_{Z1} is constant between consecutive integers (Theorems 12.5, 12.6, 6.4, 12.7);
(16) Step 3: $\zeta_{Z1} = \zeta$ implies $M(x) = O_\varepsilon(x^{1/2+\varepsilon})$ (Theorem 11.8); equivalently, the synthetic bound $M^{\mathrm{crit}}_{\mathrm{RVM}}(x) = L + O_\varepsilon(x^{1/2+\varepsilon})$ transfers to M through $1/\zeta = F_{\mathrm{RVM}} \Pi_{\mathrm{tr}}^{-1}$ (Theorem 11.9).

The computations of Section 13 with 2 001 052 zeros are consistent with each of these statements.

Remark 14.1 (Open problems). (i) Prove the identity $\zeta_{Z1} = \zeta$, equivalently condition (D), equivalently an absolutely convergent Dirichlet expansion of $\mathcal{K} = \zeta_{Z1}/\zeta_{\mathbb{Z},\mathrm{RVM}}$ in $\Re s > 1$, equivalently the product identity (12.12), equivalently the cell identity (12.18) with the jump bound of Theorem 12.7. This is the final step of the method.
(ii) Prove Hypothesis 6.7.
(iii) Determine the monotone branches of π_{RVM}, on which the real prime levels p_j^{RVM} are defined, and explain the observed proximity $p_j^{\mathrm{RVM}} \approx p_j$ for small j (Experiment 13.6).

Appendix A. Python programs

The programs below produce every number, table and figure of Section 13. They read the two data files `data/zeross.bin` (ordinates γ_k of the zeros of ζ) and `data/zLamb.bin` (midpoint ordinates τ_k), both little-endian float64 with 2 001 052 entries, and write their console output, CSV, MAT and binary files to `output/` and the figures to `figures/`. They require Python 3 with NumPy, SciPy, mpmath and Matplotlib, and are run from the directory `code/` in the order `s01`–`s07`.

A.1. **rvm_common.py.**

```
"""Common routines for the midpoint Riemann-von Mangoldt (RVM) zeta system.

Data files (little-endian float64, one ordinate per entry):
    data/zeross.bin  ordinates gamma_k of the nontrivial zeros of zeta (2,001,052 values)
    data/zLamb.bin   midpoint RVM ordinates tau_k, N_RVM(tau_k) = k - 1/2

Notation follows the paper: N_RVM is the clock, tau_k the midpoint ordinates,
gamma_k the ordinates of zeta, eta_k = tau_k - gamma_k, q(s) = s(s-1),
G(s) = s(s-1) Gamma(s/2) pi^(-s/2), zeta_lambda = P_lambda / G with ordinates
gamma_k(lambda) = (1-lambda) gamma_k + lambda tau_k.
"""
import os
import sys
import time
import numpy as np
from scipy import special

ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA = os.path.join(ROOT, "data")
OUT = os.path.join(ROOT, "output")
FIG = os.path.join(ROOT, "figures")
os.makedirs(OUT, exist_ok=True)
```

```
os.makedirs(FIG, exist_ok=True)

TWO_PI = 2.0 * np.pi


class Tee:
    """Write console output both to the screen and to output/<name>_console.txt."""

    def __init__(self, name):
        self.f = open(os.path.join(OUT, name + "_console.txt"), "w")
        self.t0 = time.time()

    def __call__(self, fmt, *args):
        s = fmt % args if args else fmt
        sys.stdout.write(s + "\n")
        sys.stdout.flush()
        self.f.write(s + "\n")
        self.f.flush()

    def close(self):
        self("total run time %.1f s" % (time.time() - self.t0))
        self.f.close()


def read_bin(name):
    return np.fromfile(os.path.join(DATA, name), dtype="<f8")


def load_ordinates():
    gam = read_bin("zeross.bin")
    tau = read_bin("zLamb.bin")
    n = min(gam.size, tau.size)
    return gam[:n], tau[:n]


def clock(T):
    """N_RVM(T) = T/(2 pi) log(T/(2 pi e)) + 7/8 (real or complex T, principal log)."""
    return T / TWO_PI * (np.log(T / TWO_PI) - 1.0) + 0.875


def tail_coefficients(U):
    """Midpoint-rule tail of sum_{k>N} log(1 + q/tau_k^2), N = N_RVM(U):
    returns (c1, c2, c3) with tail(q) = c1 q - c2 q^2 + c3 q^3."""
    LU = np.log(U / TWO_PI)
    c1 = (LU + 1.0) / U / TWO_PI
    c2 = (3.0 * LU + 1.0) / (9.0 * U ** 3) / 2.0 / TWO_PI
    c3 = (5.0 * LU + 1.0) / (25.0 * U ** 5) / 3.0 / TWO_PI
    return c1, c2, c3


def clock_inverse(y):
    """Solve N_RVM(U) = y by Newton iteration."""
    U = TWO_PI * max(y, 2.0) / np.log(max(y, 3.0))
    for _ in range(80):
        U -= (clock(U) - y) / (np.log(U / TWO_PI) / TWO_PI)
    return U


class Product:
    """log P(s) = sum_k log(1 + q(s)/(1/4 + o_k^2)) for an ordinate list o_k.

    Exact summation for k <= K1, power sums for K1 < k <= N, and the clock tail
    for k > N.  Accurate for |q(s)| << o_{K1}^2."""

    def __init__(self, ords, K1=100000, nterms=6):
```

```
        self.o = np.asarray(ords, dtype=float)
        self.N = self.o.size
        self.K1 = min(K1, self.N)
        A = 0.25 + self.o[self.K1:] ** 2
        self.pw = [np.sum(A ** (-j)) for j in range(1, nterms + 1)]
        self.A1 = 0.25 + self.o[: self.K1] ** 2
        self.tail = tail_coefficients(clock_inverse(float(self.N)))

    def logP(self, s):
        s = np.atleast_1d(np.asarray(s, dtype=complex))
        q = s * (s - 1.0)
        out = np.empty(s.shape, dtype=complex)
        for i, qi in enumerate(q):
            v = np.sum(np.log1p(qi / self.A1))
            for j, p in enumerate(self.pw, start=1):
                v += (-1) ** (j + 1) * qi ** j / j * p
            c1, c2, c3 = self.tail
            v += c1 * qi - c2 * qi ** 2 + c3 * qi ** 3
            out[i] = v
        return out


def logG(s):
    s = np.asarray(s, dtype=complex)
    return np.log(s) + np.log(s - 1.0) + special.loggamma(s / 2.0) - s / 2.0 * np.log(np.pi)


def zeta_from_product(prod, s):
    s = np.atleast_1d(np.asarray(s, dtype=complex))
    return np.exp(prod.logP(s) - logG(s))


def chi(s):
    s = np.asarray(s, dtype=complex)
    return np.exp((s - 0.5) * np.log(np.pi) + special.loggamma((1 - s) / 2) - special.loggamma(s / 2))


def w_transition(s):
    """w(s) = exp(-2 pi i N_RVM(tau_*)), tau_* = -i (s - 1/2)."""
    s = np.asarray(s, dtype=complex)
    tstar = -1j * (s - 0.5)
    return np.exp(-2j * np.pi * clock(tstar))


def theta_rs(t):
    t = np.asarray(t, dtype=float)
    return t / 2 * np.log(t / TWO_PI) - t / 2 - np.pi / 8 + 1 / (48 * t) + 7 / (5760 * t ** 3)


def gamma_lambda(gam, tau, lam):
    return (1.0 - lam) * gam + lam * tau


def ei_upper(w):
    """Ei on Im w != 0 along the horizontal line (paper, eq. Ei-def): Ei(w) = -E1(-w)."""
    return -special.exp1(-np.asarray(w, dtype=complex))


def mobius_sieve(X):
    mu = np.ones(X + 1, dtype=np.int8)
    mu[0] = 0
    is_comp = np.zeros(X + 1, dtype=bool)
    for p in range(2, X + 1):
        if not is_comp[p]:
            is_comp[2 * p::p] = True
```

```
            mu[p::p] *= -1
            if p * p <= X:
                mu[p * p::p * p] = 0
    return mu


def primes_upto(X):
    s = np.ones(X + 1, dtype=bool)
    s[:2] = False
    for p in range(2, int(X ** 0.5) + 1):
        if s[p]:
            s[p * p::p] = False
    return np.nonzero(s)[0]


def write_csv(name, header, cols):
    A = np.column_stack(cols)
    np.savetxt(os.path.join(OUT, name), A, delimiter=",", header=",".join(header), comments="", fmt="%.12g")


def savefig(fig, name):
    fig.savefig(os.path.join(FIG, name), dpi=150, bbox_inches="tight")
```

A.2. **s01_clock_and_family.py.**

```
"""Experiment 1: the midpoint clock, the displacements eta_k = tau_k - gamma_k,
the lambda-family zeta_lambda, and the numerical comparison zeta_Z1 versus zeta."""
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, clock, Product, zeta_from_product, gamma_lambda,
                        write_csv, savefig, OUT)
import os

say = Tee("s01_clock_and_family")
gam, tau = load_ordinates()
N = gam.size
k = np.arange(1, N + 1)
say("zeros loaded: N = %d, gamma_1 = %.9f, gamma_N = %.6f, tau_1 = %.9f, tau_N = %.6f",
    N, gam[0], gam[-1], tau[0], tau[-1])

# ---- clock check and displacements
clock_err = np.max(np.abs(clock(tau) - (k - 0.5)))
eta = tau - gam
say("max |N_RVM(tau_k) - (k - 1/2)| = %.3e", clock_err)
say("eta_k = tau_k - gamma_k:  max |eta| = %.6f (k = %d),  mean |eta| = %.6f,  rms = %.6f,  mean eta = %.3e",
    np.max(np.abs(eta)), int(np.argmax(np.abs(eta))) + 1, np.mean(np.abs(eta)),
    np.sqrt(np.mean(eta ** 2)), np.mean(eta))
blocks = [(1, 10), (11, 100), (101, 1000), (1001, 10 ** 4), (10 ** 4 + 1, 10 ** 5), (10 ** 5 + 1, 10 ** 6),
     (10 ** 6 + 1, N)]
rows = []
say("%12s %12s %12s %12s", "k from", "k to", "max|eta|", "rms eta")
for a, b in blocks:
    e = eta[a - 1:b]
    rows.append([a, b, np.max(np.abs(e)), np.sqrt(np.mean(e ** 2))])
    say("%12d %12d %12.6f %12.6f", a, b, rows[-1][2], rows[-1][3])
rows = np.array(rows)
write_csv("s01_eta_blocks.csv", ["k_from", "k_to", "max_abs_eta", "rms_eta"], [rows[:, i] for i in range(4)])
say("first ordinates:")
say("%4s %16s %16s %12s", "k", "gamma_k", "tau_k", "eta_k")
for i in range(12):
    say("%4d %16.9f %16.9f %12.6f", i + 1, gam[i], tau[i], eta[i])
```

```
# ---- zeta_Z1 versus zeta on vertical lines
PZ = Product(gam)
PR = Product(tau)
tline = np.concatenate([np.linspace(1.0, 100.0, 199), np.linspace(101.0, 2000.0, 300)])
mp.mp.dps = 20
cmp_rows = []
for sigma in (2.0, 1.25, 0.75):
    s = sigma + 1j * tline
    zz1 = zeta_from_product(PZ, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    rel = np.abs(zz1 / ze - 1.0)
    cmp_rows.append([sigma, np.max(rel), np.median(rel)])
    say("sigma = %.2f, t in [1,2000] (%d points): max |zeta_Z1/zeta - 1| = %.3e, median = %.3e",
        sigma, tline.size, np.max(rel), np.median(rel))
    write_csv("s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaZ1", "rel_err"],
              [tline, np.abs(ze), np.abs(zz1), rel])
cmp_rows = np.array(cmp_rows)

# ---- Hardy-type functions Z_lambda(t) = exp(i theta(t)) zeta_lambda(1/2+it) on the critical line
def theta_exact(t):
    return np.array([float(mp.siegeltheta(x)) for x in t])

tc = np.linspace(0.5, 60.0, 1200)
th = theta_exact(tc)
lams = [0.0, 0.25, 0.5, 0.75, 1.0]
Zl = {}
for lam in lams:
    P = PZ if lam == 0.0 else (PR if lam == 1.0 else Product(gamma_lambda(gam, tau, lam)))
    Zl[lam] = np.real(np.exp(1j * th) * zeta_from_product(P, 0.5 + 1j * tc))
Zhardy = np.array([float(mp.siegelz(x)) for x in tc])
dZ = np.max(np.abs(Zl[0.0] - Zhardy) / np.maximum(1.0, np.abs(Zhardy)))
say("critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = %.3e", dZ)
write_csv("s01_Z_lambda_critical_line.csv", ["t", "Z_hardy"] + ["Z_lambda_%.2f" % l for l in lams],
          [tc, Zhardy] + [Zl[l] for l in lams])

# ---- figures
dec = np.arange(0, N, 50)
fig, ax = plt.subplots(2, 2, figsize=(12, 8))
ax[0, 0].plot(k[dec], eta[dec], ".", ms=1.5, color="tab:blue")
ax[0, 0].axhline(0, color="k", lw=0.5)
ax[0, 0].set_xlabel("k")
ax[0, 0].set_ylabel(r"$\eta_k=\tau_k-\gamma_k$")
ax[0, 0].set_title(r"Bounded displacement, $N=%d$ zeros" % N)
ax[0, 1].hist(eta, bins=200, density=True, color="tab:blue")
ax[0, 1].set_xlabel(r"$\eta_k$")
ax[0, 1].set_title(r"Distribution of $\eta_k$ (all $k\leq N$)")
lam = np.linspace(0, 1, 101)
for kk in range(30):
    gl = (1 - lam) * gam[kk] + lam * tau[kk]
    ax[1, 0].plot(lam, clock(gl) - (kk + 0.5), "-", lw=1)
ax[1, 0].axhline(0, color="k", lw=0.5)
ax[1, 0].set_xlabel(r"$\lambda$")
ax[1, 0].set_ylabel(r"$\mathcal{N}_{\rm RVM}(\gamma_k(\lambda))-(k-\frac{1}{2})$")
ax[1, 0].set_title(r"Moving ordinates in clock units, $k\leq30$ ($\lambda=0$: $\zeta_{Z1}$, $\lambda=1$: $\
    zeta_{\rm RVM}$)", fontsize=9)
ax[1, 1].plot(tc, Zhardy, "k-", lw=2.2, label=r"Hardy $Z(t)$ of $\zeta$")
cols = ["tab:red", "tab:orange", "tab:green", "tab:cyan", "tab:blue"]
for lam, c in zip(lams, cols):
    ax[1, 1].plot(tc, Zl[lam], "-", color=c, lw=1.0, label=r"$Z_\lambda$, $\lambda=%.2f$" % lam)
ax[1, 1].axhline(0, color="k", lw=0.5)
ax[1, 1].set_xlabel("t")
ax[1, 1].set_title(r"$e^{i\vartheta(t)}\zeta_\lambda(\frac{1}{2}+it)$ along the family")
ax[1, 1].legend(fontsize=7, ncol=2)
fig.tight_layout()
```

```
savefig(fig, "fig01_clock_family.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
for sigma, c in zip((2.0, 1.25, 0.75), ("tab:blue", "tab:green", "tab:red")):
    d = np.loadtxt(os.path.join(OUT, "s01_zetaZ1_vs_zeta_sigma%.2f.csv" % sigma), delimiter=",", skiprows=1)
    ax[0].plot(d[:, 0], d[:, 3], ".", ms=2, color=c, label=r"$\sigma=%.2f$" % sigma)
ax[0].set_xlabel("t")
ax[0].set_ylabel(r"$|\zeta_{Z1}/\zeta-1|$")
ax[0].set_title(r"$\zeta_{Z1}$ from %d ordinates versus $\zeta$" % N)
ax[0].legend()
ax[1].plot(tc, Zhardy, "k-", lw=2.5, label=r"$Z(t)$ (mpmath)")
ax[1].plot(tc, Z1[0.0], "r--", lw=1.0, label=r"$Z_{Z1}(t)$ (product)")
ax[1].set_xlabel("t")
ax[1].legend()
ax[1].set_title(r"Critical line: $\lambda=0$ endpoint and Hardy $Z$")
fig.tight_layout()
savefig(fig, "fig02_zetaZ1_vs_zeta.png")

sio.savemat(os.path.join(OUT, "s01_results.mat"), {
    "N": N, "clock_err": clock_err, "eta_blocks": rows, "zetaZ1_cmp": cmp_rows,
    "t_critical": tc, "Z_hardy": Zhardy, "Z_lambda": np.array([Z1[l] for l in lams]), "lambdas": np.array(lams
    )})
say.close()

# ---- zoom: how the smooth clock places the midpoint ordinates next to the zeros
say2 = Tee("s01b_clock_zoom")
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
T = np.linspace(9.0, 52.0, 4000)
ax[0].step(T, np.searchsorted(gam, T, side="right"), where="post", color="k", lw=1.5, label=r"$N_\zeta(T)$")
ax[0].plot(T, clock(T), "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
for i in range(9):
    ax[0].plot([gam[i], gam[i]], [0, i + 1], "k:", lw=0.8)
ax[0].plot(tau[:9], np.arange(9) + 0.5, "ro", ms=5, label=r"$\tau_k$: $\mathcal{N}_{\rm RVM}(\tau_k)=k-\frac
    {1}{2}$")
ax[0].plot(gam[:9], np.arange(9) + 0.5, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$")
ax[0].set_xlabel("T")
ax[0].set_title("Clock and staircase at the first zeros")
ax[0].legend(loc="upper left", fontsize=8)
kz = np.arange(N - 12, N)
T3 = np.linspace(gam[kz[0]] - 0.3, gam[kz[-1]] + 0.3, 6000)
ax[1].step(T3 - 1132480, np.searchsorted(gam, T3, side="right") - 2001000, where="post", color="k", lw=1.5,
    label=r"$N_\zeta(T)$")
ax[1].plot(T3 - 1132480, clock(T3) - 2001000, "r-", lw=1.5, label=r"$\mathcal{N}_{\rm RVM}(T)$")
ax[1].plot(tau[kz] - 1132480, kz + 0.5 - 2001000, "ro", ms=5, label=r"$\tau_k$")
ax[1].plot(gam[kz] - 1132480, kz + 0.5 - 2001000, "ks", ms=4, mfc="none", label=r"$(\gamma_k,\,k-\frac{1}{2})$
    ")
ax[1].set_xlabel(r"$T-1132480$")
ax[1].set_ylabel(r"count $-\,2001000$")
ax[1].set_title(r"Zoom at the last 12 zeros of the file (mean spacing %.3f)" % (2 * np.pi / np.log(gam[-1] /
    (2 * np.pi))))
ax[1].legend(loc="upper left", fontsize=8)
fig.tight_layout()
savefig(fig, "fig00_clock_zoom.png")
fig, ax = plt.subplots(1, 1, figsize=(12, 3))
T2 = np.linspace(9.0, 200.0, 20000)
ax.plot(T2, np.searchsorted(gam, T2, side="right") - clock(T2), "k-", lw=0.8)
ax.axhline(0, color="r", lw=0.8)
ax.set_xlabel("T")
ax.set_ylabel(r"$S(T)+O(1/T)$")
ax.set_title(r"$N_\zeta(T)-\mathcal{N}_{\rm RVM}(T)$: the remainder of the Riemann-von Mangoldt formula (jump
    $+1$ at each $\gamma_k$)")
fig.tight_layout()
savefig(fig, "fig00b_remainder.png")
say2("zoom table near the top of the file:")
say2("%9s %18s %18s %10s %10s", "k", "gamma_k", "tau_k", "eta_k", "N*-N_RVM")
```

```
for i in kz:
    say2("%9d %18.6f %18.6f %10.6f %10.6f", i + 1, gam[i], tau[i], tau[i] - gam[i],
         np.searchsorted(gam, gam[i], side="right") - 0.5 - clock(gam[i]))
say2.close()
```

A.3. **s02_transition.py.**

```
"""Experiment 2: the transition theorem zeta_RVM = varpi (1 + w)(1 + O(1/t)),
the constant varpi_RVM, the vertical limit of 1/zeta_RVM, and zeta_RVM versus zeta."""
import os
import numpy as np
import mpmath as mp
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, zeta_from_product, chi, w_transition,
                        write_csv, savefig, OUT)

say = Tee("s02_transition")
gam, tau = load_ordinates()
PR = Product(tau)

# ---- varpi_RVM by Richardson extrapolation on Re s = 2
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
A = np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2])
c = np.linalg.solve(A.astype(complex), v)
varpi = c[0].real
L = 1.0 / varpi
say("varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = %.10f %+.2e i", c[0].real, c[0].imag)
say("L = 1/varpi_RVM = %.10f", L)

# ---- relative error of the transition formula
tt = np.exp(np.linspace(np.log(10.0), np.log(1.0e4), 160))
rows = [tt]
hdr = ["t"]
fits = []
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    zr = zeta_from_product(PR, s)
    err_w = np.abs(zr / (varpi * (1 + w_transition(s))) - 1)
    rows.append(err_w)
    hdr.append("err_w_sigmap%.2f" % sp)
    sel = tt > 100
    slope = np.polyfit(np.log(tt[sel]), np.log(err_w[sel]), 1)[0]
    fits.append([sp, slope, np.max(err_w[sel] * tt[sel])])
    say("sigma' = %.2f: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = %.3f, max t*err = %.4f",
        sp, slope, np.max(err_w[sel] * tt[sel]))
    if sp < 0.5:
        err_c = np.abs(zr / (varpi * (1 + chi(s))) - 1)
        rows.append(err_c)
        hdr.append("err_chi_sigmap%.2f" % sp)
        say("sigma' = %.2f: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = %.4f",
            sp, np.max(err_c[sel] * tt[sel]))
    wmod = np.abs(w_transition(s))
    rows.append(wmod)
    hdr.append("abs_w_sigmap%.2f" % sp)
    Fv = 1 / zr
    rows.append(np.abs(Fv - L))
    hdr.append("abs_F_minus_L_sigmap%.2f" % sp)
    say("          |w| * t^sigma' in [%.4f, %.4f];  |1/zeta_RVM - L| at t = 1e4: %.3e",
        np.min(wmod * tt ** sp), np.max(wmod * tt ** sp), np.abs(Fv[-1] - L))
write_csv("s02_transition_errors.csv", hdr, rows)
```

```
# ---- zeta_RVM versus zeta on two vertical lines and on the critical line
mp.mp.dps = 20
tl = np.linspace(1.0, 120.0, 900)
cmp = {}
for sigma in (2.0, 0.75):
    s = sigma + 1j * tl
    zr = zeta_from_product(PR, s)
    ze = np.array([complex(mp.zeta(complex(x))) for x in s])
    cmp[sigma] = (zr, ze)
    write_csv("s02_zetaRVM_vs_zeta_sigma%.2f.csv" % sigma, ["t", "abs_zeta", "abs_zetaRVM", "
     abs_varpi_1_plus_w"],
              [tl, np.abs(ze), np.abs(zr), np.abs(varpi * (1 + w_transition(s)))])

fig, ax = plt.subplots(2, 2, figsize=(12, 8))
cols = {0.25: "tab:red", 0.5: "tab:green", 1.5: "tab:blue"}
j = 1
for sp in (0.25, 0.5, 1.5):
    ax[0, 0].plot(tt, tt * rows[j], "-", color=cols[sp], label=r"$\sigma'=%.2f$" % sp)
    j += 4 if sp < 0.5 else 3
ax[0, 0].set_xlabel("t")
ax[0, 0].set_ylabel(r"$t\,|\zeta_{\rm RVM}/(\varpi_{\rm RVM}(1+w))-1|$")
ax[0, 0].set_title(r"Transition theorem: error times $t$ stays bounded, $\varpi_{\rm RVM}=%.10f$" % varpi,
     fontsize=9)
ax[0, 0].legend()
for sp in (0.25, 0.5, 1.5):
    s = 0.5 + sp + 1j * tt
    ax[0, 1].plot(tt, np.abs(1 / zeta_from_product(PR, s) - L) * tt ** min(sp, 1.0), "-", color=cols[sp],
     label=r"$\sigma'=%.2f$" % sp)
ax[0, 1].set_xlabel("t")
ax[0, 1].set_ylabel(r"$t^{\min(\sigma',1)}\,|1/\zeta_{\rm RVM}(\sigma+it)-L|$")
ax[0, 1].set_title(r"Vertical limit $L=1/\varpi_{\rm RVM}$, rescaled error", fontsize=9)
ax[0, 1].legend()
for a, sigma in zip((ax[1, 0], ax[1, 1]), (2.0, 0.75)):
    zr, ze = cmp[sigma]
    a.plot(tl, np.abs(ze), "k-", lw=1.2, label=r"$|\zeta(\sigma+it)|$")
    a.plot(tl, np.abs(zr), "r-", lw=1.2, label=r"$|\zeta_{\rm RVM}(\sigma+it)|$")
    a.set_xlabel("t")
    a.set_title(r"$\sigma=%.2f$" % sigma)
    a.legend()
fig.tight_layout()
savefig(fig, "fig03_transition.png")

sio.savemat(os.path.join(OUT, "s02_results.mat"), {"varpi": varpi, "L": L, "richardson": c,
                                                   "slopes": np.array(fits), "t": tt})
say.close()
```

A.4. **s03_exponential_sums.py.**

```
"""Experiment 3: exponential sums S(N;u) = sum_{k<=N} exp(i u o_k) over the midpoint
ordinates tau_k and over the ordinates gamma_k of zeta, and the height transforms M_H(x)."""
import os
import numpy as np
import scipy.io as sio
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, write_csv, savefig, OUT

say = Tee("s03_exponential_sums")
gam, tau = load_ordinates()
N = gam.size
step = 2000
Nk = np.arange(step, N + 1, step)
xs = [2.0, 3.0, 4.0, 6.0, 10.0]
Lam = {2.0: np.log(2), 3.0: np.log(3), 4.0: np.log(2), 6.0: 0.0, 10.0: 0.0}
```

```
res = {}
say("%6s %14s %14s %14s %16s %14s" % ("x", "max|S_RVM|", "max|S_RVM|/log", "|S_zeta(N)|",
                                      "Landau -T/2pi*L/sqrt(x)", "|M_H^RVM(x)|"))
for x in xs:
    u = np.log(x)
    SR = np.cumsum(np.exp(1j * u * tau))[Nk - 1]
    SZ = np.cumsum(np.exp(1j * u * gam))[Nk - 1]
    res[x] = (SR, SZ)
    landau = -gam[-1] / (2 * np.pi) * Lam[x] / np.sqrt(x)
    say("%6.1f %14.4f %14.4f %14.2f %16.2f %14.3e" % (x, np.max(np.abs(SR)), np.max(np.abs(SR) / np.log(Nk)),
                                                      np.abs(SZ[-1]), landau, np.abs(SR[-1]) / N))
    write_csv("s03_expsum_x%g.csv" % x, ["N", "re_S_RVM", "im_S_RVM", "re_S_zeta", "im_S_zeta"],
              [Nk, SR.real, SR.imag, SZ.real, SZ.imag])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
cols = {2.0: "tab:blue", 3.0: "tab:green", 4.0: "tab:purple", 6.0: "tab:orange", 10.0: "tab:red"}
for x in xs:
    ax[0].plot(Nk, np.abs(res[x][0]), "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
ax[0].plot(Nk, np.log(Nk), "k--", lw=1, label=r"$\log N$")
ax[0].set_xlabel("N")
ax[0].set_ylabel(r"$|S_{\rm RVM}(N;\log x)|$")
ax[0].set_title(r"Midpoint ordinates: $|\sum_{k\leq N}x^{i\tau_k}|$ stays $O(\log N)$")
ax[0].legend(fontsize=8, ncol=2)
for x in xs:
    ax[1].plot(Nk, res[x][1].real, "-", color=cols[x], lw=0.8, label=r"$x=%g$" % x)
T = np.interp(Nk, np.arange(1, N + 1), gam)
for x in (2.0, 3.0, 4.0):
    ax[1].plot(Nk, -T / (2 * np.pi) * Lam[x] / np.sqrt(x), ":", color=cols[x], lw=1.5)
ax[1].set_xlabel("N")
ax[1].set_ylabel(r"$\mathrm{Re}\sum_{k\leq N}x^{i\gamma_k}$")
ax[1].set_title(r"Zeros of $\zeta$: Landau term $-\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}}$ (dotted)")
ax[1].legend(fontsize=8, ncol=2)
fig.tight_layout()
savefig(fig, "fig04_exponential_sums.png")
sio.savemat(os.path.join(OUT, "s03_results.mat"), {"N": Nk, "x": np.array(xs),
                                                   "S_RVM": np.array([res[x][0] for x in xs]),
                                                   "S_zeta": np.array([res[x][1] for x in xs])})
say.close()
```

A.5. **s04_primes.py.**

```
"""Experiment 4: the prime-power count J_RVM, the synthetic prime count pi_RVM, the cell
masses c_n^RVM and the real RVM prime levels p_j^RVM, compared with J, pi(x) and the primes.

J_o(x) = li(x) - 2 Re sum_k Ei((1/2 + i o_k) log x) - log 2 + int_x^inf dt/(t(t^2-1) log t)
for an ordinate list o_k (o = tau: J_RVM; o = gamma: Riemann's explicit formula for J),
pi_o(x) = sum_{m <= log2 x} mu(m)/m J_o(x^(1/m)).
The zero sums are evaluated on a uniform grid in u = log x by the recurrence
exp(i o_k (u + du)) = exp(i o_k u) exp(i o_k du); Ei is exact (scipy exp1) for k <= 300 and
uses Ei(w) = e^w (1/w + 1/w^2 + 2/w^3 + 6/w^4) for k > 300, where |w| > 400."""
import os
import numpy as np
import scipy.io as sio
from scipy import special, integrate
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, ei_upper, mobius_sieve, primes_upto, write_csv,
                        savefig, clock_inverse, OUT)

say = Tee("s04_primes")
gam, tau = load_ordinates()
KE = 300
```

```
def zero_sum_ei(ords, u, K=None, taper=None):
    """sum_k Ei((1/2 + i o_k) u) on a uniform grid u, k <= K, optional weights."""
    o = ords if K is None else ords[:K]
    wts = np.ones(o.size) if taper is None else taper
    rho = 0.5 + 1j * o
    out = np.zeros(u.size, dtype=complex)
    # exact part
    for i in range(0, KE):
        out += wts[i] * ei_upper(rho[i] * u)
    # asymptotic part by recurrence
    r = rho[KE:]
    b = [wts[KE:] / r ** m for m in (1, 2, 3, 4)]
    du = u[1] - u[0]
    stepf = np.exp(1j * o[KE:] * du)
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * o[KE:] * u[j])
        A = [E @ bm for bm in b]
        uj = u[j]
        out[j] += np.exp(uj / 2) * (A[0] / uj + A[1] / uj ** 2 + 2 * A[2] / uj ** 3 + 6 * A[3] / uj ** 4)
        E = E * stepf
    return out


def tail_integral(x):
    return integrate.quad(lambda t: 1.0 / (t * (t * t - 1.0) * np.log(t)), x, np.inf, limit=200)[0]


def J_from(ords, u, K=None, taper=None):
    x = np.exp(u)
    S = zero_sum_ei(ords, u, K, taper)
    ti = np.array([tail_integral(v) for v in x])
    return special.expi(u) - 2 * S.real - np.log(2.0) + ti


def pi_from_J(u, J):
    mu = mobius_sieve(64)
    out = np.zeros(u.size)
    for i, ui in enumerate(u):
        m = 1
        while ui / m >= np.log(2.0) - 1e-12:
            if mu[m] != 0:
                out[i] += mu[m] / m * np.interp(ui / m, u, J)
            m += 1
    return out


def exact_J_pi(x):
    P = primes_upto(int(x.max()) + 1)
    pi = np.searchsorted(P, x, side="right").astype(float)
    J = np.zeros(x.size)
    for m in range(1, 40):
        J += np.searchsorted(P, x ** (1.0 / m) + 1e-12, side="right") / m
    return J, pi


def run(xmax, npts, K, label):
    u = np.linspace(np.log(2.0), np.log(xmax), npts)
    if K is None:
        tg = tr = None
        Kg = Kr = None
    else:
        Kg = Kr = K
        v = np.arange(K) / K
        tg = tr = np.where(v < 0.6, 1.0, np.cos(np.pi / 2 * (v - 0.6) / 0.4) ** 2)
```

```
    JR = J_from(tau, u, Kr, tr)
    JG = J_from(gam, u, Kg, tg)
    PR = pi_from_J(u, JR)
    PG = pi_from_J(u, JG)
    x = np.exp(u)
    Je, Pe = exact_J_pi(x)
    write_csv("s04_prime_counts_%s.csv" % label, ["x", "J_exact", "J_from_zeta_zeros", "J_RVM",
                                                  "pi_exact", "pi_from_zeta_zeros", "pi_RVM", "li"],
              [x, Je, JG, JR, Pe, PG, PR, special.expi(u)])
    return x, Je, JG, JR, Pe, PG, PR


# x <= 100 with all zeros; x <= 1000 with a smooth cut at K zeros
x1, Je1, JG1, JR1, Pe1, PG1, PR1 = run(100.0, 3000, None, "x100_allzeros")
say("x <= 100, all %d zeros of each list, %d grid points:" % (gam.size, x1.size))
say("  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): %.4f",
    np.max(np.abs(PG1 - Pe1)[np.min(np.abs(x1[:, None] - primes_upto(101)[None, :]), axis=1) > 0.2]))
say("  pi_RVM(100) = %.4f, pi(100) = %d, li(100) = %.4f, J_RVM(100) = %.4f, J(100) = %.4f",
    PR1[-1], Pe1[-1], special.expi(np.log(100.0)), JR1[-1], Je1[-1])
dPR = np.diff(PR1)
say("  pi_RVM monotone on [2,100]: %s (min increment on the grid %.3e)", bool(np.all(dPR > 0)), dPR.min())

tauK = 2 * np.pi * 1000.0 * 2.5
K = int(np.searchsorted(tau, tauK))
x2, Je2, JG2, JR2, Pe2, PG2, PR2 = run(1000.0, 40000, K, "x1000_K%d" % K)
say("x <= 1000, smooth cut at K = %d zeros (tau_K = %.1f), %d grid points:" % (K, tau[K - 1], x2.size))
say("  pi_RVM(1000) = %.4f, pi(1000) = %d, li(1000) = %.4f", PR2[-1], Pe2[-1], special.expi(np.log(1000.0)))

# cell masses and real prime levels from the fine x<=1000 run
n = np.arange(2, 1001)
pin = np.interp(n, x2, PR2)
c = np.diff(np.concatenate([[0.0], pin]))
P = primes_upto(1000)
isp = np.isin(n, P).astype(float)
say("cell masses c_n = pi_RVM(n) - pi_RVM(n-1):")
say("%5s %10s %10s %8s" % ("n", "c_n", "1/log n", "prime"))
for i in range(0, 40):
    say("%5d %10.5f %10.5f %8d" % (n[i], c[i], 1 / np.log(n[i]), isp[i]))
say("sum_{n<=1000} c_n = %.4f,  pi(1000) = %d", c.sum(), len(P))
levels = np.arange(1, int(np.floor(PR2.max())) + 1)
pj = np.array([x2[np.argmax(PR2 >= j)] for j in levels])
say("real RVM prime levels p_j^RVM (first crossing of level j) versus p_j:")
say("%5s %12s %8s %10s" % ("j", "p_j^RVM", "p_j", "diff"))
for j in range(min(30, levels.size)):
    say("%5d %12.4f %8d %10.4f" % (levels[j], pj[j], P[j], pj[j] - P[j]))
write_csv("s04_cell_masses.csv", ["n", "c_n_RVM", "inv_log_n", "is_prime"], [n, c, 1 / np.log(n), isp])
m = min(levels.size, P.size)
write_csv("s04_rvm_prime_levels.csv", ["j", "p_j_RVM", "p_j"], [levels[:m], pj[:m], P[:m]])

# ---- figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].step(x1, Pe1, where="post", color="k", lw=1.6, label=r"$\pi(x)$")
ax[0].plot(x1, PG1, "b-", lw=0.6, label=r"$\pi(x)$ from %d zeros of $\zeta$" % gam.size)
ax[0].plot(x1, PR1, "r-", lw=1.4, label=r"$\pi_{\rm RVM}(x)$ from %d $\tau_k$" % tau.size)
ax[0].plot(x1, special.expi(np.log(x1)) - special.expi(np.log(2.0)), "g--", lw=1, label=r"$\mathrm{li}(x)-\
    mathrm{li}(2)$")
ax[0].set_xlabel("x")
ax[0].set_title(r"Prime counts on $[2,100]$")
ax[0].legend(fontsize=7)
ax[1].step(x2, Pe2, where="post", color="k", lw=1.2, label=r"$\pi(x)$")
ax[1].plot(x2, PR2, "r-", lw=1.0, label=r"$\pi_{\rm RVM}(x)$")
ax[1].set_xlabel("x")
ax[1].set_title(r"Prime counts on $[2,1000]$ ($K=%d$ ordinates, smooth cut)" % K)
ax[1].legend(fontsize=8)
fig.tight_layout()
```

```
savefig(fig, "fig05_prime_counts.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
nn = n[:98]
ax[0].bar(nn, isp[:98], width=0.8, color="0.75", label="prime indicator")
ax[0].plot(nn, c[:98], "r.-", lw=0.8, ms=4, label=r"$c_n^{\rm RVM}$")
ax[0].plot(nn, 1 / np.log(nn), "g--", lw=1, label=r"$1/\log n$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Cell masses $c_n^{\rm RVM}=\pi_{\rm RVM}(n)-\pi_{\rm RVM}(n-1)$")
ax[0].legend(fontsize=8)
ax[1].plot(P[:m], pj[:m], "r.", ms=4, label=r"$(p_j,\,p_j^{\rm RVM})$")
ax[1].plot([0, 1000], [0, 1000], "k-", lw=0.8, label="diagonal")
ax[1].set_xlabel(r"$p_j$")
ax[1].set_ylabel(r"$p_j^{\rm RVM}$")
ax[1].set_title(r"Real RVM prime levels $\pi_{\rm RVM}(p_j^{\rm RVM})=j$ versus primes")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig06_rvm_primes.png")

sio.savemat(os.path.join(OUT, "s04_results.mat"), {"x100": x1, "pi100": Pe1, "piZeta100": PG1, "piRVM100": PR1,

                                                   "x1000": x2, "pi1000": Pe2, "piRVM1000": PR2,
                                                   "c_n": c, "p_RVM": pj, "K": K})
say.close()
```

A.6. **s05_mertens.py.**

```
"""Experiment 5: the Mertens function M, the synthetic Mertens function M_RVM^crit and the
ternary walk M_h with values in {-1,0,+1} on the squarefree support.

Explicit formulas (simple zeros, residues a = 1/(rho zeta'(rho))):
  M_zeta(x) = -2 + 2 Re sum_k a_k x^(rho_k) + trivial terms        (zeros of zeta)
  M_RVM(x)  = -2 + 2 Re sum_k b_k x^(1/2 + i tau_k) + trivial terms (b_k = 1/(rho_k zeta_RVM'(rho_k)))
Ternary walk: mu_h(n) = mu(n)^2 * sign, sign = +1 if M_h(n-1) < target(n) else -1."""
import os
import math
import numpy as np
import mpmath as mp
import scipy.io as sio
from scipy.interpolate import PchipInterpolator
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import (Tee, load_ordinates, Product, logG, clock_inverse, tail_coefficients,
                        theta_rs, mobius_sieve, write_csv, savefig, OUT)

say = Tee("s05_mertens")
gam, tau = load_ordinates()
Nz = gam.size
XMAX = 10 ** 6
K0 = 1000

# ------------------------------------------------------------------ RVM residues
c1, c2, c3 = tail_coefficients(clock_inverse(float(Nz)))
t2 = tau ** 2
zpR = np.zeros(K0, dtype=complex)
for k in range(K0):
    tk = tau[k]
    fac = (t2 - tk ** 2) / (t2 + 0.25)
    fac[k] = 1.0
    q = -0.25 - tk ** 2
    lmod = np.sum(np.log(np.abs(fac))) + c1 * q - c2 * q ** 2 + c3 * q ** 3
    rho = 0.5 + 1j * tk
    lP = np.log(2j * tk / (0.25 + tk ** 2)) + lmod
    zpR[k] = (-1) ** k * np.exp(lP - logG(rho))
```

```
ratio = zpR / np.log(tau[:K0] / (2 * np.pi))
PR = Product(tau)
Ts = 1.0e4 * np.array([0.25, 0.5, 1.0])
from rvm_common import zeta_from_product, w_transition
v = np.array([zeta_from_product(PR, 2 + 1j * T)[0] / (1 + w_transition(2 + 1j * T)) for T in Ts])
varpi = np.linalg.solve(np.column_stack([np.ones(3), 1 / Ts, 1 / Ts ** 2]).astype(complex), v)[0].real
sel = np.arange(K0 // 2, K0)
cfit = np.mean((ratio[sel] / varpi - 1) * tau[sel])
say("zeta_RVM'(rho_k) exact for k <= %d; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = %d: %.10f %+.2e i",
    K0, K0, ratio[-1].real, ratio[-1].imag)
say("varpi_RVM = %.10f; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = %.4f %+.4f i",
    varpi, cfit.real, cfit.imag)
zR = varpi * np.log(tau / (2 * np.pi)) * (1 + cfit / tau)
zR[:K0] = zpR
bR = 1.0 / ((0.5 + 1j * tau) * zR)

# ------------------------------------------------------------------ residues of zeta
def rs_Z(t):
    a = np.sqrt(t / (2 * np.pi))
    Nn = np.floor(a).astype(int)
    p = a - Nn
    nmax = Nn.max()
    n = np.arange(1, nmax + 1)
    ph = theta_rs(t)[:, None] - t[:, None] * np.log(n)[None, :]
    mask = n[None, :] <= Nn[:, None]
    Z = 2 * np.sum(np.cos(ph) * mask / np.sqrt(n)[None, :], axis=1)
    cc = np.cos(2 * np.pi * p)
    C0 = np.cos(2 * np.pi * (p ** 2 - p - 1.0 / 16)) / cc
    return Z + (-1.0) ** (Nn - 1) * (t / (2 * np.pi)) ** (-0.25) * C0


mp.mp.dps = 20
KM = 1000
zG = np.zeros(Nz, dtype=complex)
for k in range(KM + 50):
    zG[k] = complex(mp.zeta(mp.mpc(0.5, gam[k]), 1, 1))
h = 1e-5
for i0 in range(KM, Nz, 4000):
    i1 = min(Nz, i0 + 4000)
    t = gam[i0:i1]
    Zp = (rs_Z(t + h) - rs_Z(t - h)) / (2 * h)
    val = -1j * np.exp(-1j * theta_rs(t)) * Zp
    if i0 == KM:
        chk = np.max(np.abs(val[:50] / zG[KM:KM + 50] - 1))
        say("Riemann-Siegel zeta'(rho) versus mpmath at k = %d..%d: max relative difference %.2e",
            KM + 1, KM + 50, chk)
        zG[KM + 50:i1] = val[50:]
    else:
        zG[i0:i1] = val
aG = 1.0 / ((0.5 + 1j * gam) * zG)
S2G = 2 * np.sum(np.abs(aG) ** 2)
S2R = 2 * np.sum(np.abs(bR) ** 2)
say("median |zeta'(rho)| = %.4f;  spectral sums 2 sum|a|^2: zeta %.6f, RVM %.6f", np.median(np.abs(zG)), S2G,
    S2R)


def trivial_zeta(x, n=6):
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        v += (-1) ** (k - 1) * (2 * np.pi / x) ** (2 * k) / (math.factorial(2 * k) * k * float(mp.zeta(2 * k +
     1)))
    return v


def trivial_rvm(x, n=6):
```

```
    v = np.zeros_like(x)
    for k in range(1, n + 1):
        s = -2.0 * k
        lP = np.real(PR.logP(s)[0])
        v -= x ** (-2 * k) * (2 * k + 1) * np.pi ** k * 2 * (-1) ** k / (math.factorial(k) * np.exp(lP))
    return v


def explicit_uniform(ords, wts, u):
    """2 Re sqrt(x) sum_k w_k exp(i o_k u) on a uniform u grid (recurrence)."""
    out = np.zeros(u.size)
    st = np.exp(1j * ords * (u[1] - u[0]))
    for j in range(u.size):
        if j % 200 == 0:
            E = np.exp(1j * ords * u[j])
        out[j] = 2 * np.exp(u[j] / 2) * np.real(E @ wts)
        E = E * st
    return out


# ---------------------------------------------------------------- exact M by sieve
mu = mobius_sieve(XMAX)
M = np.cumsum(mu.astype(np.int64))
sqf = mu != 0
say("M(10^6) = %d;  squarefree density Q(10^6)/10^6 = %.6f (6/pi^2 = %.6f)", M[XMAX], sqf[1:].mean(), 6 / np.
    pi ** 2)

# ---------------------------------------------------------------- explicit formulas on [2, 2000]
u = np.linspace(np.log(2.0), np.log(2000.0), 2500)
xg = np.exp(u)
Mz = -2 + explicit_uniform(gam, aG, u) + trivial_zeta(xg)
Mc = -2 + explicit_uniform(tau, bR, u) + trivial_rvm(xg)
Mfl = M[np.floor(xg).astype(int)]
off = np.abs(xg - np.round(xg)) > 0.05
say("explicit formula with %d zeros of zeta on [2,2000]: rms(M_zeta - M) = %.4f off integers (|x-n|>0.05)",
    Nz, np.sqrt(np.mean((Mz - Mfl)[off] ** 2)))
write_csv("s05_explicit_formulas.csv", ["x", "M_floor", "M_zeta_explicit", "M_RVM_explicit"], [xg, Mfl, Mz, Mc
    ])

# ---------------------------------------------------------------- integer target for M_RVM^crit
def target_grid(X, div=24):
    x = [1.0]
    while x[-1] < X:
        x.append(min(X, x[-1] + max(1.0, np.floor(np.sqrt(x[-1]) / div))))
    return np.array(x)


def tapered(x, ords, a, C1=2.0, C2=4.0):
    S = np.zeros(x.size, dtype=complex)
    lx = np.log(x)
    i0 = 0
    while i0 < x.size:
        i1 = min(x.size, i0 + 200)
        K = max(1, int(np.searchsorted(ords, C2 * 2 * np.pi * np.sqrt(x[i1 - 1]))))
        uu = ords[None, :K] / (2 * np.pi * np.sqrt(x[i0:i1, None]))
        w = np.where(uu <= C1, 1.0, np.where(uu >= C2, 0.0, np.cos(np.pi / 2 * (uu - C1) / (C2 - C1)) ** 2))
        E = np.exp(1j * lx[i0:i1, None] * ords[None, :K])
        S[i0:i1] = np.sqrt(x[i0:i1]) * np.sum(E * w * a[None, :K], axis=1)
        i0 = i1
    return S


xT = target_grid(float(XMAX))
TT = -2 + 2 * np.real(tapered(xT, tau, bR)) + trivial_rvm(xT)
nn = np.arange(1, XMAX + 1)
```

```
T = np.concatenate([[0.0], PchipInterpolator(xT, TT)(nn)])
nchk = np.arange(1000, 100001, 997)
full = -2 + 2 * np.real(np.array([np.sqrt(n) * np.sum(np.exp(1j * np.log(n) * tau) * bR) for n in nchk])) +
    trivial_rvm(nchk.astype(float))
say("target on %d grid points; at %d integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = %.4f",
    xT.size, nchk.size, np.max(np.abs(full - T[nchk])))

# ------------------------------------------------------------------ ternary walk
mu_h = np.zeros(XMAX + 1, dtype=np.int8)
Mh = np.zeros(XMAX + 1)
sgn = np.zeros(XMAX + 1, dtype=np.int8)
cur = 0.0
for n in range(1, XMAX + 1):
    if sqf[n]:
        sgn[n] = 1 if cur < T[n] else -1
        cur += sgn[n]
        mu_h[n] = sgn[n]
    Mh[n] = cur
err = Mh[1:] - T[1:]
say("ternary walk: max |M_h - target| on [1,1e6] = %.4f, on [1e4,1e6] = %.4f", np.max(np.abs(err)), np.max(np.
    abs(err[9999:])))
dens = lambda a: [np.mean(a[1:] == -1), np.mean(a[1:] == 0), np.mean(a[1:] == 1)]
say("densities of (-1, 0, +1):  mu   %.6f %.6f %.6f", *dens(mu))
say("                           mu_h %.6f %.6f %.6f   (3/pi^2 = %.6f, 1 - 6/pi^2 = %.6f)", *dens(mu_h), 3 / np.
    pi ** 2, 1 - 6 / np.pi ** 2)
say("sign agreement mu_h = mu on squarefrees: %.6f;  sum mu mu_h / Q = %.6f",
    np.mean(mu_h[sqf] == mu[sqf]), np.sum(mu_h.astype(int) * mu.astype(int)) / sqf[1:].sum())
say("max |M(n)|/sqrt(n) on [10,1e6] = %.4f;  max |M_RVM(n)|/sqrt(n) = %.4f;  max |M_h(n)|/sqrt(n) = %.4f",
    np.max(np.abs(M[10:]) / np.sqrt(nn[9:])), np.max(np.abs(T[10:]) / np.sqrt(nn[9:])), np.max(np.abs(Mh[10:])
     / np.sqrt(nn[9:])))
say("\nNUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)")
say("%4s %4s %6s %10s %8s %6s %6s %6s %6s" % ("n", "mu", "mu^2", "target", "M_h(n-1)", "sign", "mu_h", "M_h",
    "M"))
rows = []
for n in range(1, 41):
    say("%4d %4d %6d %10.4f %8d %6s %6d %6d %6d" % (n, mu[n], abs(mu[n]), T[n], Mh[n - 1],
                                                    ("+1" if sgn[n] > 0 else ("-1" if sgn[n] < 0 else "-")),
                                                    mu_h[n], Mh[n], M[n]))
    rows.append([n, mu[n], abs(mu[n]), T[n], Mh[n - 1], sgn[n], mu_h[n], Mh[n], M[n]])
rows = np.array(rows, dtype=float)
write_csv("s05_ternary_example.csv", ["n", "mu", "mu2", "target", "Mh_prev", "sign", "mu_h", "Mh", "M"],
          [rows[:, i] for i in range(rows.shape[1])])

Ns = 10 ** np.arange(3, 7)
say("\nENERGY E_N = (1/N) sum_{n<=N} W(n)^2")
say("%10s %14s %14s %14s" % ("N", "E_N(M)", "E_N(M_RVM)", "E_N(M_h)"))
EN = []
for Nn in Ns:
    EN.append([Nn, np.mean(M[1:Nn + 1].astype(float) ** 2), np.mean(T[1:Nn + 1] ** 2), np.mean(Mh[1:Nn + 1] **
      2)])
    say("%10d %14.4f %14.4f %14.4f" % tuple(EN[-1]))
EN = np.array(EN)
write_csv("s05_energy.csv", ["N", "E_M", "E_MRVM", "E_Mh"], [EN[:, i] for i in range(4)])
dec = np.arange(1, XMAX + 1, 25)
write_csv("s05_walks_decimated.csv", ["n", "M", "M_RVM_target", "M_h"], [dec, M[dec], T[dec], Mh[dec]])
mu_h[1:].astype(np.int8).tofile(os.path.join(OUT, "mu_h_int8_1e6.bin"))

# ------------------------------------------------------------------ figures (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 2001)
ax[0].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[0].plot(w, T[w], "r-", lw=1.2, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[0].step(w, Mh[w], where="post", color="tab:green", lw=0.8, label=r"$M_h(n)$")
ax[0].set_xlabel("n")
ax[0].set_title(r"Mertens functions on $[1,2000]$, every integer")
```

```
ax[0].legend(fontsize=8)
nb = 2000
bw = XMAX // nb
xb = (np.arange(nb) + 0.5) * bw
Mb = M[1:].reshape(nb, bw)
Tb = T[1:].reshape(nb, bw)
ax[1].fill_between(xb, Mb.min(1), Mb.max(1), color="0.6", label=r"$M(n)$ (range per block)")
ax[1].plot(xb, Tb.mean(1), "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(n)$")
ax[1].plot(xb, np.sqrt(xb), "k--", lw=0.8, label=r"$\pm\sqrt{n}$")
ax[1].plot(xb, -np.sqrt(xb), "k--", lw=0.8)
ax[1].set_xlabel("n")
ax[1].set_title(r"$[1,10^6]$, blocks of %d integers" % bw)
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig07_mertens.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
w = np.arange(1, 61)
ax[0].vlines(w - 0.15, 0, mu[w], color="k", lw=2, label=r"$\mu(n)$")
ax[0].vlines(w + 0.15, 0, mu_h[w], color="tab:green", lw=2, label=r"$\mu_h(n)$")
ax[0].axhline(0, color="k", lw=0.5)
ax[0].set_xlabel("n")
ax[0].set_yticks([-1, 0, 1])
ax[0].set_title(r"Ternary values: same zero set $\{\mu(n)=0\}$, signs chosen by tracking")
ax[0].legend(fontsize=8)
w = np.arange(1, 301)
ax[1].step(w, M[w], where="post", color="k", lw=1.0, label=r"$M(n)$")
ax[1].plot(w, T[w], "r.-", lw=0.8, ms=3, label=r"target $M_{\rm RVM}^{\rm crit}(n)$")
ax[1].step(w, Mh[w], where="post", color="tab:green", lw=1.2, label=r"$M_h(n)$")
ax[1].set_xlabel("n")
ax[1].set_title(r"The walk $M_h$ follows $M_{\rm RVM}^{\rm crit}$ within $\pm1$ step")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig08_ternary_walk.png")

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
sel = xg <= 300
ax[0].step(np.arange(1, 301), M[1:301], where="post", color="k", lw=1.0, label=r"$M(x)$")
ax[0].plot(xg[sel], Mz[sel], "b-", lw=0.7, label=r"explicit formula, %d zeros of $\zeta$" % Nz)
ax[0].plot(xg[sel], Mc[sel], "r-", lw=1.0, label=r"$M_{\rm RVM}^{\rm crit}(x)$, %d $\tau_k$" % Nz)
ax[0].set_xlabel("x")
ax[0].set_title("Explicit formulas on [2,300]")
ax[0].legend(fontsize=7)
kk = np.arange(5000)
ax[1].plot(gam[kk], np.abs(zG[kk]) / np.log(gam[kk] / (2 * np.pi)), "b.", ms=1.5, label=r"$|\zeta'(\rho_k)|/\
    log(\gamma_k/2\pi)$")
ax[1].plot(tau[kk], np.abs(zR[kk]) / np.log(tau[kk] / (2 * np.pi)), "r.", ms=1.5, label=r"$|\zeta_{\rm RVM}'(\
    rho^{\rm RVM}_k)|/\log(\tau_k/2\pi)$")
ax[1].axhline(varpi, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[1].set_xlabel("ordinate")
ax[1].set_ylim(0, 4)
ax[1].set_title(r"Derivatives at the zeros, $k\leq5000$")
ax[1].legend(fontsize=7)
fig.tight_layout()
savefig(fig, "fig09_explicit_weights.png")

sio.savemat(os.path.join(OUT, "s05_results.mat"), {"varpi": varpi, "cfit": cfit, "S2G": S2G, "S2R": S2R,
                                                   "energy": EN, "ternary_example": rows,
                                                   "x_explicit": xg, "M_zeta_explicit": Mz, "M_RVM_explicit":
    Mc})
say.close()
```

A.7. **s06_prime_ratios.py.**

```
"""Experiment 6: the three ratios li/pi_RVM, pi/li, pi/pi_RVM and the square-root estimate
```

```
pi_RVM(x) - li(x) = O(sqrt(x) (log log x)^2 / log x), from the prime counts written by s04_primes.py.

Input: output/s04_prime_counts_x1000_K17061.csv (columns x, pi_exact, pi_RVM, li, ...).
B(x) = sqrt(x) (log log x)^2 / log x is the scale of the square-root estimate."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, savefig, write_csv, OUT

say = Tee("s06_prime_ratios")
D = np.genfromtxt(os.path.join(OUT, "s04_prime_counts_x1000_K17061.csv"), delimiter=",", names=True)
x, pe, pr, li = D["x"], D["pi_exact"], D["pi_RVM"], D["li"]
say("grid: %d points on [%.0f, %.0f]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)"
    % (x.size, x[0], x[-1]))

sel = x >= 10
r1, r2, r3 = li[sel] / pr[sel], pe[sel] / li[sel], pe[sel] / pr[sel]
xs = x[sel]
say("")
say("ratios at selected x")
say("%8s %12s %12s %12s %12s %12s %12s" % ("x", "pi", "pi_RVM", "li", "li/pi_RVM", "pi/li", "pi/pi_RVM"))
for X in [10, 20, 50, 100, 200, 500, 1000]:
    i = int(np.argmin(np.abs(x - X)))
    say("%8.1f %12.4f %12.4f %12.4f %12.6f %12.6f %12.6f"
        % (x[i], pe[i], pr[i], li[i], li[i] / pr[i], pe[i] / li[i], pe[i] / pr[i]))

s8 = x >= 8
B = np.sqrt(x[s8]) * np.log(np.log(x[s8])) ** 2 / np.log(x[s8])
e_rvm = (pr[s8] - li[s8]) / B
e_pi = (pe[s8] - li[s8]) / B
say("")
say("square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]")
say("max |pi_RVM - li| / B = %.4f   (at x = %.2f)" % (np.max(np.abs(e_rvm)), x[s8][np.argmax(np.abs(e_rvm))]))
say("max |pi - li| / B     = %.4f   (at x = %.2f)" % (np.max(np.abs(e_pi)), x[s8][np.argmax(np.abs(e_pi))]))
for X in [100, 1000]:
    m = x[s8] >= X / 2
    say("max |pi_RVM - li| / B on [%d, 1000] = %.4f" % (X // 2, np.max(np.abs(e_rvm[m]))))

# CSV files: every 10th grid point
write_csv("s06_prime_ratios.csv", ["x", "li_over_pi_RVM", "pi_over_li", "pi_over_pi_RVM"],
          [xs[::10], r1[::10], r2[::10], r3[::10]])
write_csv("s06_square_root_scale.csv", ["x", "pi_RVM_minus_li_over_B", "pi_minus_li_over_B"],
          [x[s8][::10], e_rvm[::10], e_pi[::10]])

# ---- figure (linear axes)
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].plot(xs, r3, "k-", lw=0.8, label=r"$\pi(x)/\pi_{\rm RVM}(x)$")
ax[0].plot(xs, r2, "b-", lw=0.8, label=r"$\pi(x)/\mathrm{li}(x)$")
ax[0].plot(xs, r1, "r-", lw=1.4, label=r"$\mathrm{li}(x)/\pi_{\rm RVM}(x)$")
ax[0].axhline(1.0, color="0.5", lw=0.8, ls="--")
ax[0].set_xlabel("x")
ax[0].set_title(r"The three ratios on $[10,1000]$")
ax[0].legend(fontsize=8)
ax[1].plot(x[s8], e_pi, "b-", lw=0.6, label=r"$(\pi-\mathrm{li})/B$")
ax[1].plot(x[s8], e_rvm, "r-", lw=1.4, label=r"$(\pi_{\rm RVM}-\mathrm{li})/B$")
ax[1].axhline(0.0, color="0.5", lw=0.8, ls="--")
ax[1].set_xlabel("x")
ax[1].set_title(r"Square-root scale $B(x)=\sqrt{x}\,(\log\log x)^2/\log x$")
ax[1].legend(fontsize=8)
fig.tight_layout()
savefig(fig, "fig10_prime_ratios.png")
say.close()
```

A.8. **s07_product_identity.py.**

```
"""Experiment 7: the value at infinity of the endpoint.

varpi_Z1 = lim_{sigma -> +inf} zeta_Z1(sigma) = varpi_RVM * prod_k (1/4 + tau_k^2) / (1/4 + gamma_k^2),
so zeta_Z1 = zeta holds exactly when the ordered partial products
P_K = prod_{k <= K} (1/4 + gamma_k^2) / (1/4 + tau_k^2)
converge to varpi_RVM.  The partial products are computed from data/zeross.bin (gamma_k) and
data/zLamb.bin (tau_k) and compared with varpi_RVM from s02_transition.py."""
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from rvm_common import Tee, load_ordinates, savefig, write_csv

say = Tee("s07_product_identity")
gam, tau = load_ordinates()
VARPI = 0.9619415807          # Richardson value of varpi_RVM (s02_transition.py)
N = gam.size
logP = np.cumsum(np.log1p(0.25 / gam ** 2) + 2 * np.log(gam) - np.log1p(0.25 / tau ** 2) - 2 * np.log(tau))
P = np.exp(logP)
say("N = %d ordinates; varpi_RVM = %.10f" % (N, VARPI))
say("%10s %16s %14s" % ("K", "P_K", "P_K - varpi"))
for K in [10, 100, 1000, 10 ** 4, 10 ** 5, 10 ** 6, N]:
    say("%10d %16.10f %14.3e" % (K, P[K - 1], P[K - 1] - VARPI))
for K0 in [10 ** 4, 10 ** 5, 10 ** 6]:
    seg = P[K0 - 1:]
    say("K >= %7d: min P_K = %.10f, max P_K = %.10f, mean = %.10f" % (K0, seg.min(), seg.max(), seg.mean()))
step = 50
idx = np.arange(step - 1, N, step)
write_csv("s07_partial_products.csv", ["K", "P_K"], [idx + 1, P[idx]])

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
k1 = np.arange(1, 10 ** 4 + 1)
ax[0].plot(k1, P[:10 ** 4], "r-", lw=0.6, label=r"$P_K$")
ax[0].axhline(VARPI, color="k", lw=0.8, ls="--", label=r"$\varpi_{\rm RVM}$")
ax[0].set_xlabel("K")
ax[0].set_title(r"Partial products $P_K$, $K\leq 10^4$")
ax[0].legend(fontsize=8)
sel = idx >= 10 ** 5 - 1
ax[1].plot(idx[sel] + 1, (P[idx[sel]] - VARPI) * 1e6, "r-", lw=0.6)
ax[1].axhline(0.0, color="k", lw=0.8, ls="--")
ax[1].set_xlabel("K")
ax[1].set_ylabel(r"$(P_K-\varpi_{\rm RVM})\cdot 10^{6}$")
ax[1].set_title(r"$10^5\leq K\leq %d$" % N)
fig.tight_layout()
savefig(fig, "fig11_product_identity.png")
say.close()
```

Appendix B. Console output of the programs

The complete console output of each program, as written to `output/*_console.txt`.

B.1. **s01_clock_and_family.**

```
zeros loaded: N = 2001052, gamma_1 = 14.134725142, gamma_N = 1132490.658714, tau_1 = 14.521346953, tau_N =
    1132490.722588
max |N_RVM(tau_k) - (k - 1/2)| = 6.985e-10
eta_k = tau_k - gamma_k:  max |eta| = 0.993711 (k = 34),  mean |eta| = 0.151209,  rms = 0.188066,  mean eta =
    1.731e-07
     k from         k to     max|eta|      rms eta
          1           10     0.840022     0.545768
         11          100     0.993711     0.453166
        101         1000     0.915450     0.345177
       1001        10000     0.833881     0.274177
      10001       100000     0.785835     0.226352
```

```
      100001      1000000     0.762816     0.192411
     1000001      2001052     0.717359     0.178900
first ordinates:
   k          gamma_k           tau_k        eta_k
   1     14.134725142     14.521346953     0.386622
   2     21.022039639     20.655740356    -0.366299
   3     25.010857580     25.492675432     0.481818
   4     30.424876126     29.739411632    -0.685464
   5     32.935061588     33.624531889     0.689470
   6     37.586178159     37.257370087    -0.328808
   7     40.918719012     40.700551604    -0.218167
   8     43.327073281     43.994013953     0.666941
   9     48.005150881     47.165128427    -0.840022
  10     49.773832478     50.233653391     0.459821
  11     52.970321478     53.214422874     0.244101
  12     56.446247697     56.118921948    -0.327326
sigma = 2.00, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 8.706e-10, median = 2.685e-10
sigma = 1.25, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 1.128e-09, median = 3.693e-10
sigma = 0.75, t in [1,2000] (499 points): max |zeta_Z1/zeta - 1| = 2.629e-09, median = 5.179e-10
critical line, t in [0.5,60]: max |Z_Z1(t) - Z(t)| / max(1,|Z(t)|) = 8.370e-10
total run time 30.6 s
```

B.2. **s01b_clock_zoom.**

```
zoom table near the top of the file:
        k            gamma_k              tau_k      eta_k   N*-N_RVM
  2001041     1132485.303923     1132485.011569  -0.292355  -0.563104
  2001042     1132485.662638     1132485.530752  -0.131886  -0.254025
  2001043     1132486.244063     1132486.049936  -0.194127  -0.373908
  2001044     1132486.392256     1132486.569120   0.176863   0.340656
  2001045     1132486.874881     1132487.088303   0.213422   0.411072
  2001046     1132487.571868     1132487.607487   0.035619   0.068605
  2001047     1132487.950694     1132488.126670   0.175977   0.338949
  2001048     1132488.905328     1132488.645854  -0.259474  -0.499774
  2001049     1132489.162780     1132489.165037   0.002257   0.004348
  2001050     1132489.585215     1132489.684221   0.099006   0.190696
  2001051     1132490.165305     1132490.203404   0.038099   0.073383
  2001052     1132490.658714     1132490.722588   0.063873   0.123026
total run time 0.6 s
```

B.3. **s02_transition.**

```
varpi_RVM (Richardson, sigma = 2, T = 2500, 5000, 10000) = 0.9619415807 -1.28e-10 i
L = 1/varpi_RVM = 1.0395641690
sigma' = 0.25: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0617
sigma' = 0.25: |zeta_RVM/(varpi(1+chi)) - 1|: max t*err (t>100) = 0.0790
          |w| * t^sigma' in [1.5832, 1.5832];  |1/zeta_RVM - L| at t = 1e4: 1.425e-01
sigma' = 0.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.181, max t*err = 0.0618
          |w| * t^sigma' in [2.5061, 2.5066];  |1/zeta_RVM - L| at t = 1e4: 2.543e-02
sigma' = 1.50: |zeta_RVM/(varpi(1+w)) - 1|: log-log slope on t>100 = -1.182, max t*err = 0.0620
          |w| * t^sigma' in [15.6619, 15.7496];  |1/zeta_RVM - L| at t = 1e4: 1.598e-05
total run time 11.5 s
```

B.4. **s03_exponential_sums.**

```
     x     max|S_RVM| max|S_RVM|/log    |S_zeta(N)| Landau -T/2pi*L/sqrt(x)   |M_H^RVM(x)|
   2.0         3.5745         0.2695       88300.77        -88341.59      1.722e-06
   3.0         3.9750         0.3759      114267.92       -114324.31      1.175e-06
   4.0         3.9523         0.3732       62435.29        -62466.94      8.004e-07
   6.0         4.5264         0.4547           0.26            -0.00      2.262e-06
  10.0         9.6989         1.1495           2.42            -0.00      4.031e-06
total run time 1.7 s
```

B.5. **s04_primes.**

```
x <= 100, all 2001052 zeros of each list, 3000 grid points:
  max |pi_from_zeta_zeros - pi| away from primes (distance > 0.2): 0.4035
  pi_RVM(100) = 26.0782, pi(100) = 25, li(100) = 30.1261, J_RVM(100) = 29.5889, J(100) = 28.5333
  pi_RVM monotone on [2,100]: False (min increment on the grid -2.528e-01)
x <= 1000, smooth cut at K = 17061 zeros (tau_K = 15707.6), 40000 grid points:
  pi_RVM(1000) = 169.9750, pi(1000) = 168, li(1000) = 177.6097
cell masses c_n = pi_RVM(n) - pi_RVM(n-1):
    n        c_n    1/log n    prime
    2    0.52944    1.44270        1
    3    1.04188    0.91024        1
    4    0.53593    0.72135        0
    5    0.46739    0.62133        1
    6    0.36089    0.55811        0
    7    0.63254    0.51390        1
    8    0.62393    0.48090        0
    9   -0.37669    0.45512        0
   10    0.17151    0.43429        0
   11    0.69779    0.41703        1
   12    0.32703    0.40243        0
   13    0.36227    0.38987        1
   14    0.57662    0.37892        0
   15    0.19671    0.36927        0
   16   -0.05284    0.36067        0
   17    0.39958    0.35296        1
   18    0.76865    0.34598        0
   19    0.43500    0.33962        1
   20   -0.03342    0.33381        0
   21    0.03009    0.32846        0
   22    0.37466    0.32352        0
   23    0.40277    0.31893        1
   24    0.11768    0.31466        0
   25    0.01006    0.31067        0
   26    0.27350    0.30693        0
   27    0.55508    0.30341        0
   28    0.47627    0.30010        0
   29    0.12577    0.29697        1
   30   -0.08367    0.29401        0
   31    0.07329    0.29121        1
   32    0.38040    0.28854        0
   33    0.43927    0.28600        0
   34    0.34619    0.28358        0
   35    0.06419    0.28127        0
   36   -0.02475    0.27906        0
   37    0.15448    0.27694        1
   38    0.40992    0.27491        0
   39    0.49628    0.27296        0
   40    0.35226    0.27109        0
   41    0.14030    0.26928        1
sum_{n<=1000} c_n = 169.9750,  pi(1000) = 168
real RVM prime levels p_j^RVM (first crossing of level j) versus p_j:
    j      p_j^RVM      p_j       diff
    1       2.2139        2     0.2139
    2       3.1550        3     0.1550
    3       5.1887        5     0.1887
    4       7.1593        7     0.1593
    5      11.1406       11     0.1406
    6      14.0208       13     1.0208
    7      17.2552       17     0.2552
    8      19.1781       19     0.1781
    9      26.0536       23     3.0536
   10      28.0447       29    -0.9553
   11      33.0707       31     2.0707
   12      38.0931       37     1.0931
   13      40.2282       41    -0.7718
```

```
   14      44.1861       43     1.1861
   15      46.1221       47    -0.8779
   16      57.1779       53     4.1779
   17      59.1201       59     0.1201
   18      61.0809       61     0.0809
   19      67.0384       67     0.0384
   20      74.0931       71     3.0931
   21      76.1235       73     3.1235
   22      83.0822       79     4.0822
   23      86.0513       83     3.0513
   24      92.0825       89     3.0825
   25      94.0781       97    -2.9219
   26      96.1319      101    -4.8681
   27     103.2219      103     0.2219
   28     112.1163      107     5.1163
   29     115.0277      109     6.0277
   30     118.1431      113     5.1431
total run time 107.7 s
```

B.6. **s05_mertens.**

```
zeta_RVM'(rho_k) exact for k <= 1000; zeta_RVM'(rho_k)/log(tau_k/2pi) at k = 1000: 0.9619388429 -1.41e-05 i
varpi_RVM = 0.9619415807; fit zeta_RVM'(rho_k) = varpi log(tau_k/2pi)(1 + c/tau_k), c = -0.0052 -0.0208 i
Riemann-Siegel zeta'(rho) versus mpmath at k = 1001..1050: max relative difference 1.19e-05
median |zeta'(rho)| = 7.8899;  spectral sums 2 sum|a|^2: zeta 0.029034, RVM 0.027203
M(10^6) = 212;  squarefree density Q(10^6)/10^6 = 0.607926 (6/pi^2 = 0.607927)
explicit formula with 2001052 zeros of zeta on [2,2000]: rms(M_zeta - M) = 0.0005 off integers (|x-n|>0.05)
target on 49896 grid points; at 100 integers <= 1e5: max |M_RVM(n) (all zeros) - target(n)| = 0.0191
ternary walk: max |M_h - target| on [1,1e6] = 2.6155, on [1e4,1e6] = 1.6539
densities of (-1, 0, +1):  mu   0.303857 0.392074 0.304069
                           mu_h 0.304003 0.392074 0.303923   (3/pi^2 = 0.303964, 1 - 6/pi^2 = 0.392073)
sign agreement mu_h = mu on squarefrees: 0.500344;  sum mu mu_h / Q = 0.000688
max |M(n)|/sqrt(n) on [10,1e6] = 0.8321;  max |M_RVM(n)|/sqrt(n) = 0.7541;  max |M_h(n)|/sqrt(n) = 0.6882

NUMERICAL EXAMPLE OF THE TERNARY DIVISION (first 40 integers)
   n   mu   mu^2     target M_h(n-1)   sign   mu_h    M_h      M
   1    1      1     0.2925        0     +1      1      1      1
   2   -1      1     0.4563        1     -1     -1      0      0
   3   -1      1    -0.6523        0     -1     -1     -1     -1
   4    0      0    -0.8720       -1      -      0     -1     -1
   5   -1      1    -1.7255       -1     -1     -1     -2     -2
   6    1      1    -1.3063       -2     +1      1     -1     -1
   7   -1      1    -1.5931       -1     -1     -1     -2     -2
   8    0      0    -2.4029       -2      -      0     -2     -2
   9    0      0    -1.3292       -2      -      0     -2     -2
  10    1      1    -1.3280       -2     +1      1     -1     -1
  11   -1      1    -2.3272       -1     -1     -1     -2     -2
  12    0      0    -2.2067       -2      -      0     -2     -2
  13   -1      1    -1.8468       -2     +1      1     -1     -3
  14    1      1    -1.9324       -1     -1     -1     -2     -2
  15    1      1    -1.5435       -2     +1      1     -1     -1
  16    0      0    -1.0414       -1      -      0     -1     -1
  17   -1      1    -1.6970       -1     -1     -1     -2     -2
  18    0      0    -3.0095       -2      -      0     -2     -2
  19   -1      1    -3.2869       -2     -1     -1     -3     -3
  20    0      0    -2.2604       -3      -      0     -3     -3
  21    1      1    -1.2515       -3     +1      1     -2     -2
  22    1      1    -1.1120       -2     +1      1     -1     -1
  23   -1      1    -1.3607       -1     -1     -1     -2     -2
  24    0      0    -1.3669       -2      -      0     -2     -2
  25    0      0    -1.3810       -2      -      0     -2     -2
  26    1      1    -1.9805       -2     +1      1     -1     -1
  27    0      0    -3.0141       -1      -      0     -1     -1
  28    0      0    -3.6155       -1      -      0     -1     -1
  29   -1      1    -3.2066       -1     -1     -1     -2     -2
```

```
  30   -1      1    -2.1599       -2     -1     -1     -3     -3
  31   -1      1    -1.3649       -3     +1      1     -2     -4
  32    0      0    -1.2973       -2      -      0     -2     -4
  33    1      1    -1.6435       -2     +1      1     -1     -3
  34    1      1    -1.7783       -1     -1     -1     -2     -2
  35    1      1    -1.4711       -2     +1      1     -1     -1
  36    0      0    -1.0723       -1      -      0     -1     -1
  37   -1      1    -1.0774       -1     -1     -1     -2     -2
  38    1      1    -1.5897       -2     +1      1     -1     -1
  39    1      1    -2.2227       -1     -1     -1     -2      0
  40    0      0    -2.4881       -2      -      0     -2      0

ENERGY E_N = (1/N) sum_{n<=N} W(n)^2
         N         E_N(M)     E_N(M_RVM)       E_N(M_h)
      1000        14.6400        18.6280        18.5960
     10000       163.6747       195.6965       196.0683
    100000      1622.3189      1308.3041      1308.8186
   1000000     14851.0857     12537.9008     12538.1174
total run time 132.6 s
```

B.7. **s06_prime_ratios.**

```
grid: 40000 points on [2, 1000]; pi_RVM from the first 17061 ordinates tau_k (smooth cut)

ratios at selected x
       x           pi       pi_RVM           li    li/pi_RVM        pi/li    pi/pi_RVM
    10.0       4.0000       3.9876       6.1657     1.546218     0.648747     1.003105
    20.0       8.0000       7.6636       9.9052     1.292496     0.807657     1.043893
    50.0      15.0000      14.7686      18.4695     1.250592     0.812151     1.015670
   100.0      25.0000      26.0813      30.1265     1.155098     0.829835     0.958541
   200.0      46.0000      43.3656      50.1910     1.157393     0.916499     1.060749
   500.0      95.0000      93.7489     101.7975     1.085854     0.933225     1.013346
  1000.0     168.0000     169.9750     177.6097     1.044916     0.945895     0.988381

square-root scale B(x) = sqrt(x) (log log x)^2 / log x on [8, 1000]
max |pi_RVM - li| / B = 2.7213   (at x = 9.73)
max |pi - li| / B     = 2.4490   (at x = 11.00)
max |pi_RVM - li| / B on [50, 1000] = 1.3455
max |pi_RVM - li| / B on [500, 1000] = 0.7216
total run time 1.0 s
```

B.8. **s07_product_identity.**

```
N = 2001052 ordinates; varpi_RVM = 0.9619415807
        K              P_K    P_K - varpi
       10     0.9625640008      6.224e-04
      100     0.9624787401      5.372e-04
     1000     0.9622891707      3.476e-04
    10000     0.9619513787      9.798e-06
   100000     0.9619301865     -1.139e-05
  1000000     0.9619418189      2.382e-07
  2001052     0.9619415390     -4.166e-08
K >=   10000: min P_K = 0.9618023968, max P_K = 0.9620641927, mean = 0.9619418623
K >=  100000: min P_K = 0.9619210208, max P_K = 0.9619605076, mean = 0.9619417591
K >= 1000000: min P_K = 0.9619389488, max P_K = 0.9619439825, mean = 0.9619416633
total run time 1.2 s
```

References

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